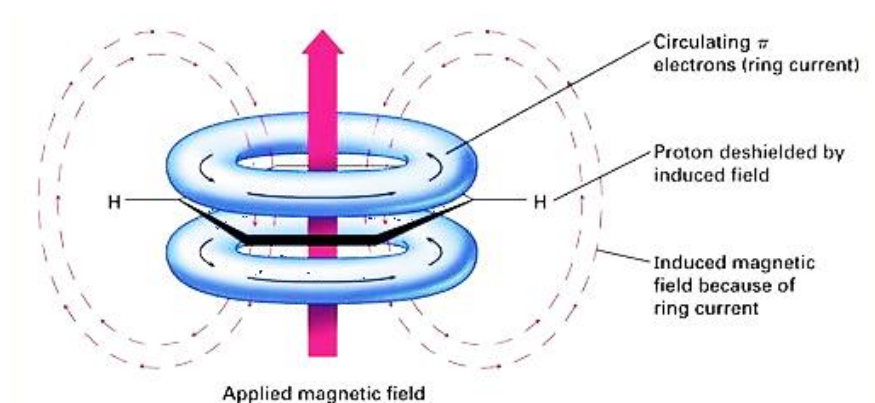


INTERPRETIVE ORGANIC SPECTROSCOPY

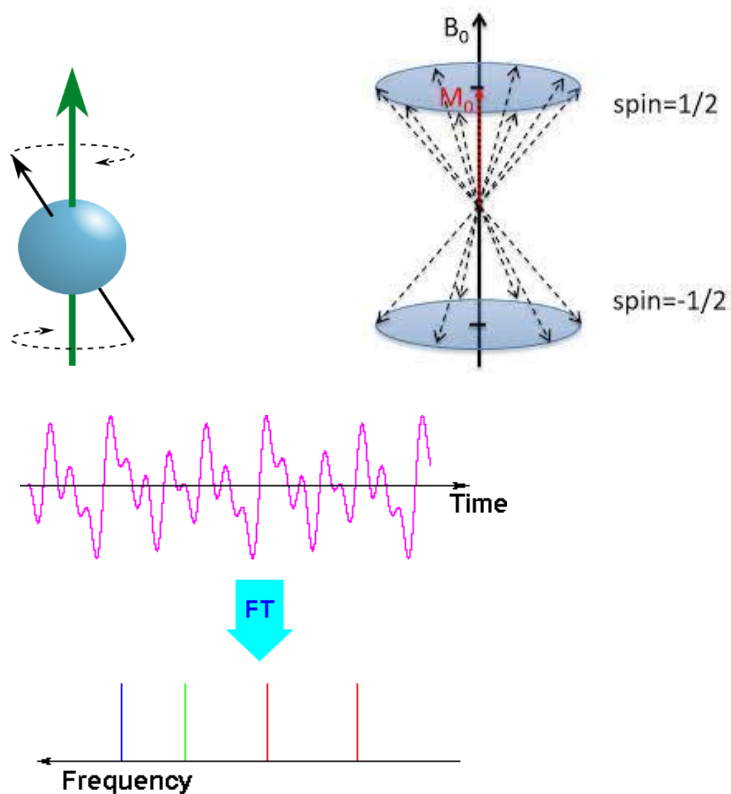


Dr. Gul Nabi

PhD (Jilin University P.R. China)

Associate professor of Chemistry

Government Postgraduate College Charsadda



GOVERNMENT POSTGRADUATE COLLEGE CHARSADDA

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*Chapter 1***NMR INSTRUMENTATION****INTRODUCTION**

Nuclear magnetic resonance spectroscopy, most commonly known as NMR spectroscopy or magnetic resonance spectroscopy (MRS), is a spectroscopic technique to observe local magnetic fields around atomic nuclei.

It is a spectroscopy technique that is based on the absorption of electromagnetic radiation in the radiofrequency region 4 to 900 MHz by nuclei of the atoms.

The sample is placed in a magnetic field and the NMR signal is produced by excitation of the nuclei sample with radio waves into nuclear magnetic resonance, which is detected with sensitive radio receivers.

NMR INSTRUMENTATION

There are two types of NMR spectrometers:

- I. Continuous wave (CW), and
- II. Pulsed Fourier transform (FT)

I. Continuous Wave NMR

Cw-NMR spectrometers have largely been replaced with pulsed FT-NMR instruments. However due to the lower maintenance and operating costs of cw instruments, they are still commonly used for routine ^1H NMR spectroscopy.

In low-resolution cw instruments electromagnets are cooled with water while in FT-NMR spectrometers are cooled with liquid helium.

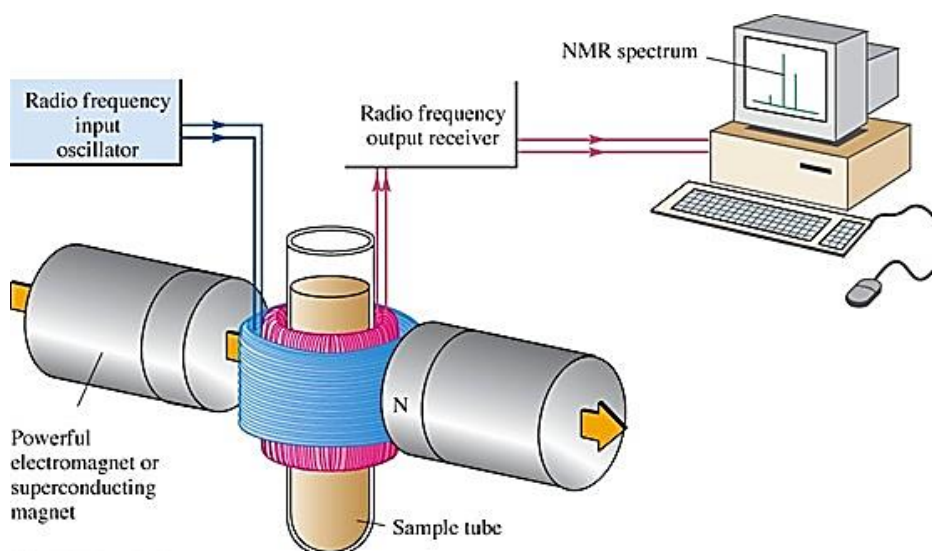
Construction

A CW-NMR spectrometer consists of a control console, magnet, and two orthogonal coils of wire that serve as antennas for radiofrequency (RF) radiation. One coil is attached to an RF generator and serves as a transmitter. This coil supplies the electromagnetic energy to change the spin orientation of

proton. Perpendicular to the RF oscillator coil is a detector coil also called the RF pick-up coil and is attached to the detection electronics.

Since the two coils are orthogonal, the pick-up coil cannot directly receive any radiation from the generator coil. When a nucleus absorbs RF radiation, it can become reoriented due to its normal movement in solution and re-emit the RF radiation in a direction that can be received by the pick-up coil.

A typical CW-spectrometer is shown in the following diagram.



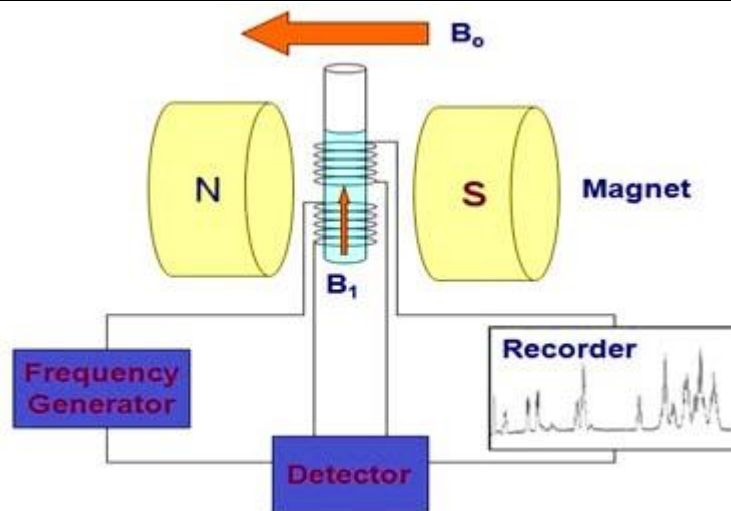
The instrument responds by recording this as a resonance signal or peak. This orthogonal coil arrangement greatly increases the sensitivity of NMR spectroscopy. Spectra are obtained by scanning the magnet and recording the pick-up coil signal on paper at the control console.

Working

A solution of the sample in a uniform 5 mm glass tube is spun between the poles of a powerful magnet. Two versions of CW spectrometers are known. In one type the spectrum is scanned by the field-sweep method while in other type by the frequency-sweep method.

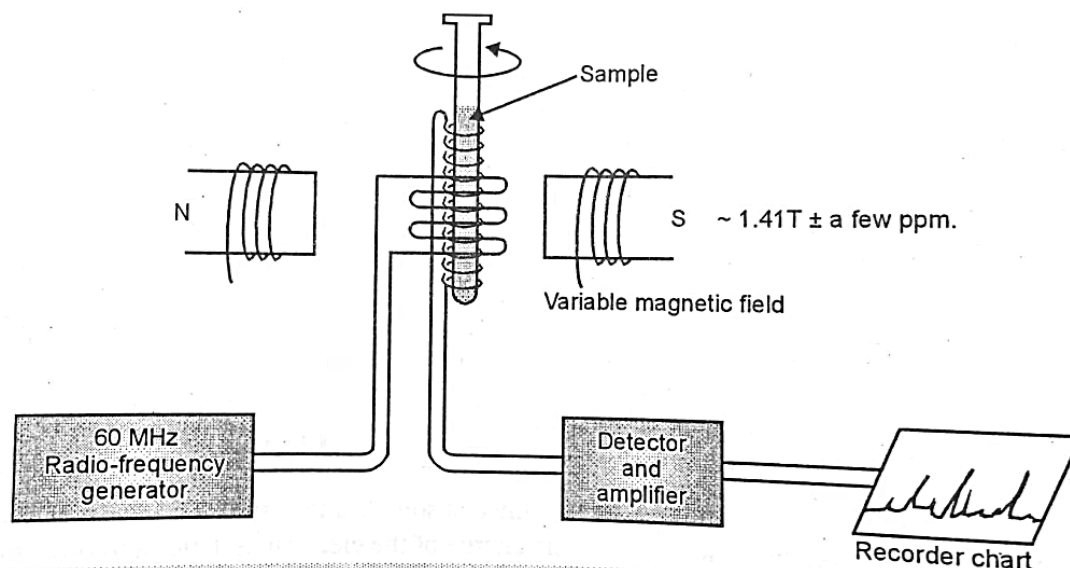
a. Frequency Sweep Method

In the frequency-sweep method, the magnetic field is held constant, which keeps the nuclear spin energy levels constant. Radio frequency radiation of appropriate energy is broadcasted into the sample from an antenna coil (transmitter). A receiver coil surrounds the sample tube, and emission of absorbed rf energy is monitored by dedicated electronic devices and a computer.



b. Field Sweep Method

The typical CW NMR spectrometer uses a constant frequency RF signal and varies the magnetic field strength. As the magnetic field strength is increased, the precessional frequencies of all the protons increase. When the precessional frequency of a given type of proton reaches 60 MHz, it has resonance. The magnet that is varied is actually a two-part device: (i) the main magnet with a constant strength (about 1.41 tesla in 60 MHz spectrometer), (ii) which is capped by electromagnetic pole pieces. By varying the current through the pole pieces, the worker can increase the main field strength by as much as 20 ppm. Changing the field in this way systematically brings all different types of protons of the sample into resonance.

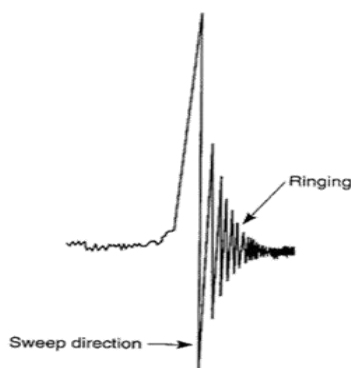


Instruments which vary the magnetic field in a continuous fashion scanning from the downfield end to the upfield end of the spectrum called **continuous-wave (cw) instruments**. Because the

chemical shifts of the peaks in this spectrum are calculated from frequency differences from TMS this type of spectrum is known as **frequency domain spectrum**.

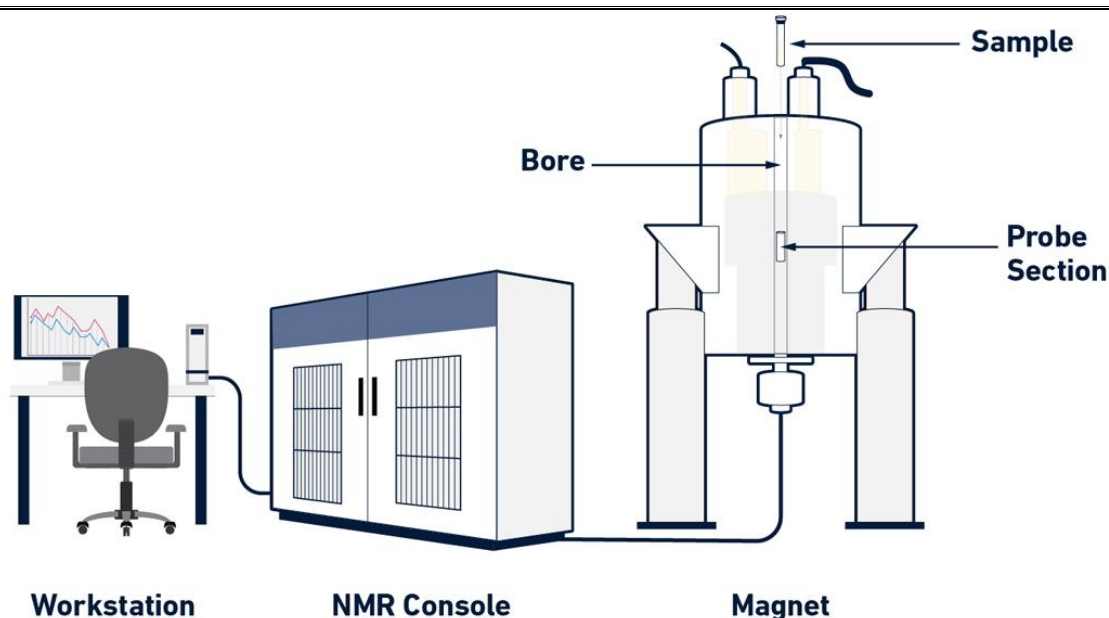
Ringling

A distinctive characteristic enables one to recognize a CW spectrum. Peaks generated by a CW instrument have ringling, a decreasing series of oscillations that occurs after the instrument has scanned through the peak. Ringling occurs because the excited nuclei do not have time to relax back to their equilibrium state before the field, and pen, of the instrument have advanced to a new position. The excited nuclei have a relaxation rate that is slower than the rate of scan. As a result, they are still emitting an oscillating, rapidly decaying signal, which is recorded as ringling. Ringling is desirable in a CW instrument and is considered to indicate that the field homogeneity is well adjusted. Ringling is most noticeable when a peak is a sharp singlet (a single, isolated peak).



II. FT-NMR

An FT-NMR spectrometer consists of a control console, magnet, and a coil of wire that serves as the antenna for transmitting and receiving the RF radiation. Only one coil is necessary because signal reception does not begin until after the end of the excitation pulse.



Fourier-transform NMR spectrometers use a powerful but short burst of energy, called a **pulse**, that excites all of the magnetic nuclei in the molecule simultaneously and flip into the higher-energy alignment. An instrument with a 2.1 Tesla magnetic field uses a short ($1\text{--}10\ \mu\text{s}$) burst of 90 MHz energy to accomplish it. Due to the Heisenberg uncertainty principle, this frequency width of the RF pulse ($1\text{--}10\ \mu\text{s}$) is wide enough to simultaneously excite nuclei in all local environments.

A shorter pulse may not be able to establish a solid fundamental frequency. Therefore, the pulse actually contains a range of frequencies centered about the fundamental. This range of frequencies is great enough to excite all the distinct types of hydrogens in the molecule at once with this single burst of energy.

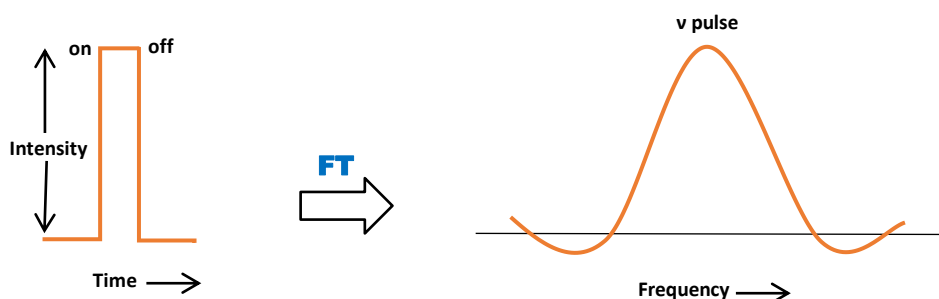


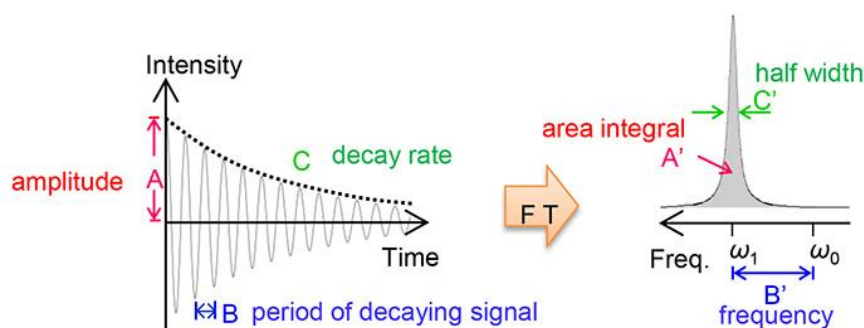
Figure: A short Pulse : (a) the original pulse (b) the frequency content of the same pulse

When the pulse is discontinued, the excited nuclei begin to lose their excitation energy and return to their original spin state, or relax. All of the nuclei will re-emit RF radiation at their respective resonance frequencies, creating an interference pattern in the resulting RF emission versus time. This emission is called **free induction decay (FID)** signal.

The intensity of the FID decays with time as all of the nuclei eventually lose their excitation. The FID is a superimposed combination of all the frequencies emitted and can be quite complex. Usually the individual frequencies due to different nuclei are extracted by using a computer and a mathematical method called a Fourier transform (FT) analysis.

FID can be detected with a radio-receiver coil that is perpendicular to the static magnetic field. The FID signal is digitized and stored in a computer for data processing. Ordinarily the time-domain decay signals from numerous successive pulses can be summed and averaged to improve the signal-to-noise ratio. The result is then converted to a frequency-domain signal by a Fourier transformation. The resulting frequency-domain output is similar to the spectrum produced by a scanning continuous-wave experiment.

The signal decays exponentially with time as the nuclei relax and their signal diminishes. Since the horizontal axis on this signal is time, the FID is sometimes called a **time-domain signal**. If the signal did not decay in intensity, it would appear as a sine (or cosine) wave of constant intensity, as shown in Figure.



Benefits of FT–NMR

a). Time-Saving

NMR active nuclei present in the molecule are scanned simultaneously and this results in greater useful structural information from samples containing nuclei other than ^1H .

Five to 10 minutes are required to scan and record a CW spectrum; a pulsed experiment is much faster, and a measurement of an FID can be performed in a few seconds.

b). Higher Sensitivity

Co-addition of spectral response over a given time period improves signal-to-noise ratios and this leads to improved sensitivity. Unlike CW–NMR which is used primarily for ^1H studies NMR information on other nuclei like ^{13}C , ^{31}P and ^{19}F give poor response due to their lower isotopic abundance. FT- NMR affords greater sensitivity for studies on such nuclei due to signal averaging.

Remarkable increase in sensitivity often results in useful information from the limited available sample amounts.

Pulsed FT-NMR is therefore especially suitable for the examination of nuclei that are not very abundant in nature, nuclei that are not strongly magnetic or very dilute samples.

FT-NMR instruments though expensive to operate and maintain yield a wealth of information which was not possible with CW instruments. Today most manufacturers have discontinued manufacture of CW instruments.

The most modern NMR spectrometers use superconducting magnets, which can have field strengths as high as 14 Tesla and operate at 600 MHz. A superconducting magnet is made of special alloys and must be cooled to liquid helium temperatures. The magnet is usually surrounded by a Dewar flask (an insulated chamber) containing liquid helium; in turn, this chamber is surrounded by another one containing liquid nitrogen. Instruments operating at frequencies above 100 MHz have superconducting magnets. NMR spectrometers with frequencies of 200 MHz, 300 MHz, and 400 MHz are now common in chemistry; instruments with frequencies of 900 MHz are used for special research projects.



THE COMPONENTS OF AN NMR INSTRUMENT

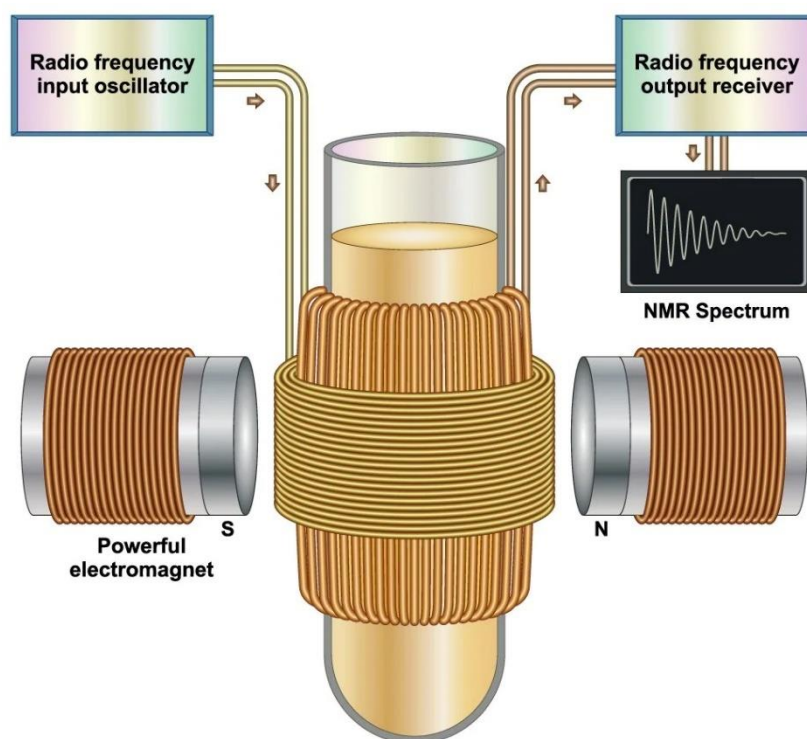
A typical NMR instrument consists mainly of the following parts:

1. The magnet
2. Probehead(s)

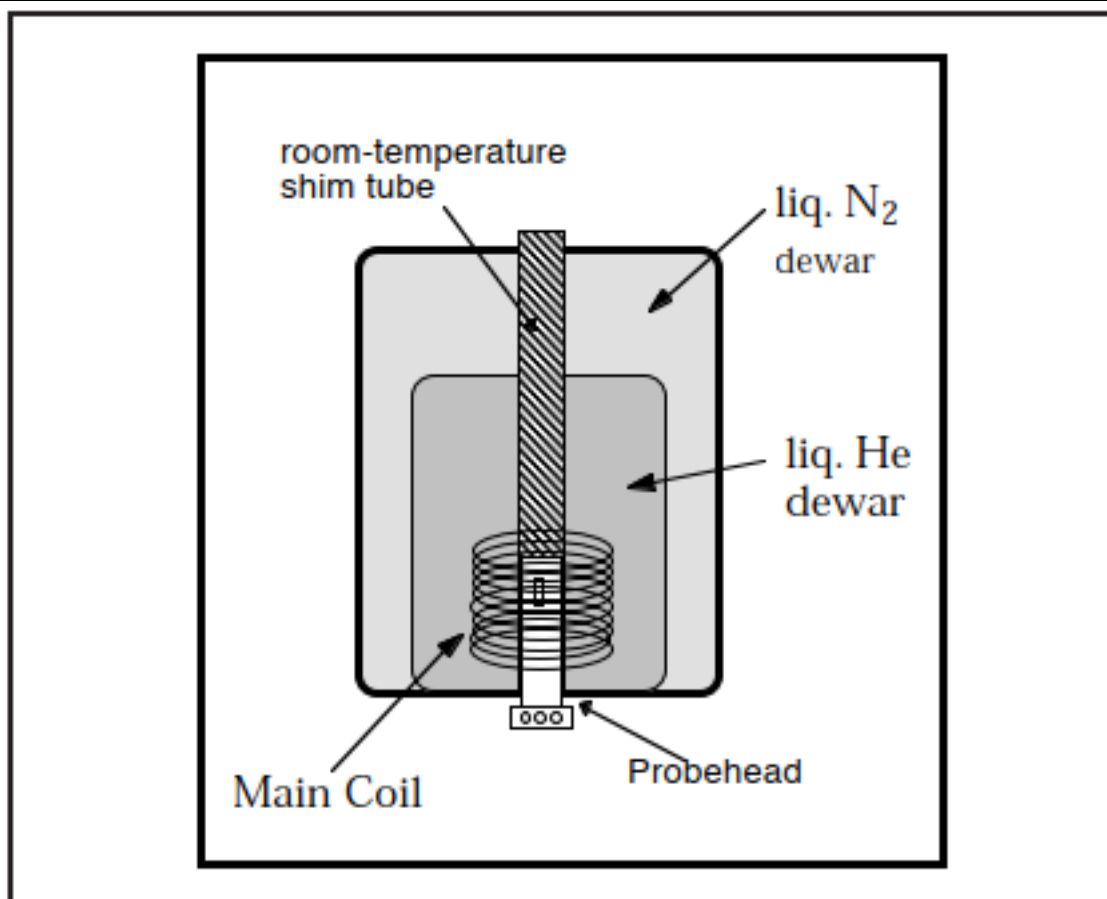
3. MR transceiver
4. The lock-system
5. The shim system
6. A computer system

1. The Magnet System

The magnet is the most important part of the NMR instrument. NMR uses a large magnet to probe the intrinsic spin properties of atomic nuclei. Both superconducting and electromagnets magnets are comprised of loops of wires with a current passing through, which results in the formation of a magnetic field.



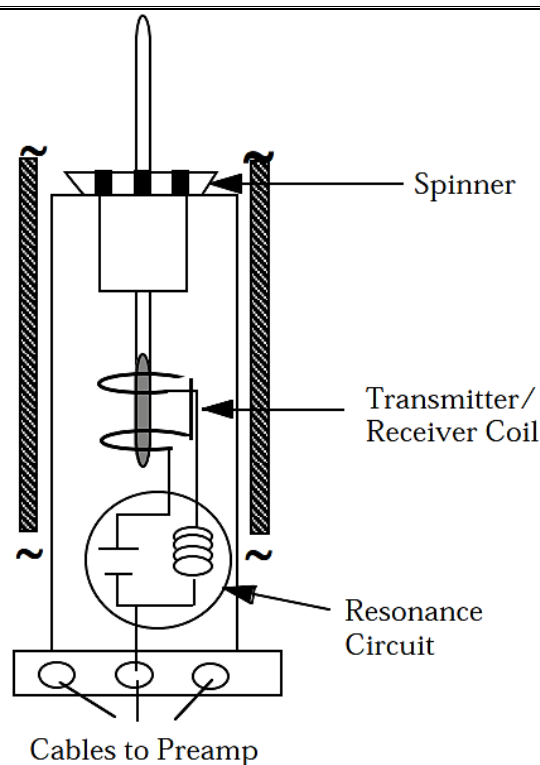
The magnet is typically composed of an alloy that is superconducting at low temperature. This alloy is made into a wire which is wrapped into a coil. The wires are kept cooled with liquid helium at 4K, which significantly reduces the resistance of the wire to nearly zero. The liquid **helium Dewar** is insulated by a vacuum chamber and then a liquid **nitrogen dewar** which itself is insulated by another vacuum chamber.



2. The Probe

The probe is a specialized piece of hardware that contains the coils and sample and therefore is responsible for tuning to the correct frequency, applying RF pulses, receiving the signal from the nuclei. More specialized probes can also spin the sample, vary the temperature, and change the angle of the sample and coil.

The probe goes into the center of the magnetic field, and the sample is inserted into the probe to perform the NMR experiment. The probe contains the **radiofrequency (Rf) coils**, tuned at specific frequencies for specific nuclei in a given magnetic field. The probe also contains the necessary hardware to control the sample temperature.



3. MR Transceiver

The MR transceiver contains two major subsystems: The transmitter and the receiver subsystems

- i) **Transmitter subsystem:** It consists of the *RF synthesizers* and *amplifiers*. This subsystem is responsible for generating pulse sequences containing RF pulses of specified frequency, amplitude, phase, shape and duration at specified times. Multiple RF synthesizers are required because many MR experiments require simultaneous application of RF pulses of different frequencies.
- ii) **Receiver subsystem:** This consists of the components: Preamplifier, Amplifier, Mixer and the Analog-to-Digital converter (ADC). The Mixer is used to subtract a reference frequency of specified phase from the observed signal, resulting a signal of lower frequency that can be easily digitized.

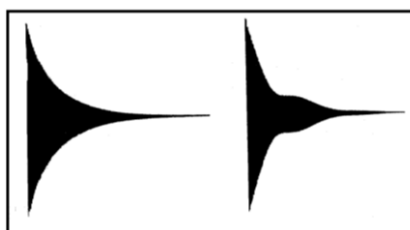
Transmitting and receiving is done on the same coil. The signal coming from the coil is blanked for a short time after the pulses (the so-called pre-scan delay) and then used for detection. The transmitter/receiver system contains mainly analog parts.

4. The Shim System

A shim is a device used to adjust the homogeneity of a magnetic field. Shimming is a process that is carried out to correct any inhomogeneities in the applied magnetic field during a nuclear magnetic resonance (NMR) experiment. To obtain a high-resolution spectrum, a magnetic field must be homogeneous to 1 part in 10^9 across the sample.

The name shims are after the thin metal pieces that NMR technicians would insert into the magnetic field to manually make a homogenous magnetic field. Nowadays, the shims are coils of wire which pass current to generate magnetic fields in certain directions. These shims vary across a variety of planes to ensure a very homogenous field is achieved.

When the field is highly homogenous, the FID should fall smoothly following an exponential decay. The resolution of the signal determines how long the FID lasts:



5. The Lock-System

The **lock system** is essentially a separate spectrometer designed to observe deuterium. Some deuterium must be introduced into samples that are under analysis. This can most easily be done by dissolving the sample in a deuterated solvent. The frequency of the signals emitted by deuterium for a particular size of magnet is precisely known. Therefore, if the magnetic field strength is correct, any deuterium nucleus in the sample should emit this exact frequency. If the magnet strength varies so will the deuterium frequency. The lock system uses a receiver to monitor this deuterium frequency and makes adjustments to the magnetic field strength accordingly.

The system has a feedback loop, which generates corrections to the magnetic field strength B_0 , such that the resonance frequency of the solvent deuterium line remains constant. If the field strength varies (drift), a current is changed in a special coil (the H_0 coil) located within the shim system of the magnet, which has the effect of returning the field strength to the correct value. As long as the system is locked, the user can be confident that the field is maintained at a constant strength during acquisition.

6. The Computer System

The NMR signals are received and recorded by using host computers attached to the spectrometer.

Chapter 2

THE NMR BASICS

INTRODUCTION

Nuclear Magnetic Resonance Spectroscopy is a powerful analytical technique used to characterize organic molecules by identifying carbon-hydrogen frameworks within molecules. The source of energy in NMR is radio waves which have long wavelengths, and thus low energy and frequency. When low-energy radio waves interact with a molecule, they can change the nuclear spins of some elements, including ^1H and ^{13}C .

NUCLEAR SPIN

An atomic nucleus behaves as tiny magnet. A nucleus consisting of p protons and n neutrons will have a total spin which is a vector combination of $p + n$ spins, all equal to $\frac{1}{2}$. The spin of a nucleus is denoted by the symbol I , called the **nuclear spin quantum number**. Unlike the electron, for which the **spin quantum number** s can take only one value, $\frac{1}{2}$ as each nucleus has a different combination of p and n and hence a different total spin value. I can be ≥ 0 and any multiple of $\frac{1}{2}$. Thus, nuclear spin can take on integral and half-integral values and even zero.

Proton	Neutron (n)	Nuclear (I) spin	Nuclei	Examples
Even	Even	0	Inactive	$^{16}_8\text{O}$
Odd	Odd	Full integer	Active	$^{14}_7\text{N}$, $^{12}_6\text{C}$
Even	Odd	Half integer	Active	$^{13}_6\text{C}$
Odd	Even	Half integer	Active	^1_1H , $^{31}_{15}\text{P}$

Based on experimental observations, some empirical rules regarding the value of the resultant spin quantum number (I) of nuclei have been formulated. These empirical rules are as follows:

- i) *Nuclei having even number of protons and neutrons have zero resultant nuclear spin quantum number (I).*

Examples: $^4_2\text{He} \Rightarrow (p = 2, n = 2) : I = 0$

$^{12}_6\text{C} \Rightarrow (p = 6, n = 6) : I = 0$

$$^{16}_8\text{O} \Rightarrow (p = 8, n = 8) : I = 0$$

ii) Nuclei with both odd number of protons and neutrons have integral nuclear spin quantum number (I).

Examples: $^2_1\text{H} \Rightarrow (p = 1, n = 1) : I = 1$

$$^{14}_7\text{N} \Rightarrow (p = 7, n = 7) : I = 1$$

$$^{10}_5\text{B} \Rightarrow (p = 5, n = 5) : I = 3$$

iii) Nuclei having odd-even combination of protons and neutrons have half integral resultant nuclear spin quantum number (I). Such nuclei have odd mass number and even or odd atomic number.

Example: $^1_1\text{H} \Rightarrow (p = 1, n = 0) : I = \frac{1}{2}$

$$^{13}_6\text{C} \Rightarrow (p = 6, n = 7) : I = \frac{1}{2}$$

$$^{19}_9\text{F} \Rightarrow (p = 9, n = 10) : I = \frac{1}{2}$$

$$^{35}_{17}\text{Cl} \Rightarrow (p = 17, n = 18) : I = \frac{3}{2}$$

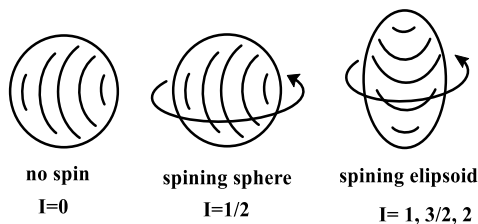
iv) Nuclei with $I = 0$ are nonmagnetic and are '**NMR silent**'.

v) Nuclei having non-zero value of resultant nuclear spin quantum number ($I \neq 0$) are said to be magnetic nuclei or nmr active nuclei.

vi) Spin $\frac{1}{2}$ nuclei have a **spherical nuclear charge distribution**, and their NMR behaviour is the easiest to understand.

vii) Other spin nuclei have **non-spherical charge distributions** and may be analyzed as **prolate** or **oblate** spinning bodies.

viii) All nuclei with non-zero spins have magnetic moments (μ^0), but the non-spherical nuclei also have an **electric quadrupole moment** (e_Q) due to **non-spherical nuclear charge distribution**.



ANGULAR MOMENTUM OF THE NUCLEUS

The magnitude of angular momentum vector (L) associated with the spinning nucleus having resultant nuclear spin quantum number I is given by:

$$L = \sqrt{I(I+1)} \frac{h}{2\pi}$$

The angular momentum vector associated with the spinning nucleus cannot orient in any arbitrary direction in space. It can orient itself in those directions only in which its components along the direction of an applied magnetic field have quantized values: integral (when I is integral) or half-integral (with I is half-integral). The direction of applied magnetic field is usually taken along Z -axis (reference direction). The allowed values of components of the angular momentum vector along the z -axis are given by the expression:

$$m_I \cdot \frac{h}{2\pi}$$

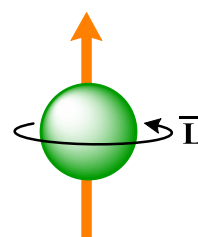
m_I can take the following values:

(a) When I is integral

$$m_I = I, (I-1), \dots, 0, \dots, -(I-1), -I$$

(b) When I is half integer

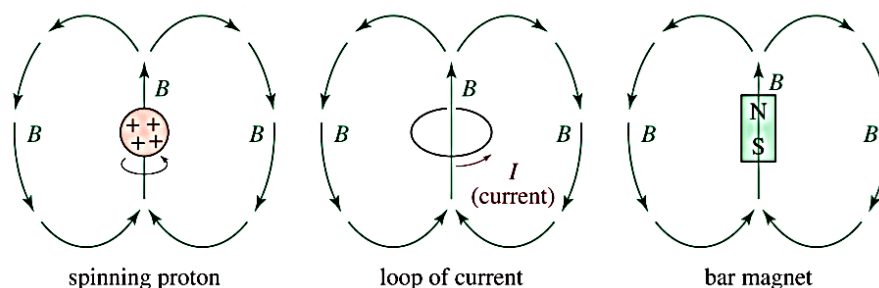
$$m_I = I, (I-1), \dots, +\frac{1}{2}, -\frac{1}{2}, \dots, -(I-1), -I$$



Thus, there are a total of $(2I+1)$ orientations that the nuclear spin angular momentum vector can take in an applied magnetic field. These orientations are degenerate, i.e., have same energy in the absence of an external magnetic field. However, the degeneracy of these orientations is destroyed in the presence of the applied magnetic field. The NMR spectrum is basically the result of the lifting up of the degeneracy of these energy states.

THE SPINNING CHARGED PARTICLE

A charged particle spinning about an axis generates a circular electric current, which in turn produces a magnetic dipole. In other words, the spinning particle behaves as a **tiny bar magnet** placed along the spin axis.



The size of the dipole (i.e. **strength of the magnet**) is proportional to the spin angular momentum L .

The value of dipole moment (u), representing the strength of the magnet, for a point charge (q) is given by:

$$u = \frac{q}{2m} L$$

$$u = \frac{q}{2m} \times \sqrt{I(I+1)} \frac{h}{2\pi}$$

$$u = \frac{qh}{4\pi m} \sqrt{I(I+1)} \text{ JT}^{-1}$$

Where γ is the **magnetogyric ratio** of the nucleus and has characteristic value for each nucleus.

Nuclei	Unpaired Protons	Unpaired Neutrons	Net Spin	γ (MHz/T)
^1H	1	0	1/2	42.58
^2H	1	1	1	6.54
^{31}P	0	1	$\frac{1}{2}$	17.25
^{23}Na	2	1	3/2	11.27
^{15}N	1	1	1	3.08
^{13}C	0	1	$\frac{1}{2}$	10.71
^{19}F	0	1	$\frac{1}{2}$	40.08

The net magnetization for a sample is the sum of the individual magnetic moments in the sample.

$$M = \sum_i \mu_i$$

The magnetization can then be written as

$$M = \gamma J$$

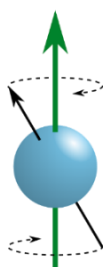
where J is the net spin angular momentum. The angular momentum J is quantized and can only take discrete values.

The spin states of the nucleus are quantified:

$$I, (I - 1), (I - 2), \dots, -I$$

THE LARMOR PRECESSION

When magnetic field is applied around the nucleus, the rotating axis of the spinning nucleus follows a **cone shape** about the axis of the applied magnetic field. Such motion is called **Larmor precession**. The precession angular frequency, ω is called its **Larmor frequency** and is directly proportional to the strength of the applied magnetic field, B_0 . The higher the applied magnetic field, the higher the rate of the Larmor frequency.



For a proton, if the applied field is 1.41 Tesla (14,100 Gauss), the frequency of precession is approximately 60 MHz.

$$\omega = 2\pi\nu \quad (i)$$

$$\omega = \gamma B_0 (\text{radian}) \quad (ii)$$

Comparing the two equations:

$$2\pi\nu = \gamma B_0$$

$$\nu = \frac{\gamma B_0}{2\pi} \quad \text{Larmor Frequency}$$

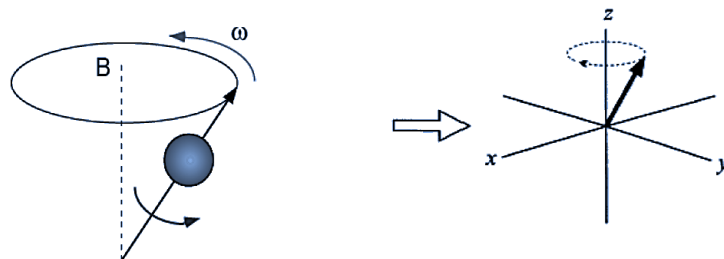
where

ν = Frequency in cycles per sec or Hz

B_0 = Strength of the magnetic field in gauss

γ = Nuclear constant or Gyromagnetic ratio

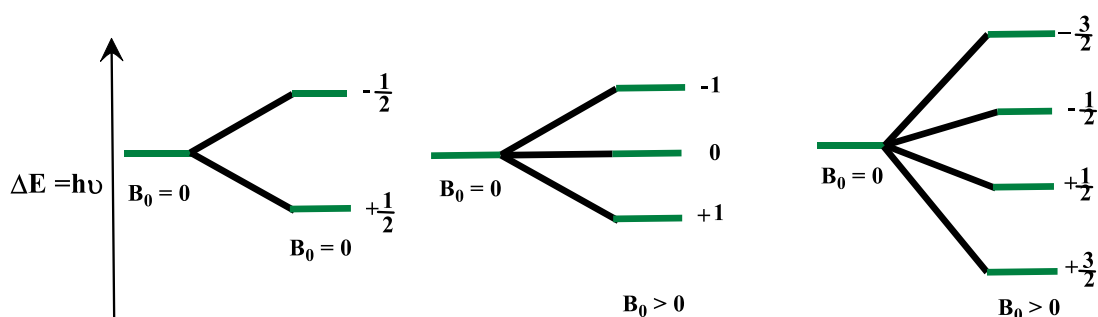
The direction of motion can be **clockwise** or **counterclockwise** and is determined by the sign of γ . By convention, the field is applied along the z- axis of a Cartesian Coordinate Frame.



THE ZEEMAN EFFECT

When a spinning nucleus is placed in a strong external magnetic field, its magnetic moment will either partially become parallel (*α spin state*) or antiparallel to the external magnetic field (*β spin state*). The two spin states now have different energies; the one parallel to the magnetic field is lower in energy while the other having antiparallel spin is higher in energy. This splitting of energy levels in the applied magnetic field is called ***Zeeman Effect***.

The total number of spin states are given by the relation $2I+1$.



The energy difference ΔE , which corresponds to the two states with $m = \pm 1/2$, depends upon the strength of the external magnetic field. According to Planks' equation, $\Delta E = h\nu$

The angular frequency may be written as:

$$\omega = 2\pi\nu \quad (\text{ii})$$

By rearranging

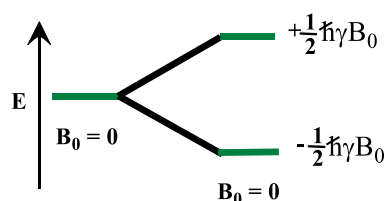
$$\nu = \frac{\omega}{2\pi} \quad (\text{iii})$$

Putting the value of ν into equation (i), we get

$$\Delta E = h \times \frac{\omega}{2\pi} = \omega \frac{h}{2\pi} = \omega \hbar \quad \left(\hbar = \frac{h}{2\pi} \right)$$

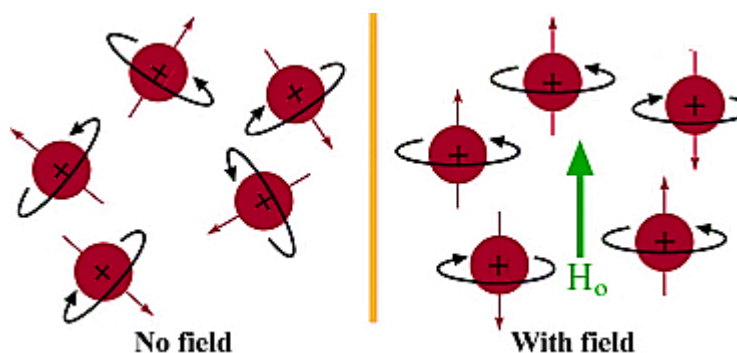
The angular frequency is given by the equation, $\omega = \gamma B_0$

$$\Delta E = \gamma B_0 \hbar$$



BOLTZMANN DISTRIBUTION OF SPIN STATES

When a charged particle such as a proton spins on its axis, it creates a magnetic field. Thus, the nucleus can be considered to be a tiny bar magnet. Normally, these tiny bar magnets are randomly oriented in space. However, in the presence of a magnetic field B_0 , they are oriented with or against this applied field. More nuclei are oriented with the applied field because this arrangement is lower in energy.



In a given sample of a specific nucleus, the nuclei will be distributed throughout the various spin states available. Because the energy separation between these states is comparatively small, energy from thermal collisions is sufficient to place many nuclei into higher energy spin states. The numbers of nuclei in each spin state are described by the Boltzman distribution

$$\frac{N_\alpha}{N_\beta} = e^{\Delta E/kT}$$

where N values are the numbers of nuclei in the respective spin states, ΔE is the difference in energy between the α and β spin states, k is the Boltzmann constant, and T is the temperature in degrees kelvin. Because $\Delta E/kT$ is very nearly zero, mathematically, this equal distribution raise e (2.718.) to the power of almost 0; thus, showing that the ratio of the populations of the two spin states is almost 1:1. In other

words, the upper and lower energy spin states are almost equally populated with only a very small excess in the lower energy state.

Example: Determine the relative population distribution between $m = -\frac{1}{2}$ and $m = +\frac{1}{2}$ for a proton at room temperature for the following NMR spectrometers:

(a) 60 MHz (b) 200 MHz (c) 300 MHz (d) 900 MHz

Solution:

(b) A 200 MHz field means that $\nu = 200 \times 10^6 \text{ s}^{-1}$

Convert radio frequency to angular frequency:

$$\omega = 2\pi\nu = 2\pi(200 \times 10^6 \text{ s}^{-1}) = 1.256 \times 10^9 \text{ s}^{-1}$$

Next to calculate B_0 , use the following equation

$$\omega = \gamma B_0$$

Now substitute, using the correct value for the gyromagnetic constant

for the ^1H nucleus, to determine B_0 :

$$1.256 \times 10^9 \text{ s}^{-1} = (26.75 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}) B_0$$

$$B_0 = 4.698 \text{ T}$$

Next use the following Equation to determine the energy gap (ΔE)

$$\Delta E = \frac{h}{2\pi} \gamma B_0 = \frac{6.6262 \times 10^{-34} \text{ Js}}{2 \times 3.14} (26.75 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}) (4.698 \text{ T}) = 1.325 \times 10^{-25} \text{ J}$$

Finally using the equation for the Boltzman distribution, we calculate the $\frac{N_\alpha}{N_\beta}$ ratio for the energy gap.

$$\frac{N_\alpha}{N_\beta} = e^{-\Delta E/kT}$$

$$\frac{N_\alpha}{N_\beta} = e^{-\frac{1.325 \times 10^{-25} \text{ J}}{1.38 \times 10^{-23} \text{ Js}^{-1} \times 298.15 \text{ K}}} = 0.9999678$$

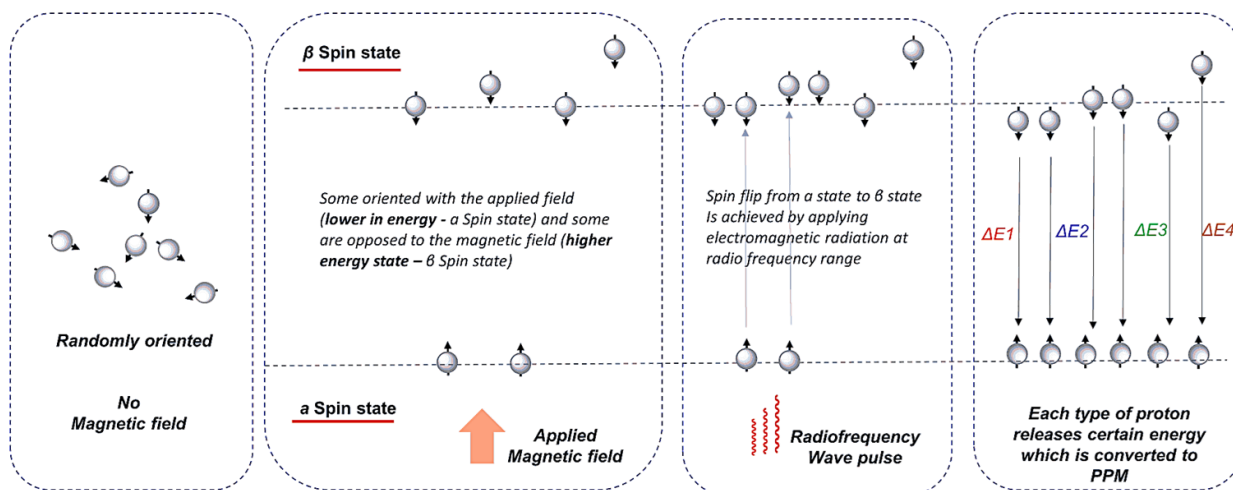
Taking the inverse: $N_\alpha/N_\beta = 1.000032$.

This tells us that there is a 32 ppm excess in ground state.

Saturation

Saturation is the condition that exists when the upper and lower energy states of nuclei are equal (no observed signal by NMR).

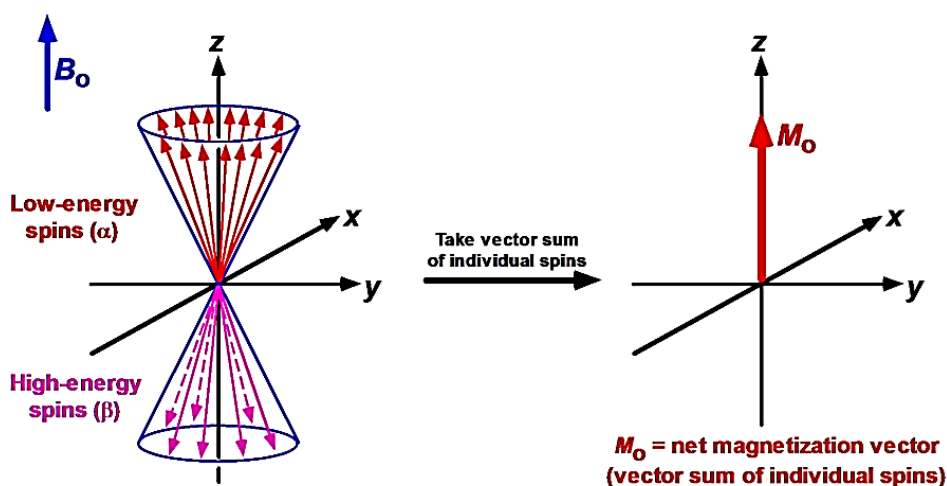
The Basic few-steps principle of NMR spectroscopy



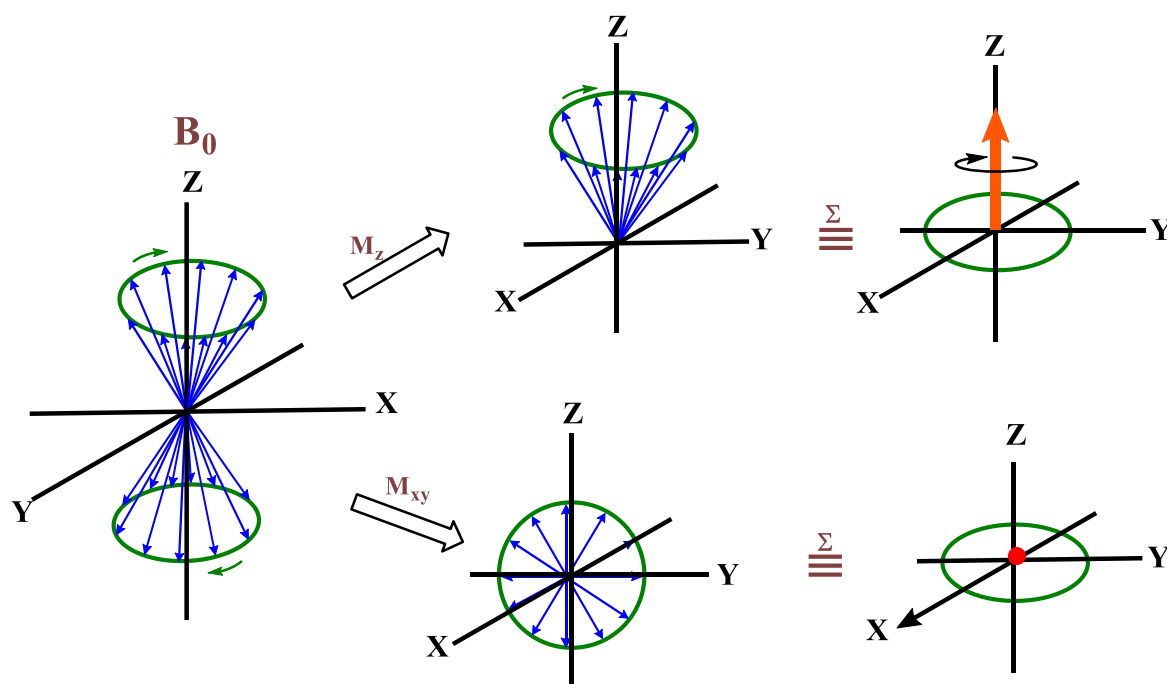
BULK MAGNETIZATION (M_0)

In an NMR experiment, we do not observe just one nucleus but a very large number of nuclei that are spinning. In the absence of external magnetic field, the magnetic moments of these nuclei are randomly arranged. However, in the presence of magnetic field at equilibrium the magnetic moments are not aligned randomly, but in such a way that when their contributions are all added up there is a net magnetic field along the direction of the applied field (B_0). This is called the **bulk magnetization** of the sample. Thus, **the sum of the dipole moments of all nuclei is called bulk magnetization**.

The magnetization can be represented by a vector – called the magnetization vector – pointing along the direction of the applied field (z).



Similarly, for each vector pointing in one direction of the transverse plane a corresponding vector can be found which points into the *opposite* direction:



THE NMR PULSE

A pulse is a burst of radiofrequency energy that may be applied by switching on the Rf transmitter.

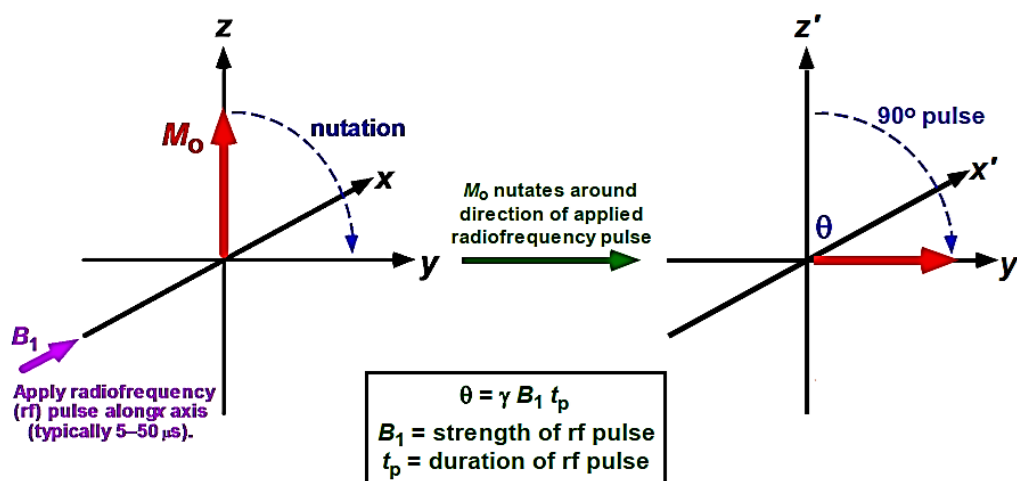
As long as the pulse is on, a constant force is exerted on the sample magnetization, causing it to precess about the Rf vector. The slight Boltzmann excess of nuclei aligned with the external magnetic field corresponds to a net magnetization pointing towards the +z- axis.

We can control the extent by which the +z-magnetization is bent by choosing the duration for which the pulse is applied. Thus, the term "90° pulse" actually refers to the time period for which the pulse has to be applied to bend the magnetization by 90°.

The duration for which the pulse is applied is inversely proportional to the bandwidth; i.e., if we wish to stimulate nuclei in a large frequency range, then we must apply a pulse of a short duration. Nuclear excitation will only occur if the magnitude of the B_1 field is large enough to produce the required tip angle. A "soft" pulse is one that has low power or a long duration (milliseconds rather than microseconds), and such pulses can be used to excite nuclei selectively in specific regions of the spectrum.

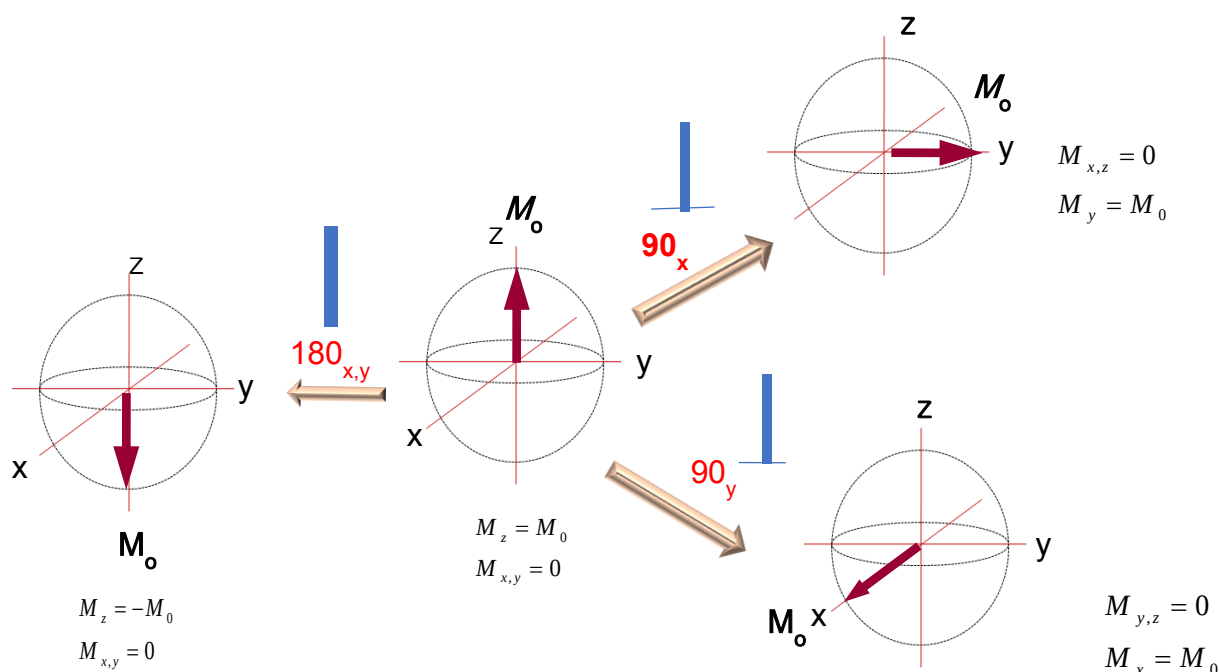
The direction in which the nuclear magnetization is bent by a particular pulse is controlled by the direction in which the pulse is applied. For example, a pulse along the x-axis will rotate the

magnetization around the x-axis in the yz-plane. Similarly, a pulse along the y-axis will rotate the magnetization in the xz-plane. The final position of the magnetization will depend on the time t_1 for which the Rf pulse is applied.



The angle (flip angle) by which the magnetization vector is bent may be calculated as $\theta = \gamma B_1 t_p$, where B_1 is the applied field and t_p is the duration of the pulse. Since the spectrometers are normally set up to detect the component lying along the y-axis, only this component of the total magnetization will be recorded as signals. The pulse, therefore, serves to convert longitudinal magnetization, or z-magnetization, to transverse, or detectable, magnetization along the y-axis.

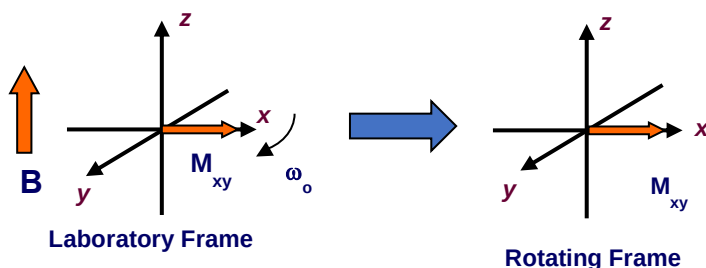
The magnitude of the magnetization component along the y-axis is given by $M_0 \sin \theta$, where θ is the angle by which the magnetization is bent away from the z-axis.



LABORATORY FRAME VS ROTATING FRAME

After the RF pulse, M^0 will be tipped along the (xy)-plane and will rotate at the Larmor frequency. Analysis of a rapidly moving vector is very cumbersome. This is because in real time, the motion of the magnetic moment is so fast that it seems blur. The reason M^0 appears blurred in real time is that we are viewing the system from a stationary point of view, called the **laboratory frame of reference**.

For convenience, it is preferred to use the rotating frame of reference. When using the rotating frame, the axes are designated with primes ('), i.e., x' and y' ; the z-axis is sometimes designated as z' . Simply, our axis now rotates at the Larmor frequency (ω_0). In the absence of any other factors, M_{xy} will stay on the x-axis. If the system is viewed in rotating frame where the frame rotation matches the Larmor frequency. This produces a "stop action" photo that allows us to clearly appreciate the direct effect B_1 has on M_0 .



Three very Interesting features about the system in the rotating frame should be noted:

a) The effect of the main field (B_0) has disappeared

In the rotating frame the precession of M (and that of its component individual spins) around B_0 is no longer occurring. Everything is 'frozen' in place so it is as though B_0 no longer exists.

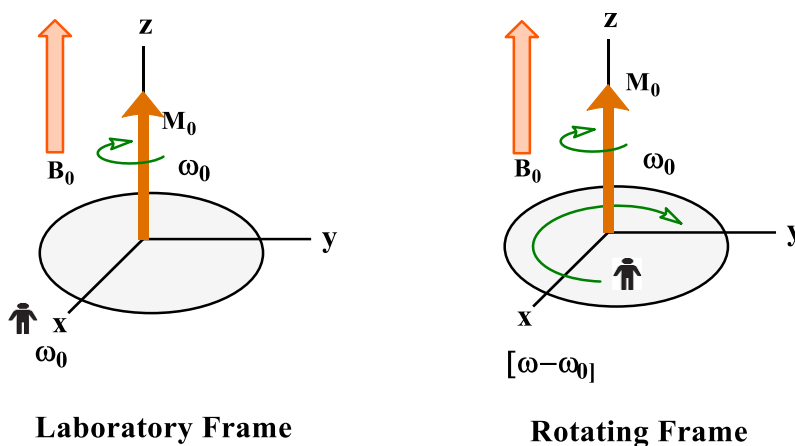
b) The B_1 field is no longer rotating but appears static

B_1 is locked in a fixed relationship to the rotating frame.

c) M has begun a new precession around B_1

Since B_0 no longer exists in this frame of reference, the only magnetic field acting on M is the now stationary B_1 field. So, M will precess around B_1 with frequency $f_1 = \gamma B_1$.

The new precession of M around B_1 is called nutation.



RELAXATION PROCESS

When the system is perturbed from equilibrium by application of B_1 , it will relax back toward equilibrium. This process is called **relaxation**. Relaxation brings the magnetization back to equilibrium. It generates the nuclear magnetization from which all NMR signals are derived.

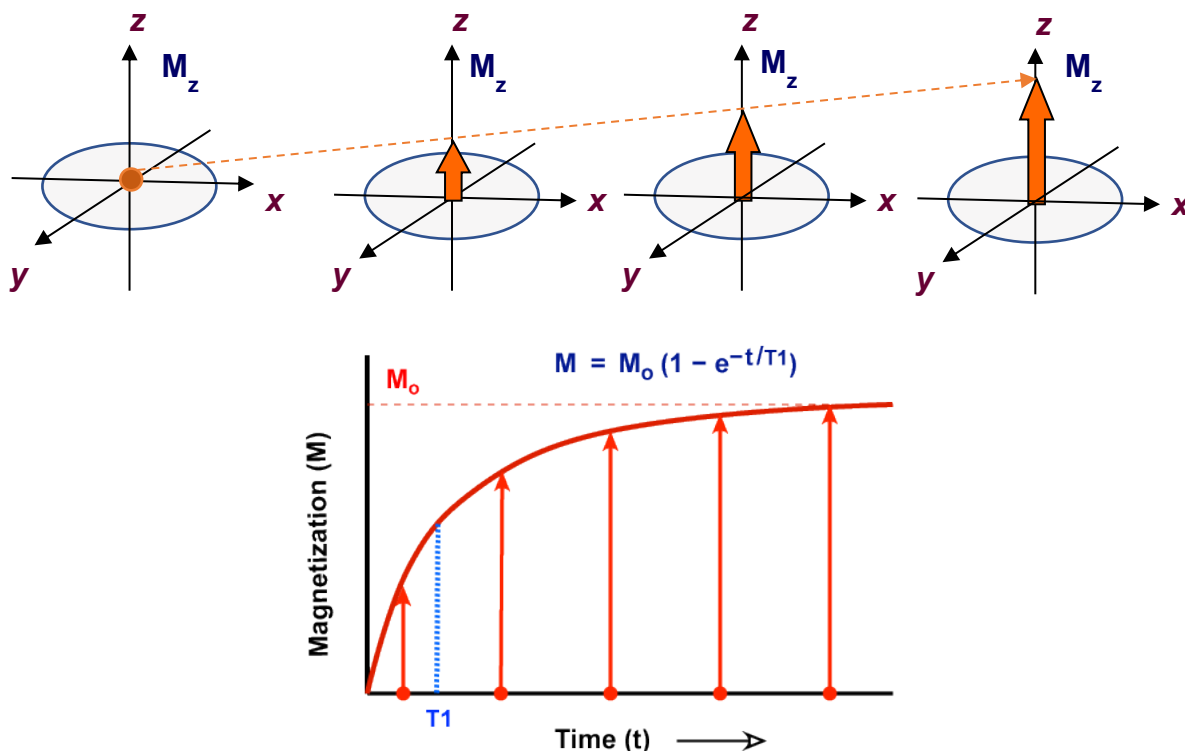
Relaxation in NMR is a phenomenon describing the time dependence involved in signal intensity after an induced radiofrequency pulse is applied. In NMR, this process can be very slow, requiring between a fraction of a second to many minutes, depending on the sample and the environmental conditions.

There are three types of relaxation, and all are time-dependent exponential decay processes.

1. Longitudinal or Spin-Lattice Relaxation (T_1)

After application of a 90° RF pulse, the bulk magnetization, M_0 will move to the transverse (xy) plane and will gradually return to its original equilibrium position along the longitudinal (z) axis. T_1 relaxation is the process by which the net magnetization (M) grows/returns to its initial maximum value (M_0) parallel to B_0 . it is also called **longitudinal relaxation**, **thermal relaxation** and **spin-lattice relaxation**.

Graphically, T_1 can be viewed as the time required for the z-component of M to reach $(1-1/e)$ or about 63% of its maximum value (M_0).

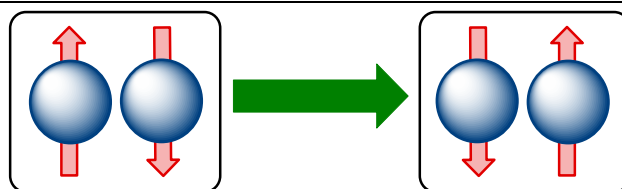


Since T_1 is a measure of the time required for magnetization vector to return to its equilibrium length and direction, it directly affects when the next radiofrequency pulse can be applied. As the longitudinal component of M (M_z) grows toward M_0 , the energy of the spin system decreases. This is because statistically more spins are "favoring" the lower energy (spin-up, parallel) orientation. Energy must therefore leave the spin system for T_1 relaxation to occur. This energy loss is unrecoverable and represents the transfer of heat. This energy must be transferred somewhere into nearby nuclei, atoms, and molecules through collisions, rotations, or electromagnetic interactions. At the most basic level, therefore, T_1 relaxation is simply an energy flow between spins and their external environment.

In general, the shorter T_1 the relaxation time the less intense the peak signal will be and the broader the signal linewidth. An efficient relaxation process involves a short time and results in the broadening of absorption peaks. Smaller the time of the excited state, greater is the line width. This mechanism is not effective in solids. This process keeps the excess of nuclei in the lower energy state which is a necessary condition for nuclear magnetic resonance phenomenon.

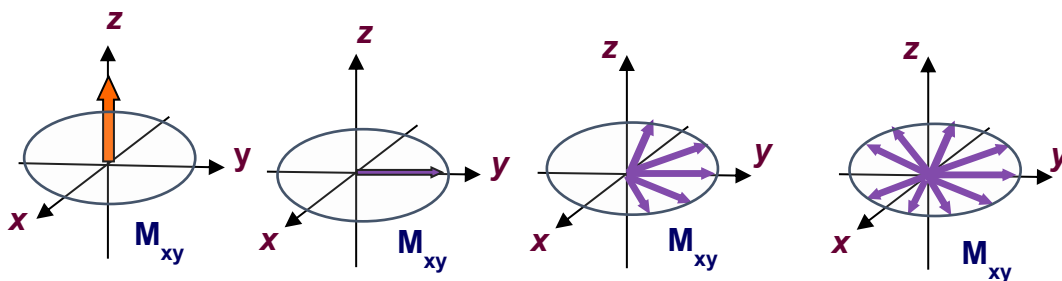
2. Transverse or Spin-Spin relaxation (T_2)

*T_2 relaxation is the process by which the transverse components of magnetization (M_{xy}) decay or dephase. T_2 relaxation is also called **transverse relaxation** and **spin-spin relaxation**.*



Spin-Spin Relaxation

T_2 relaxation follows first order kinetics, resulting in a simple exponential decay with time constant T_2 . Thus, T_2 is the time required for the transverse magnetization to fall to approximately 37% of its

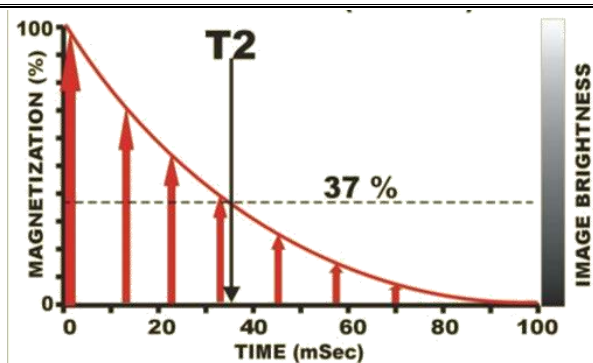


initial value.

Since T_2 is a measure of the time required for the magnetization vector to leave the xy -plane, it is a direct measure of the length during which a signal can be observed. The magnitude of T_2 therefore affects the length for which an FID should be recorded.

In transverse relaxation energy is transferred to a neighboring nucleus. The half-life for this process is called the ***spin-spin relaxation time*** (T_2). This process exchanges the spin of nucleus A with the spin of nucleus B. There is no net change in spin for this process. Inhomogeneity of the magnetic field or the presence of paramagnetic materials can be a large contributor to the value observed for transverse relaxation.

The peak widths in an NMR spectrum are inversely proportional to the lifetime (due to the Heisenberg uncertainty principle) and depend on both T_1 and T_2 . For most organic solutions, T_1 and T_2 are long enough to result in spectra with sharp lines. However, if magnetic field homogeneity is poor or paramagnetic material (such as iron) is present the NMR signals can be broadened to the extent that the signal is destroyed or unusable.



3. Quadrupole Relaxation

It is a prominent relaxation process for nuclei having $I > \frac{1}{2}$. The nuclei (such as ^{14}N , ^{17}O , ^{11}B , etc) due to anisotropic interaction between non-spherical, electrically quadrupole nuclei and the electric field gradients at the nucleus caused by electric environments possess an asymmetric positive charge distribution on the nuclei. Hence, these nuclei exhibit electric quadrupole moments and relax rapidly as display very broad signals. This process of deactivation of nuclei is called electric quadrupole relaxation. In case of chlorine, bromine and iodine, this mechanism is very fast and these nuclei became practically non-magnetic for the purpose of high resolution nuclear magnetic resonance spectroscopy. But for nitrogen (^{14}N) and deuterium (^2H), quadrupole relaxation is less effective and their resonance lines are observed more easily.

FACTORS AFFECTING SENSITIVITY

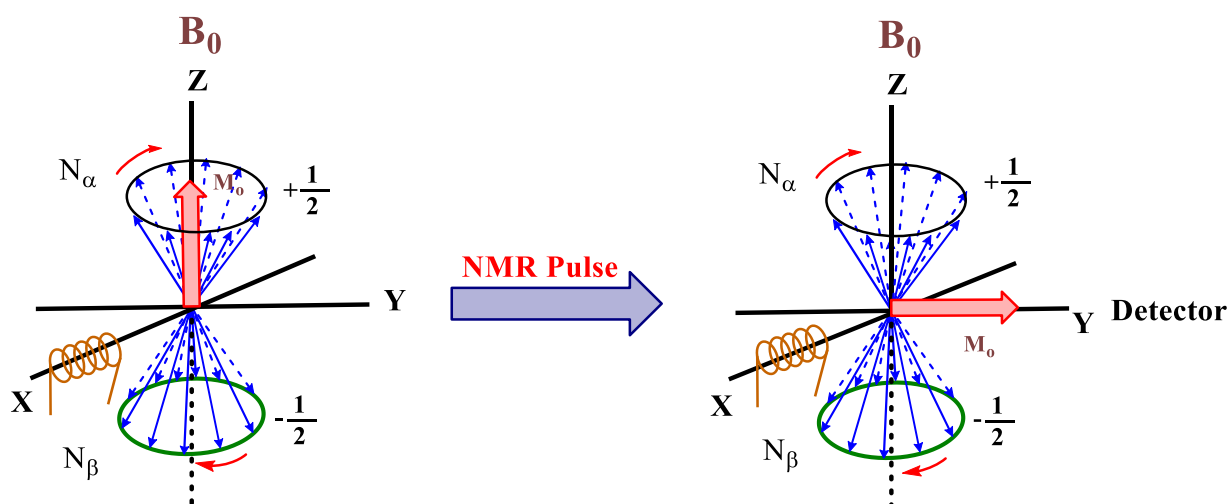
The NMR signal depends linearly on the magnitude of magnetization in the xy -plane, $\left[\vec{M}_{xy} \right]$. This quantity is governed by four factors for a given nucleus.

- 1 First, the tip angle of the magnetization vector affects $\left[\vec{M}_{xy} \right]$. The closer the tip angle is to 90° , the larger the magnitude of the magnetization vector is in the xy -plane.
- 2 Second, the external magnetic field strength affects the magnitude the magnetization vector $\left[\vec{M}_{xy} \right]$, and consequently its projection in the xy -plane.
- 3 Third, the sample temperature also affects magnetization vector $\left[\vec{M}_{xy} \right]$, as a colder sample results in a larger population difference between spin states.
- 4 Fourth, the number of nuclei in the coil, which is usually controlled by sample concentration, also directly affects the magnetization vector $\left[\vec{M}_{xy} \right]$.

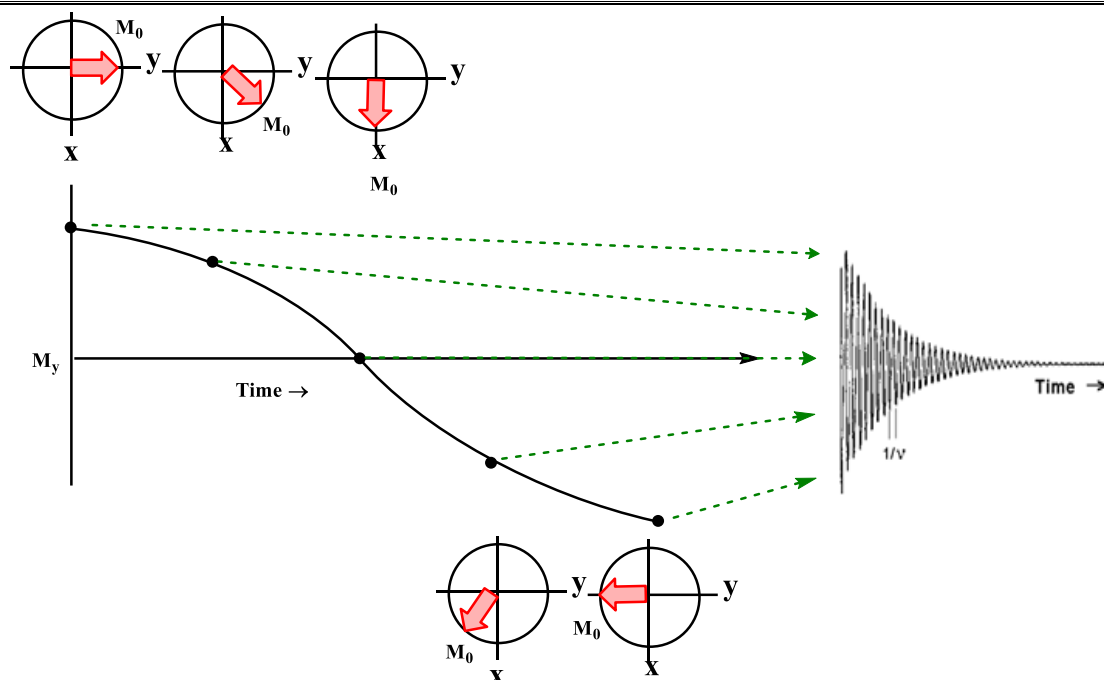
- 5 The NMR signal also depends on the magnetogyric ratio γ . This strong dependence arises from the magnetic moment depending linearly on γ , the population difference between spin states depending linearly on γ , and the efficiency of signal generation depending on the Larmor frequency ω , which is proportional to γ .

FREE INDUCTION DECAY (FID)

By convention the pulse is implemented along the x-axis. This brings M into the y-axis which is the observation axis (detector).



Immediately after the RF pulse is turned off, the spins relax back to its equilibrium condition; the spin population returns to its equilibrium distribution, and the spins lose coherence. As the B_1 field is turned off and M_{xy} continues to precess about B_0 at frequency ω_0 . The oscillation of M_{xy} generates a fluctuating magnetic field which can be used to generate a current in a receiver coil to detect the NMR signal. The FID reflects the change in the magnitude of M_{xy} as the signal is changing relative to the receiver along the y-axis. Since the detector is aligned along the y-axis, the receiver detects it is the M_y component of M which is detected as the FID signal. The FID is a time-domain signal: the nucleus precesses at its resonance frequency ν over time.



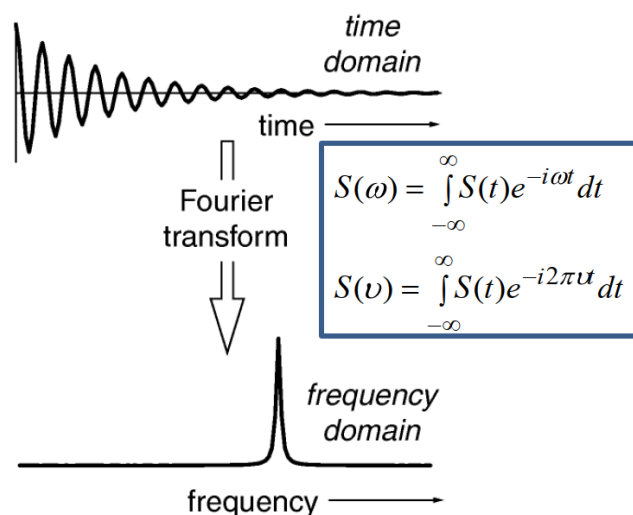
The FID follows an exponential decay which is proportional to $[\exp (-t / T_2)]$.

FOURIER TRANSFORM

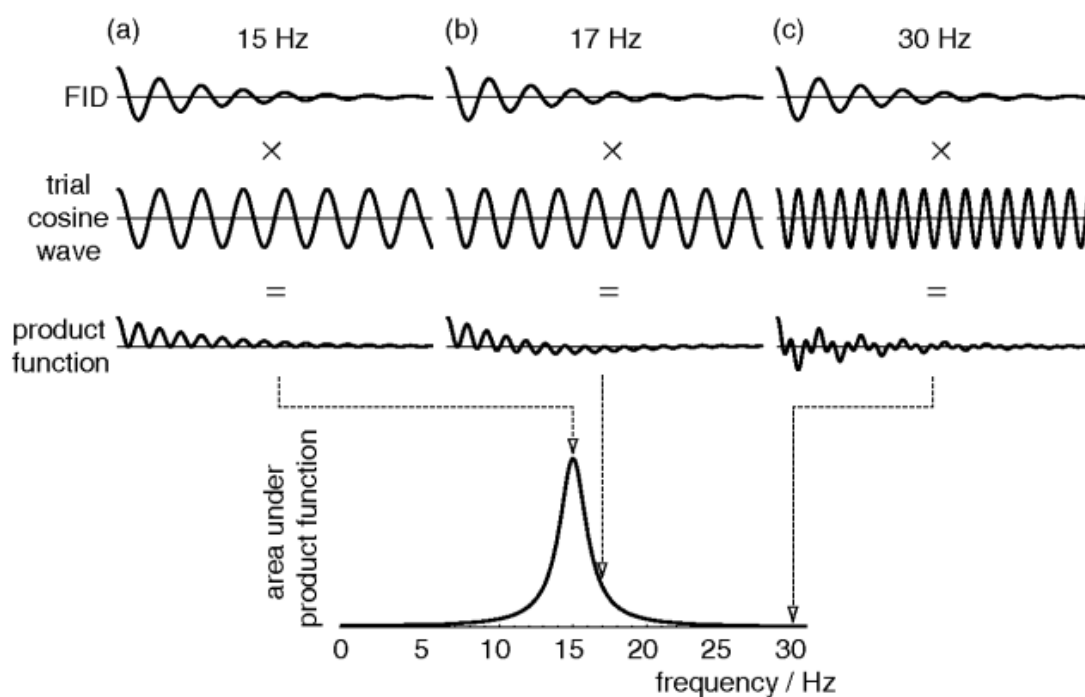
Fourier transform is a mathematical technique that converts the time-domain FID into a frequency domain spectrum.

The measured signal in modern NMR is in time domain. This is a major difference compared to other kinds of spectroscopy. The time domain signal is of limited value. Thus, it is essential to present a spectrum i.e. frequency vs intensity plot. Fourier transform performs this conversion from the time domain signal or FID.

Mathematically, the Fourier transform of a time domain signal can be expressed as an integral of the product of the time domain signal and a sinusoidal signal. The result can be expressed in either rad/sec or in Hz units.

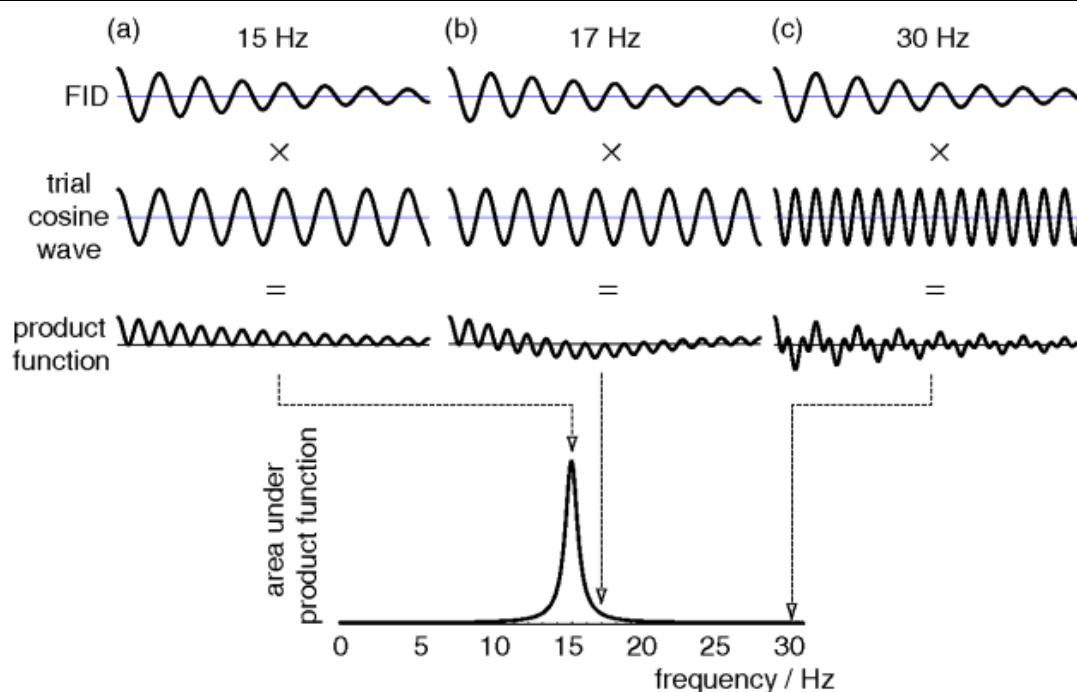


To understand FT qualitatively with a specific case. Consider a single FID for analysis. We multiply this FID with 3 trial cosine signals of (a) 15Hz, (b) 17Hz, and (c) 30Hz. We take the product signals compute area under these and plot them as a function of the reference cosine wave frequency.



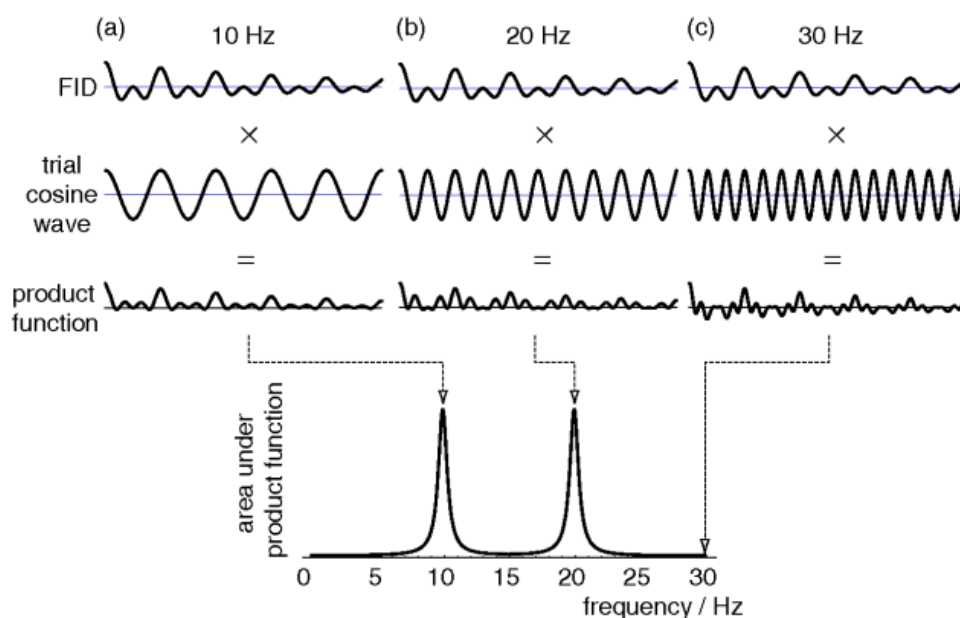
In (a) the product is always positive and the area is shown by the arrow. In (b) the signal is positive for most but less than that of (a) and so the area is less than (a). In (c) the product signal oscillates rapidly and the area under the product signal is zero. The spectrum is plotted as area under the product signal vs the reference cosine wave frequency.

In cases of slower FID decay, same analysis as before may be observed. This is shown below:



The resulting spectrum is narrower than the one before. These analyses illustrate the FID is at a frequency of 15 Hz.

The same analysis can be performed even if the FID arises from more than one resonance.



Thus, FT is a procedure in which the intensity at a frequency f Hz is calculated as the area under the product of the FID and a cosine wave at that frequency f .

Since there is no signal before time $t=0$ the FT integral can be written as

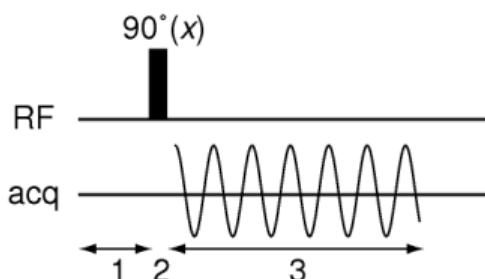
$$S(\omega) = \int_0^{\infty} S(t) e^{-i\omega t} dt$$

$$S(\nu) = \int_0^{\infty} S(t) e^{-i2\pi \nu t} dt$$

In any spectrometer the FID is not detected as a continuous signal (a) but as a discrete set of N digital points - ith point at time t_i (b) and then the spectrum is computed as:

$$S(\nu) = \sum_{i=1}^{i=N} S_{FID}(t) e^{-i2\pi \nu t_i}$$

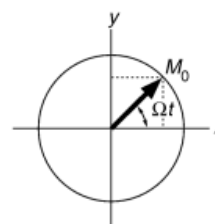
Let's look at the simple 1D – pulse-acquire experiment



The $90^\circ(x)$ pulse rotate M^0 to $-y$ axis. The x and y component of the magnetizations are then given as

$$M_x = M_0 \sin \Omega t \quad M_y = -M_0 \cos \Omega t$$

Instead, if we use $90^\circ(y)$ pulse then M_0 will go to x-axis and then the x and y components are



$$M_x = M_0 \cos \Omega t \quad M_y = M_0 \sin \Omega t$$

The precession of the magnetization in the xy-plane induces a voltage (signal S) in a coil which will be written as

$$S_x = S_0 \cos \Omega t \quad S_y = S_0 \sin \Omega t$$

We should also take in to account that the signal decay over time and “model” this decay as an exponential decay

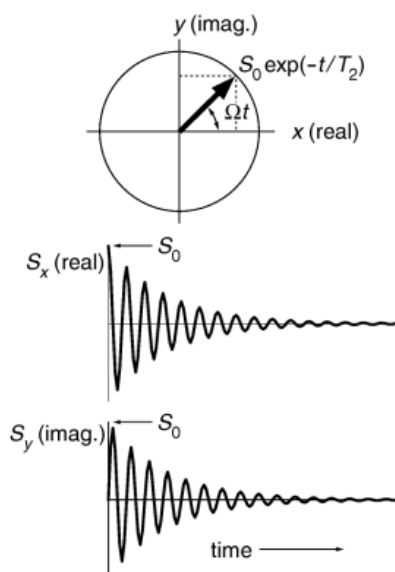
$$S_x = S_0 \cos \Omega t \times e^{-\frac{t}{T_2}} \quad S_y = S_0 \sin \Omega t \times e^{-\frac{t}{T_2}}$$

T_2 is a time constant characterizing the decay. Combining $S_x(t)$ and $S_y(t)$

$$S(t) = S_x + iS_y = S_0 (\cos \Omega t + i \sin \Omega t) \exp\left(-\frac{t}{T_2}\right)$$

Thus, the time domain signal is represented as a complex function with a decay constant T_2 means that the vector S_0 rotates in the xy plane while its length shrinks as time goes by. The x and y components of this rotating vector is the real and imaginary part of the signal.

$$S(t) = S_0 \exp(i\Omega t) \exp\left(-\frac{t}{T_2}\right)$$



Demonstration

In order to understand how the Fourier transformation works, converting the FID information into the spectrum consider an FID composed of only two frequencies (2 non-equivalent protons) for simplicity, with values of 1 and 5 Hz. The FID will be the sum of all frequencies, and the Fourier transformation will need to deal with it.

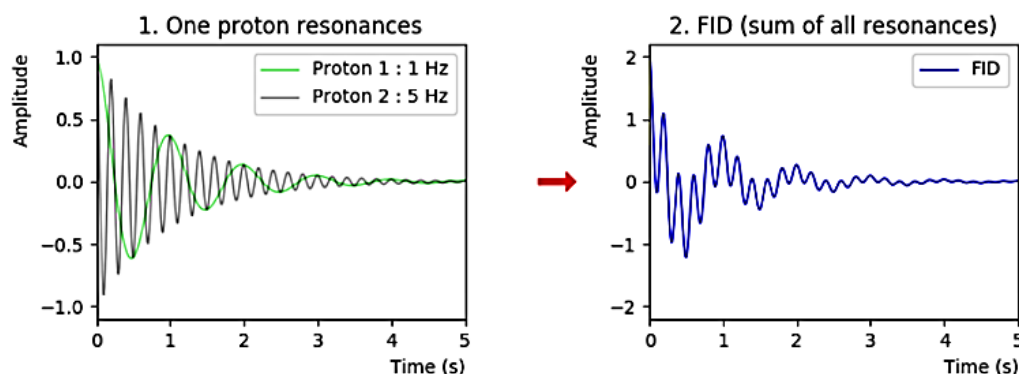
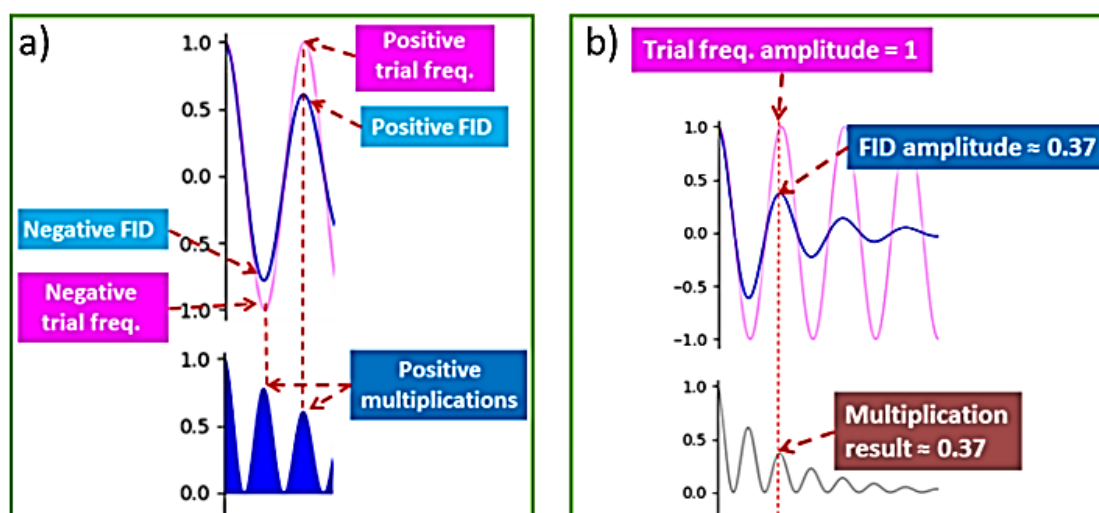


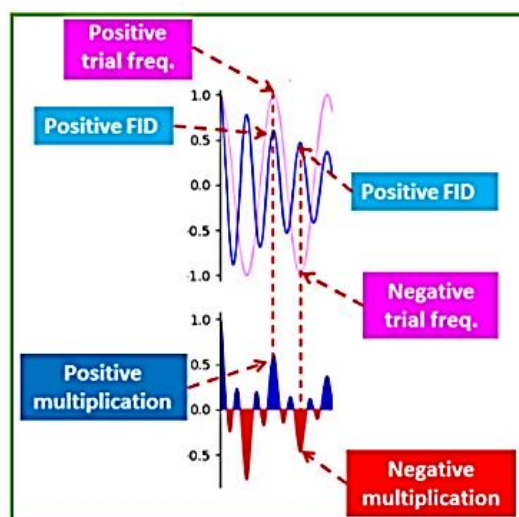
Figure: (a) Representation of both cosine waves (1 and 5 Hz); (b) Representation of the FID containing both frequencies (1 and 5 Hz)

To extract the frequencies embedded in the FID, the computer will run the Fourier transformation (FT), which is a multiplication of the FID by cosine functions with increasing frequencies, covering all the frequency range of spectral width. These cosine functions are called as "Trial Frequencies".

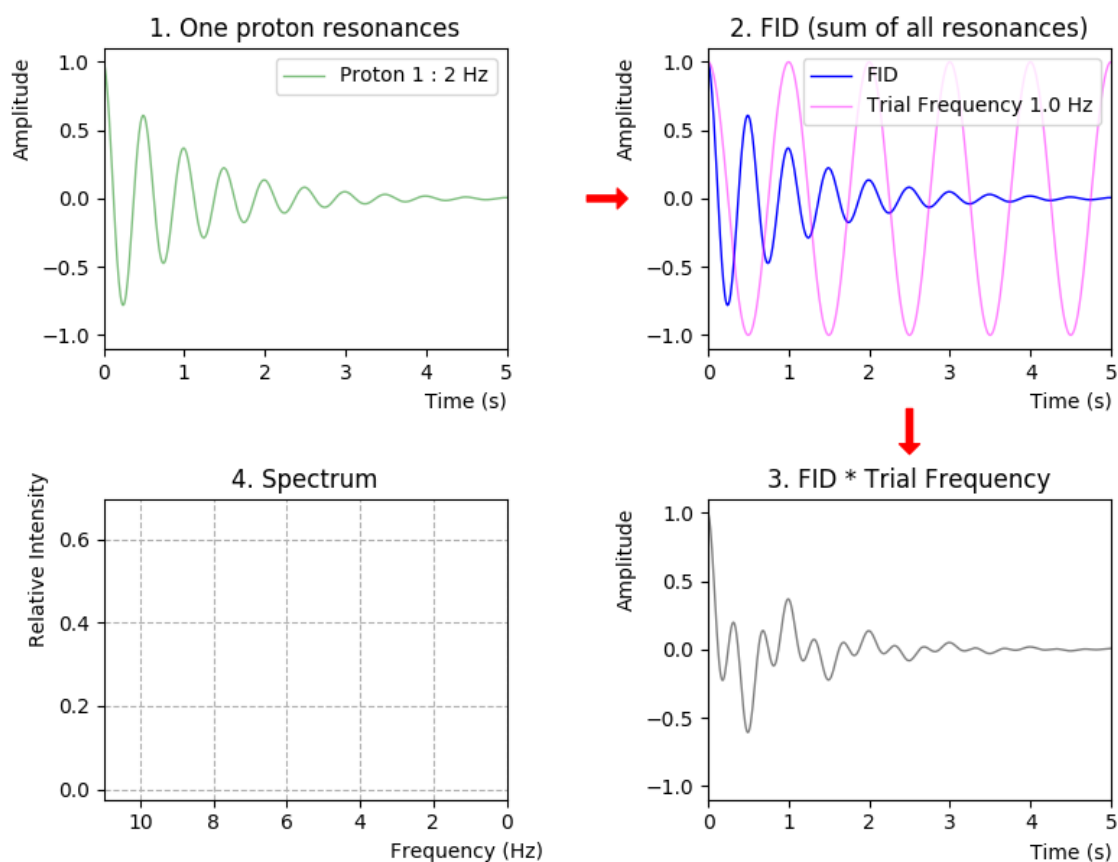
Case 1: Using a trial frequency that matches with a frequency in the FID, such as 1 Hz: the multiplication of both functions will always lead to a positive value since the trial frequency, and the FID will be positive or negative at the same time, as both have the same frequency. In this case, the total area under the multiplication curve will always be positive (Figure a). Also, as the trial frequency has an amplitude equal to 1, the intensities of the original FID are preserved.



Case 2: If the trial frequency doesn't correspond to any embedded frequency in the FID, the multiplication curve will regularly oscillate between positive and negative values. The total area under the curve will be close to zero.



Each of these multiplications (trial frequency and FID) corresponds to a point in the spectrum with its respective intensity. This process is performed very quickly by the computer and will originate the well-known NMR spectrum.:

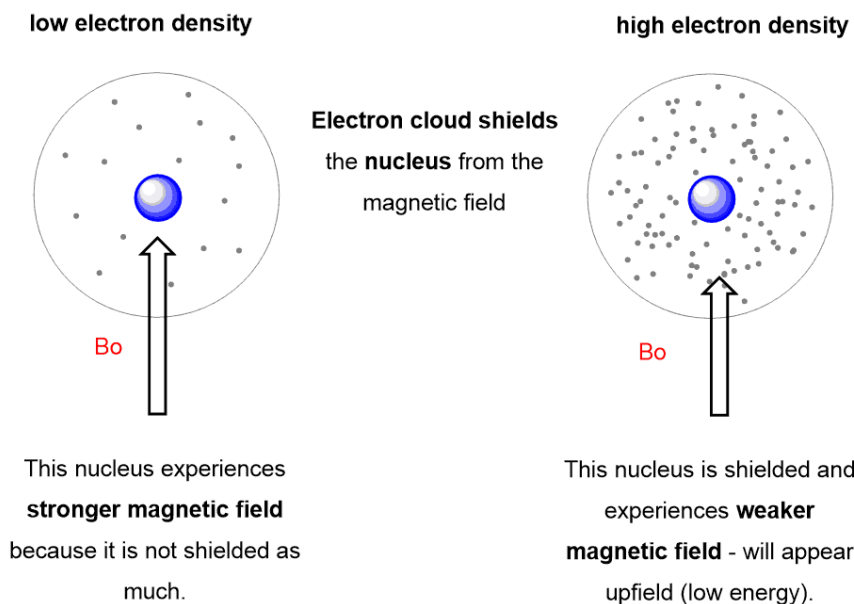


*Chapter 3***CHEMICAL SHIFTS IN NMR SPECTRA**

The positions of the signals in the spectrum help to understand the nature of protons, viz., aromatic, aliphatic, acetylenic, adjacent to some electron attracting or electron releasing group, etc. Each of these types of protons will have different chemical environments, and thus, they absorb at slightly different applied field strengths.

The signal frequency that is detected in nuclear magnetic resonance (NMR) spectroscopy is proportional to the magnetic field applied to the nucleus. Since all protons have the same magnetic moment it might be expected that all hydrogen atoms will resonate at the same frequency and only give rise to a single signal. Fortunately, this is not the case and different protons will give rise to slightly (ppm) different frequencies. The reason is that the effective magnetic field B_{eff} that a nuclei experiences is equal to the very large static magnetic field (B_0) plus a much smaller shielding magnetic field which arises from electrons surrounding the nucleus.

When a molecule is placed in a magnetic field, its electrons are caused to circulate and thus, they produce secondary magnetic fields i.e., **induced magnetic field (B_i)**. Rotation of electrons about the nearby nuclei generates a field that can either oppose or reinforce the applied field at the proton. If the induced field opposes the applied field, then proton is said to be shielded. But if the induced field reinforces the applied field, the proton feels a higher field strength and thus, such a proton is said to be deshielded. Shielding shifts the absorption upfield and deshielding shifts the absorption downfield to get an effective field strength necessary for absorption. *Such shifts (compared with a standard reference) in the positions of nmr absorptions which arise due to the shielding or deshielding of protons by the electrons are called chemical shifts.*



The effective magnetic field at the nucleus can be expressed in terms of the externally applied field B_0 by the expression

$$B_{\text{eff}} = B_0(1 - \sigma)$$

where σ is called the shielding factor or screening factor. The factor σ is small – typically 10^{-5} for protons and $<10^{-3}$ for other nuclei. The magnitude of the shift depends upon the type of nucleus and the details of the electron motion in the nearby atoms and molecules.

The shielding constant, σ , for a particular nucleus in a particular environment can be expressed by rearranging the Larmor equation.

$$\nu = \frac{\gamma}{2\pi} B_0(1 - \sigma)$$

On rearranging, this equation gives

$$\sigma = 1 - \left(\frac{2\pi\nu}{\gamma B_0} \right)$$

The signal shift is very small, parts per million, but the great precision with which frequencies can be measured permits the determination of chemical shift to three or more significant figures.

INTERNAL STANDARDS

Each different proton in a molecule is in a slightly different chemical environment, and consequently has a slightly different amount of electronic shielding, which results in a slightly different resonance frequency. These differences in resonance frequency are very small.

In a magnetic field strength of 1.4 T (60 MHz), the resonant frequencies for different protons (as many it can have) present in a compound differ only by 600 Hz. Thus, the resonant frequencies of different protons lie in the range 60000000-60000600 Hz in a magnetic field strength of 1.4T.

Therefore, it is extremely difficult to measure exact resonant frequencies of different protons in a molecule to a high precision. Hence, no attempt is made to measure the exact resonance frequency of any proton. Instead, a reference compound, called an internal standard, is placed in the solution of the substance to be measured, and the resonance frequency of each proton in the sample is measured relative to the resonance frequency of the protons of the reference substance. In other words, the frequency difference is measured directly under similar condition.

It is usual to calibrate chemical shifts by using a suitable compound as a marker resonance. In order to establish a chemical shift scale for a given nucleus, it is necessary to choose some substance as a standard and define its chemical shift as zero.

Thus, if we have unknown and marker resonances in different environments with screening constants σ_1 , and σ_2 , then the two nuclear frequencies in a given magnetic field B_0 are

$$\nu_1 = \frac{\gamma}{2\pi} B_0 (1 - \sigma_1) \quad (i)$$

$$\nu_2 = \frac{\gamma}{2\pi} B_0 (1 - \sigma_2) \quad (ii)$$

Whence

$$\nu_2 - \nu_1 = \frac{\gamma}{2\pi} B_0 (\sigma_2 - \sigma_1) \quad (iii)$$

It is possible to measure a frequency with very high precision but not a magnetic field strength so this expression is not of great use. We thus eliminate magnetic field, B_0 from the equation by dividing through by ν_1 . This gives us the frequency change as a fraction of ν_1 .

$$\frac{\nu_2 - \nu_1}{\nu_1} = \frac{(\sigma_2 - \sigma_1)}{(1 - \sigma_1)}$$

since $\sigma \ll 1$ in many cases reduces to

$$\frac{\nu_2 - \nu_1}{\nu_1} = \sigma_2 - \sigma_1$$

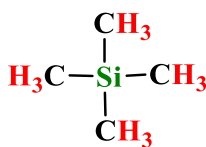
Thus, the fractional frequency change is the same as the difference in screening in the two nuclear environments. **This is called the chemical shift and is given the symbol δ .** Its value is expressed in parts per million (ppm). It can be determined with high accuracy since it is possible to resolve shifts of 0.001 ppm for spin $-\frac{1}{2}$ nuclei, or even less in favourable cases.

A. Internal Standards for Protons and Similar Nuclei

Reference materials which are commonly employed as internal standards for proton and similar nuclei are:

1. Tetramethyl Silane (TMS)

The usual, almost universal, standard for the four nuclei, the proton (^1H), deuterium (^2H), carbon (^{13}C) and silicon (^{29}Si), is tetramethylsilane, $(\text{CH}_3)_4\text{Si}$, usually called TMS.



Tetramethylsilane(TMS)

The TMS is a rare example of a carbon connected with a less electronegative element, silicon, which makes it shielded and therefore appear at lower ppm where other protons do not give signal and therefore 0 ppm value is assigned to TMS peak.

The signals of all the other protons are reported in terms of how far (in Hertz) they are shifted from TMS signal and the chemical shift value (δ) is measured by the ratio of this shift in Hertz and the operational frequency of the spectrometer in MHz:

$$\delta = \frac{\text{(shift in Hz from TMS)}}{\text{(spectrometer frequency in MHz)}}$$

The operational frequency of most NMR instruments is in the MHz region and this is why the units are given in parts per million (ppm).

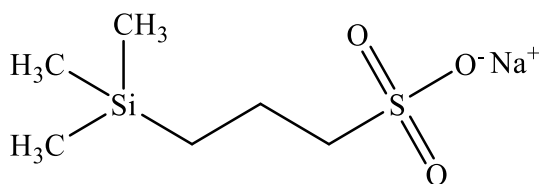
It is due to the following facts:

- i) *It is chemically inert and miscible with a large range of solvents.,*
- ii) *Its twelve protons are all magnetically equivalent.*

- iii) *It is highly volatile with a boiling point 27°C, and can be easily removed to get back the sample. However, for a contemporary NMR spectrometer (including a benchtop NMR), it is no longer necessary to actually add TMS since the computer can calibrate the chemical shift electronically based on the resonance frequencies of the solvent used.*
- iv) *It does not take part in intermolecular associations with the sample.*
- v) *Its resonance position is far away from absorptions due to protons in most organic molecules. Thus, assigning TMS = 0(δ), a scale can be devised in which most of the proton resonances are of the same sign.*

2. Sodium 2,2-dimethyl-2-silapentane-5-sulphonate (DSS)

Since TMS is insoluble in deuterium oxide, it cannot be used with this solvent. A suitable reference for aqueous solutions appears to be the methyl groups of sodium 2,2-dimethyl-2-silapentane-5-sulphonate (DSS). This standard has the disadvantage that it is non-volatile and has absorptions other than $\text{CH}_3\text{-Si}$ (4 signals in total).



Sodium 2,2-dimethyl-2-silapentane-5-sulphonate

B. Internal Standards for Other Nuclei

The conventions adopted for other nuclei are less firm. The standard is assigned 0 ppm and the δ scales are then plotted. An aqueous salt solution is often used as standard for groups 1, 2, 3, and 17: ${}^7\text{Li}^+(\text{aq})$ for lithium, $[{}^{27}\text{Al}(\text{H}_2\text{O})_6]^{3+}$ for aluminium and ${}^{35}\text{Cl}^-(\text{aq})$ for chlorine, for instance. Some much used references are given in the following table:

Spectroscopy	Reference
${}^{11}\text{B}$	$(\text{CH}_3\text{O})_3\text{B}$ and $(\text{CH}_3\text{CH}_2)_2\text{O} \rightarrow \text{BF}_3$
${}^{14}\text{N}$	CH_3NO_2 , or nitrate ion, $[\text{NO}_3]^-$
${}^{17}\text{O}$	H_2O
${}^{31}\text{P}$	85% orthophosphoric acid, H_3PO_4
${}^{19}\text{F}$	CFCl_3 , hexafluoro benzene, C_6F_6 , or trifluoroacetic acid.

THE NMR SCALES

The positions of the signals in the spectrum help us to know the nature of protons. Different protons have different electronic environments and thus, they absorb at different applied field strengths or at a steady external field, different sets of protons will absorb or come to resonance at different values of the radio-frequencies.

The NMR spectra is displayed as a plot of the applied radio frequency versus the absorption. The applied frequency increases from left to right, thus the left side of the plot is the low field, **downfield** or **deshielded side** and the right side of the plot is the **high field, upfield** or **shielded side**.

It is a unitless number (actually a ratio, in which the units cancel), but we assign ‘units’ of ppm. Shielded protons have a lower resonance frequency and smaller chemical shift (δ) values. Deshielded protons have a higher resonance frequency and larger chemical shift (δ) values.

1. The Delta (δ) Scale

In practice the chemical shift is usually indicated by a symbol δ which is defined in terms of a standard reference. This scale increases to the left and is usually reported in units of parts per million (ppm).

The value for $\nu_{\text{sample}} - \nu_{\text{reference}}$ can be expressed in the corresponding field strength in gauss. Since, the operating frequency of the instrument is directly proportional to the strength of the magnetic field, we can define δ as:

$$\delta_{\text{sample}} = \frac{\nu_{\text{sample}} - \nu_{\text{reference}}}{\text{operating frequency in mega cycles per second}}$$
$$\delta_{\text{sample}} = \frac{\Delta\nu}{\text{operating frequency in mega cycles per second}}$$

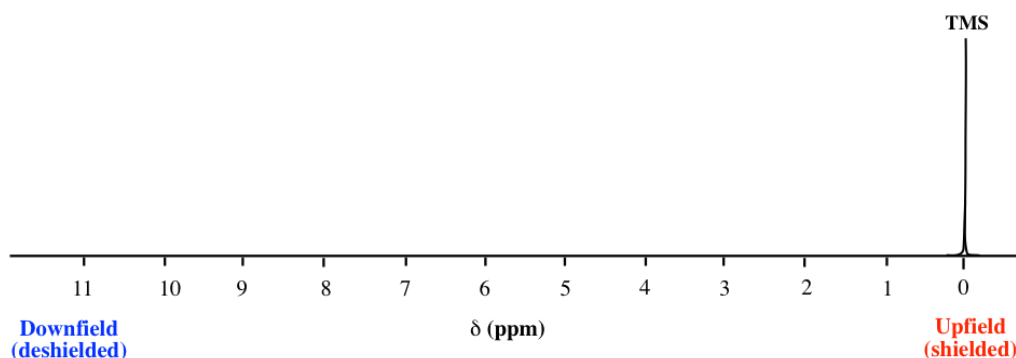
where $\Delta\nu$ is the frequency shift. The value of ν is expressed in parts per million (ppm). Most chemical shifts have ν values between 0 and 10 or 12.

The chemical shift of a resonance in a sample, in ppm is defined as follows:

$$\delta_{\text{sample}} = \frac{\nu_{\text{sample}} - \nu_{\text{reference}}}{\nu_{\text{reference}}} \times 10^6$$

where ν_{sample} is the absolute frequency of the sample resonance and $\nu_{\text{reference}}$ is the absolute frequency of an agreed upon reference compound.

When the scale is plotted in this manner, the peak positions are relative to that of the standard compound. This scale is particularly useful, as it is independent of a **single** absolute frequency and therefore does not depend on the magnetic field strength, which varies from laboratory to laboratory. Thus, spectra recorded using magnets of unequal field strength can be compared more easily.



The energy axis is called a δ (**delta**) axis and the units are given in **part per million (ppm)**. Most often the signal area for organic compounds ranges from **0-12 ppm**.

The ppm value of a given proton depends on the following factors:

- The energy difference of its α and β states.
- The extent to which each proton is exposed to magnetic field the field.
- The electron cloud around them.

2. The τ Scale

In the τ scale, signal for the standard reference, TMS is taken as 10 ppm. It has been found that signals for fluorine resonances (absorptions due to fluorine nuclei) do not appear in this range but are observed downfield by about 50 ppm or even more.

The two scales are related by the expression:

$$\tau = 10 - \gamma$$

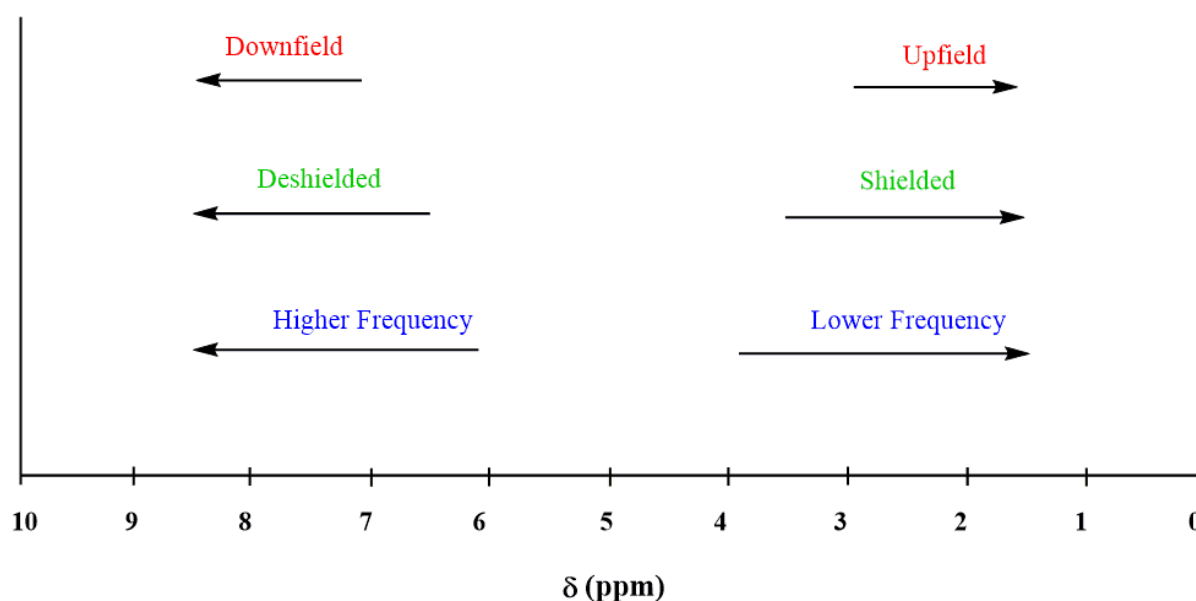
UPFIELD AND DOWNFIELD

The terms **upfield** and **downfield** refer to the **low** and **high** energy of the signals respectively. The wording has a **historical origin**. The first-generation NMR spectrometers, called “**continuous wave field sweep**” in which the frequency of rf radiation was kept constant and the magnetic field strength was slowly increased to detect which field strengths produce a signal. The more **shielded** protons required a stronger magnetic field to resonate and therefore the high energy was on the

right side thus making in upfield and the terms upfield and downfield indicated high energy and low energy.

However, modern NMR instruments are designed to work by pulsed **Fourier-transform NMR (FT-NMR)** where the **magnetic field is held constant** and a short pulse covering the **entire range of relevant radio frequencies is irradiated**. So, the **protons that are shielded appear to be at lower energy level**, therefore resonate at lower frequency and the ones **deshielded experience stronger magnetic field and resonate at higher energy** which is opposite to what happened in older instruments. This doesn't mean that the results were different, no they produce identical data, but the wording is mixed.

- Downfield means higher energy – left side of the spectrum (higher ppm)
- Upfield means lower energy – right side of the spectrum (lower ppm)



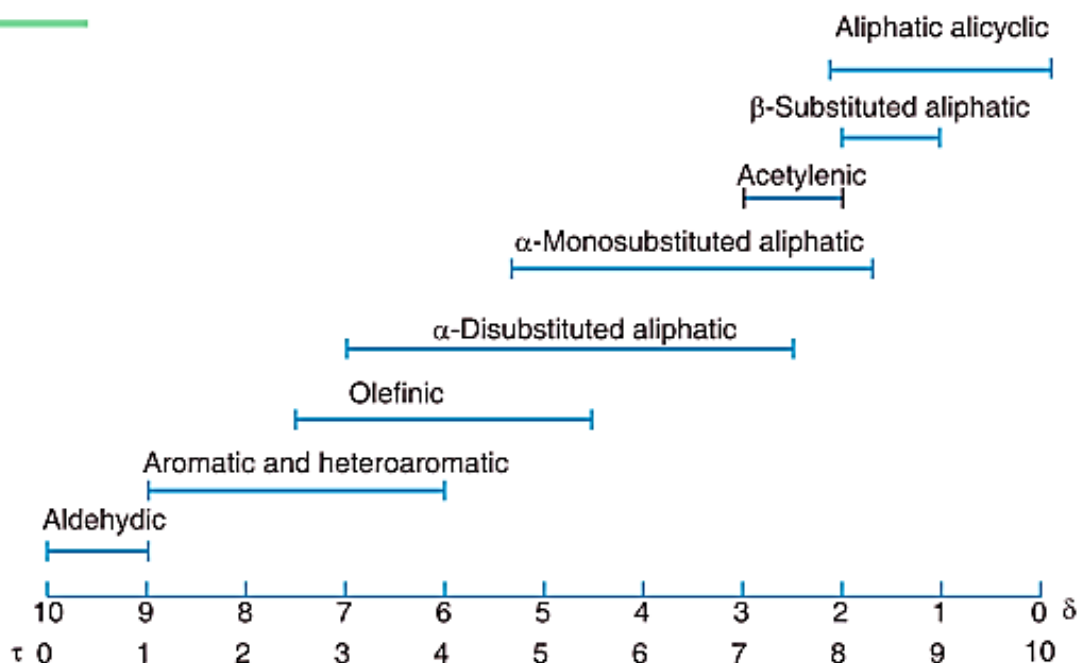


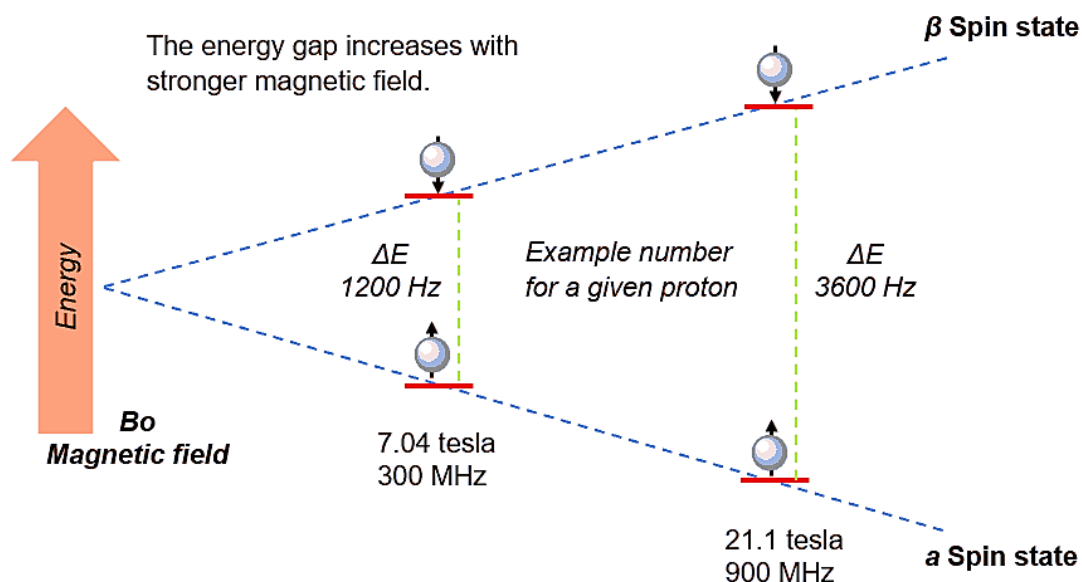
Fig. 5.15. General regions of chemical shifts.

CALCULATION OF THE CHEMICAL SHIFTS

Calculation of a chemical shift is a simple matter. Modern spectrometers give the frequency separation of signals and it is only necessary to divide by the spectrometer operating frequency. If in a given solution the signal of benzene is 2901 Hz higher in frequency than TMS and the frequency of TMS is 400.134 394 MHz (i.e. $B_0 = 9.39$ T), then the chemical shift is

$$\delta = \frac{2901}{400.134394} \times 10^6 = 7.25 \text{ ppm}$$

Or, it is the frequency difference in hertz divided by the frequency of the reference in megahertz. We see also that the frequency separations will be different in spectrometers operating at different frequencies. Thus, in a 200 MHz spectrometer, the frequency separation above will be 1450 Hz.



However, ppm stays the same because it is the ratio of the frequencies of the signal and the instrument.

$$\delta = \frac{1200 \text{ Hz}}{300 \text{ MHz}} = 4 \text{ ppm}$$

$$\delta = \frac{3600 \text{ Hz}}{900 \text{ MHz}} = 4 \text{ ppm}$$

EQUIVALENT PROTONS AND NONEQUIVALENT PROTONS

1. Equivalent Protons

The protons which are in an identical chemical environment in a compound and resonate at the same frequency are called equivalent protons. These protons are identical in every way. They have the same magnetic environment, and they appear at the same position in the proton NMR spectrum in the compound. Equivalent protons in a compound show single signal.

All the six protons in ethane, four protons in ethylene, two protons in acetylene, and six protons in benzene are all equivalent.

2. Nonequivalent Protons

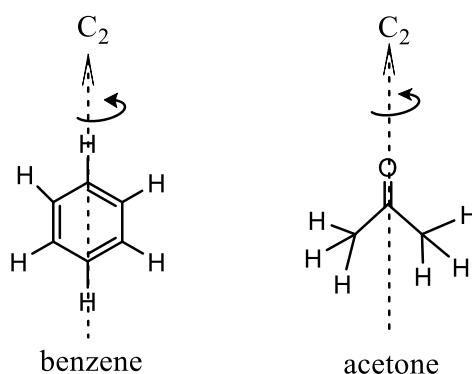
The protons which are in a very different chemical environments in a molecule, each gives rise to an absorption and the absorptions are widely separated in the spectrum are called nonequivalent protons. Nonequivalent protons have the different magnetic environment. These protons of a

compound resonate at different frequencies. They appear at different positions in the proton NMR spectrum.

NUMBER OF SIGNALS IN A COMPOUND

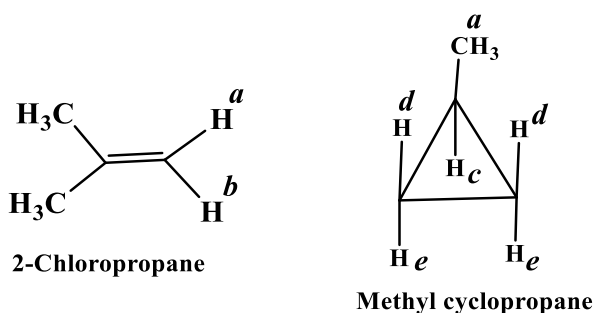
The number of signals in the proton NMR spectrum of a compound is generally equivalent to the number of different sets of equivalent protons in it. Each signal corresponds to a set of equivalent proton/s. All of the protons found in chemically identical environments within a molecule are chemically equivalent.

All the protons in tetramethyl silane [TMS, $(\text{CH}_3)_4\text{Si}$] or the protons in benzene, or in acetone are equivalent by symmetry considerations (Figure).



Each such compound gives rise to a single absorption peak in its proton NMR spectrum. These “**homotopic**” protons in each compound are said to be chemically equivalent. On the other hand, a molecule that has sets of protons that are chemically distinct from one another may give rise to a different absorption peak from each set, in which case the sets of protons are chemically nonequivalent.

Two olefinic protons (H_a and H_b) in 2-chloropropene are diastereotopic. They reside in a different chemical environment, and absorb separately. These two protons are, therefore, chemically nonequivalent.



The protons H_d and H_e in methyl cyclopropane) are also nonequivalent for the same reason. Consequently, it shows four signals in 1H -NMR spectrum.

CHOICE OF SOLVENTS

The non-viscous samples give the sharpest NMR spectra. It is usually necessary to record the spectra of organic compounds in solution. If the compound, itself is a non-viscous liquid, the neat liquid can be used. The solvent should be capable of dissolving at least 10% of the substance under investigation. It is an advantage to use the aprotic solvents which do not themselves give an NMR spectrum to superimpose on that of the sample.

The commonly used NMR solvents are normal organic solvents in which the hydrogen has been replaced by deuterium.

Choice of Solvents

Most FT NMR instruments maintain frequency accuracy by “locking” simultaneously to the deuterium NMR frequency, and in these instruments a deuterium containing solvent is essential. The common deuterated solvents used for organic compounds are $CDCl_3$, CD_3CN , CD_3OD , $(CD_3)_2CO$, C_6D_6 , $(CD_3)_2SO$, D_2O , etc.

These solvents differ considerably as regards their polarity. NMR spectrum of a compound measured in one solvent may be slightly different from that measured in another solvent of different polarity. Hence, it is important to mention the solvent used in an NMR spectrum.

There is always some residual $CHCl_3$ (~0.01%) in $CDCl_3$. A small $CHCl_3$ peak at δ 7.26 ppm from the $CDCl_3$ solvent will cause no difficulty. This solvent signal usually does not overlap with the signal of the target sample in most cases.

Properties of the Selected Solvent

1. It should be chemically inert, and magnetically isotropic.
2. It should be devoid of hydrogen atom. Generally, deuterated solvent is used.
3. It should be inexpensive.
4. It should have low boiling point.

5. It should dissolve the sample to a reasonable extent (at least 10% of the substance under investigation).

It has been found that the NMR spectrum of even completely deuterated solvent does show one or more peaks. It is due to the presence of minor impurities in the solvent.

Deuterated Solvents

Deuterated Solvents are used in NMR Experiments for the following reasons:

1). To avoid swamping by the solvent signal

There is usually much more solvent than sample in an NMR tube. Therefore, an ordinary proton containing solvent would give a huge solvent absorption that would dominate and spoil the ^1H -NMR spectrum.

2). To stabilize the magnetic field strength

The field strength of superconducting magnets tends to drift slowly. Modern NMR spectrometers measure the deuterium absorption of the solvent, and adjust the magnetic field strength to keep the resonance frequency constant.

The deuterium lock prevents changes in the static field (B_0) and radiofrequency (B_1) by maintaining a constant ratio between the two. It, therefore, ensures long-term stability of the magnetic field. If the ^2H lock is not applied, a drastic deterioration in the shape of the NMR lines is observed, due to magnetic field and radiofrequency inhomogeneities.

3). To accurately define 0 ppm

The deuterium signal is used to accurately define 0 ppm as the resonant frequency of the lock solvent. Deuterated solvents are used in NMR experiments.

CDCl_3 as solvent

CDCl_3 as Common Solvent in NMR. CDCl_3 is a common solvent used for NMR analysis. It is used because:

1. Most compounds dissolves in it.

2. It is volatile (b. p.: 61 °C), and therefore easy to get rid of from the precious compound of the sample.
 3. It is non-reactive.
 4. It does not exchange its deuterium with protons in the molecule being studied.
-

FACTORS INFLUENCING CHEMICAL SHIFT

There are a number of factors which influence the chemical shift at which proton resonances occur. These may be categorized as

(a) local diamagnetic shielding effects and

(b) the effect of neighbouring atoms or groups in the molecule

Each proton in a molecule is considered to share two bonding electrons, which are induced to circulate by the influence of the external magnetic field in a manner so as to produce local secondary fields which oppose the applied field.

These local diamagnetic shielding effects are dependent on the electron density around the atom and on the extent to which the C-H bond may be polarized.

The major factors affecting chemical shifts of the protons are the following:

1. Inductive Effects
2. Hybridisation Effects
3. Diamagnetic Anisotropy
4. van der Waals Deshielding
5. Solvent Effects
6. Concentration
7. Temperature
8. Hydrogen Bonding

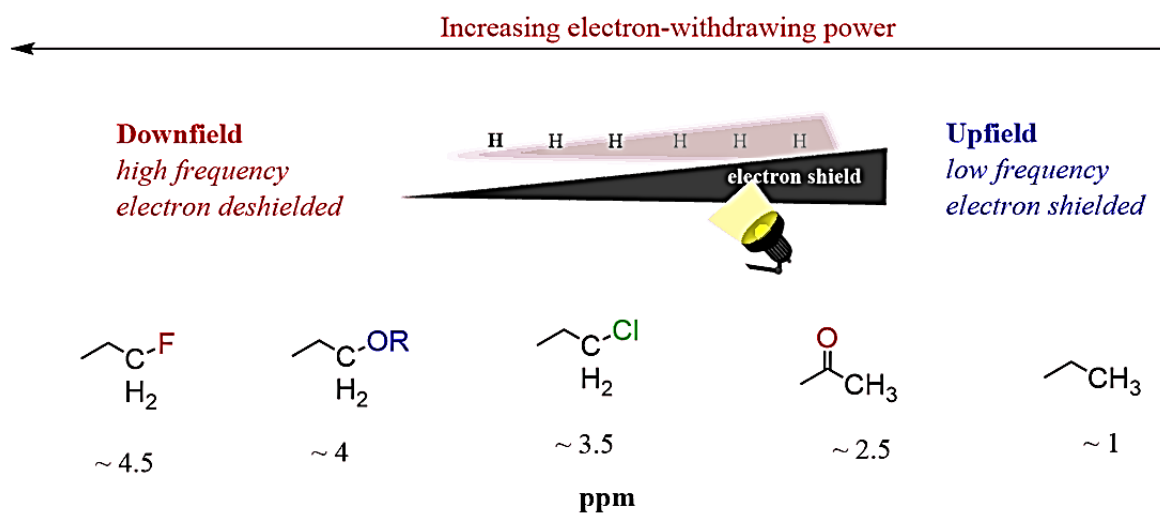
1. Inductive Effects

When a nucleus is placed in a uniform external magnetic field, the electrons surrounding it circulate in such a manner as to produce a secondary field which reduces the influence of the external magnetic

field on the nucleus. This shielding influence of the surrounding electrons decreases if another electronegative atom is attached to the nucleus because the protective electronic clouds are pulled away towards the neighbouring electronegative atom, causing it to come to resonance at a lower frequency.

On the other hand, groups which donate electrons will tend to increase the electron density around the nucleus, and hence a higher frequency is required for resonance to occur since the nucleus is more shielded from the influence of the external magnetic field.

The majority of alkanes show resonances between 0.8 and 1.6 ppm. Methyl groups attached to saturated carbon atoms are the most shielded of aliphatic protons and absorb in the region 0.6-1.2 ppm.



Methylene protons are less shielded and resonate around 1.3 ppm in saturated hydrocarbons while the methine protons resonate even further downfield (around 1.6 ppm). The methylene protons of cycloalkanes absorb at similar positions to those of alkanes except in cyclopropanes in which the methylenes are significantly shielded (chemical shift: 0.1-0.5 ppm) because of the anisotropy of the cyclopropane C-C bonds.

A proton will be deshielded if it is attached with an electronegative atom or group. Greater the electronegativity of the atom, greater is the deshielding caused to the proton. If the deshielding is more for a proton, then its δ value will also be more. Therefore, the chemical shift of a proton increases as the electronegativity of the attached element increases.

When methyl groups are attached to oxygen atoms, the methyl protons resonate at about 3.3 ppm, but if methyl groups attached to nitrogen, they experience a relatively higher shielding and resonate at around 2.3 ppm. Methyl groups attached to olefinic carbon atoms resonate around 1.6-1.8 ppm but if attached to aromatic rings they are deshielded and resonate around 2.4 ppm.

Multiple substituents have a stronger effect than a single substituent. The influence of the substituent drops off rapidly with distance, an electronegative element having little effect on protons that are more than three carbons distant. Table 1 and Table 2 illustrate these effects for the protons.

Table: Dependence of the Proton Chemical Shift (in ppm) of CH ₃ X on the Element X							
Compound CH ₃ X	CH ₃ F	CH ₃ OH	CH ₃ Cl	CH ₃ Br	CH ₃ I	CH ₄	(CH ₃) ₄ Si
δ value	4.26	3.40	3.05	2.68	2.16	0.23	0

Table: Substitution Effect					
Compound	CH ₃ Cl	CH ₂ Cl ₂	CHCl ₃	CH ₃ CH ₂ Br	CH ₃ CH ₂ Br
δ value	3.05	5.30	7.26	3.34	1.69

2. Steric effect

In overcrowded molecules, it is possible that some protons may occupy sterically hindered position resulting in van der Waal's repulsion. In such molecules the electron cloud of a bulk group (hindering group) will tend to repel the electron cloud surrounding the proton. Thus such a proton will be deshielded and will resonate at slightly higher value of chemical shift than the expected in the absence of this effect.

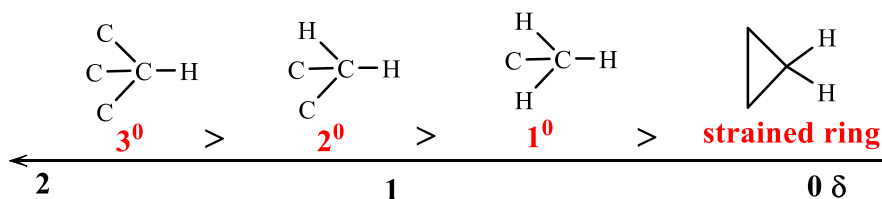
3. Hybridisation Effects

The state of hybridisation of the carbon atom to which a proton is attached significantly influences the chemical shift of the proton. With increasing s-character of the carbon, the bonding electrons will be drawn closer towards the carbon and away from the hydrogen, thus decreasing the electron density around the proton. This will result in the “**deshielding**” of the proton and it will therefore come to resonance at higher frequency.

i) *sp*³ Hydrogens

All hydrogens attached to purely *sp*³ carbon atoms (C-CH₃, C-CH₂-C, cycloalkanes) have resonance in the limited range from 0 to 1.6 ppm, provided that no electronegative elements or π -bonded groups are nearby.

Methylene hydrogens (attached to sp^3 carbons) appear at greater chemical shifts (near 1.2 to 1.4 ppm) than do methyl hydrogens. Methine hydrogens occur at higher chemical shift than methylene hydrogens. The following diagram illustrates these relationships:



Hydrogens on an sp^3 carbon that is directly attached to a heteroatom ($-O-CH_2-$, and so on) or to an unsaturated carbon ($-C=C-CH_2-$) do not fall in this region, but have greater chemical shifts.

ii). Sp^2 Hybridisation

Simple olefinic hydrogens ($-C=C-H$) have resonance in the range from 5 to 6 ppm. However, different substituents shift the olefinic proton resonances to a higher or to a lower frequency. In an $C_{sp^2}-H$ bond, the carbon atom has more s-character (33%), which effectively renders it “more electronegative” than an sp^3 carbon (25% s-character).

The s-orbitals hold electrons closer to the nucleus than do the carbon p-orbitals. As the sp^2 carbon atom holds its electrons more tightly, this results in less shielding for the attached hydrogen nucleus than do an sp^3 carbon atom. Thus, vinyl hydrogens have a greater chemical shift (5 to 6 ppm) than aliphatic hydrogens on sp^3 carbons (1 to 4 ppm).

Protons which are a part of a conjugated system appear downfield than when attached to a non-conjugated system. Aromatic hydrogens appear in a range further downfield (7 to 8 ppm) than to most olefinic protons.

Aldehydic protons (also attached to sp^2 carbon) appear even farther downfield (9 to 10 ppm) than aromatic protons since the inductive effect of the electronegative oxygen atom further decreases the electron density on the attached proton.

The downfield positions of ethylenic, aromatic, and aldehydic proton resonances are, however, greater than one would expect based on these hybridisation differences. Another effect, called **anisotropy**, is responsible for the largest part of these shifts.

iii). sp Hydrogens

An sp carbon (50% s-character) should behave as if it were more electronegative than an sp^2 carbon. On the basis of hybridization alone, one would expect the acetylenic proton to have a chemical shift greater than that of the ethylenic proton.

However, this is the opposite of what is actually observed. Acetylenic hydrogens ($C_{sp}\text{-H}$) appear anomalously at 2 to 3 ppm compared to olefinic protons which appear in the range 5-7 ppm. This sharp difference can be explained in terms of diamagnetic anisotropic effect involving the circulation of π electrons of carbon-carbon triple bond in an applied magnetic field. This effect depends on the orientation of the molecule in an applied magnetic field.

4. Diamagnetic Anisotropy

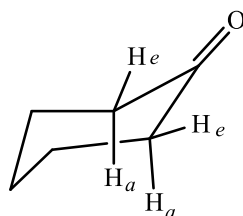
Magnetic anisotropy is the *uneven magnetic field* due to the circulation of π or even σ electrons. Electrons in π systems (e.g. aromatics, alkenes, alkynes, carbonyls etc.) interact with the applied field which induces a magnetic field that causes the anisotropy. As a result, the nearby protons will experience 3 fields: the applied field, the shielding field of the valence electrons and the field due to the π system. Depending on the position of the proton in this third field, it can be either shielded (smaller δ) or deshielded (larger δ).

The anisotropic effect depends upon the orientation of the molecule with respect to the applied field.

i. Single Bonds

The electronic clouds around protons attached to carbon atoms do not have spherical symmetry, but are asymmetrically distributed. The resulting secondary fields from protons are, therefore, “**anisotropic**”. A random tumbling of molecules will accordingly not result in their averaging to zero, but will reinforce or diminish the applied field, depending on the orientation of the bonds.

Thus, in a rigid cyclohexane ring or in cyclohexanone, the axial protons are more shielded than the equatorial protons because of the anisotropy of the carbon framework, so that they appear about 0.5 ppm further upfield than their equatorial counterparts.



ii. Double Bonds ($\delta 5\text{-}\delta 6$)

Anisotropic effects are even more marked in the case of double bonds such as $C=C$, $C=O$, and $C=N$. Thus, in a double bond, the applied magnetic field causes an induced circulation of π electrons when the double bond is aligned at right angles to the applied field which tends to oppose the applied magnetic field at the centre of the double bond, but reinforces it at the ends.

In solution, the tumbling molecules adopt anyone of a large number of orientations, and the average of these various orientations would afford two shielding cones with their apexes meeting at the centre of the double bond as shown in Figure

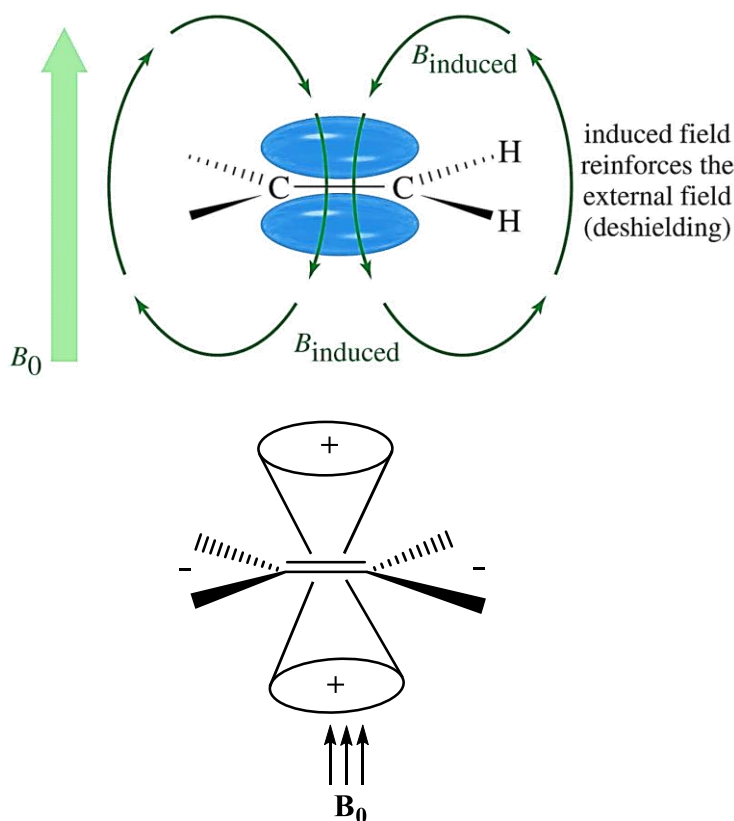


Figure: Magnetic isotropy of π electrons

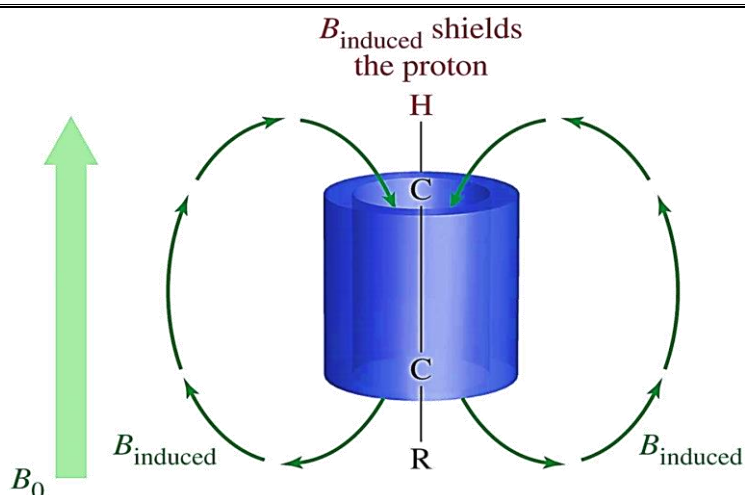
+ : shielding region
- : deshielding region

Any protons which fall in the regions of these cones (+zone) would be shielded whereas protons falling in the regions beyond the ends of the double bonds (-zone) would experience a deshielding influence. Similar situations would hold for $C=O$, $C=S$, $C=N$, and NO_2 groups. A striking example of such effects is provided by the aldehydic proton which lies in the deshielding zone of the carbonyl group, and is therefore shifted considerably downfield and appear even farther downfield (9-10 δ)

iii. Triple Bonds

Triple bonds, such as those encountered in acetylenes or nitriles, have axial symmetry. The induced circulation of π electrons in acetylenes or nitriles results in the generation of a secondary magnetic field which tends to oppose the applied field at the ends but reinforces it at a certain distance along the axis.

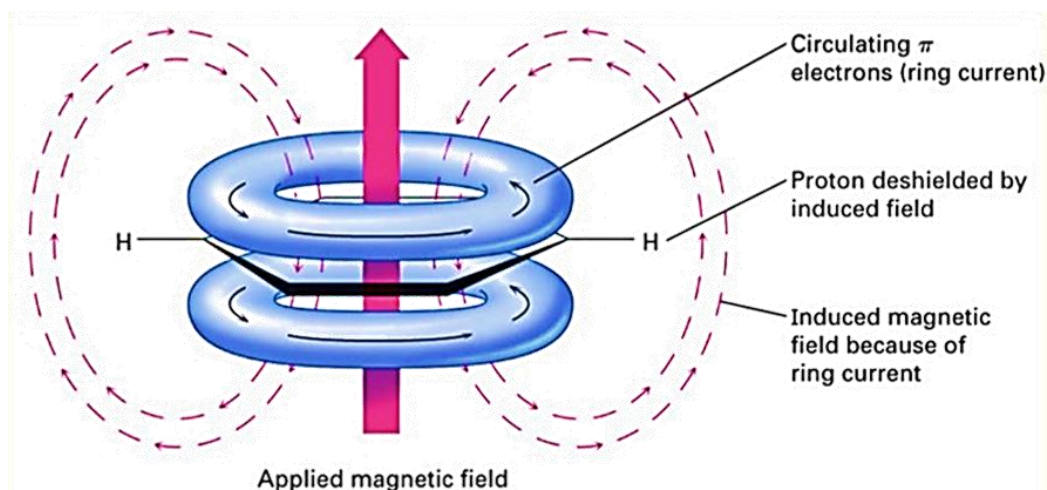
Since acetylenic protons fall in the shielding zone of the cone they are shifted significantly upfield and resonate at a value of δ 1.88, in contrast to ethylenic protons which resonate at δ 5.25.



Methyl protons in propyne ($\text{H}_3\text{C}-\text{C}\equiv\text{CH}$) appears somewhere outside this shielding zone, and hence resonated at δ 1.80.

iv. Aromatic Rings

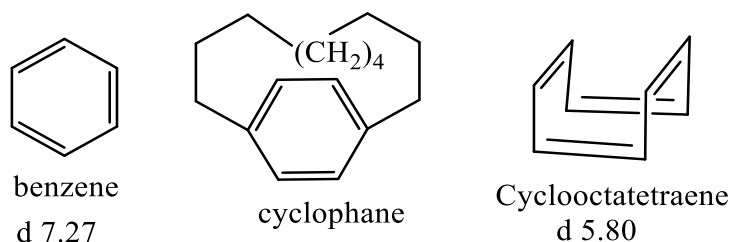
In aromatic rings, the π electrons delocalized over the ring atoms are induced to circulate above and below the plane of the ring (in two loops on the two sides of the σ bond framework). This circulation is such as to oppose the applied magnetic field at the centre of the ring but reinforce it in the region where the aromatic protons are located. This results in a strong deshielding of the aromatic protons.



On the other hand, if a nucleus is held above the centre of the aromatic ring, it will be strongly shielded. Thus, in cyclophane (Figure), the central methylene protons are located in the shielding region of the aromatic ring, and are therefore, shifted upfield by 0.45 ppm as compared to the methylene protons in a normal aliphatic chain.

The ring current effects are only discernible in planar conjugated cyclic molecules. In nonplanar molecules, e.g., cyclooctatetraene, which is tub-shaped, the delocalisation of π electrons is severely

restricted, and the protons of cyclooctatetraene, therefore, resonate in the normal olefinic region (δ 5.80, cf. 5.85 in cyclohexadiene).

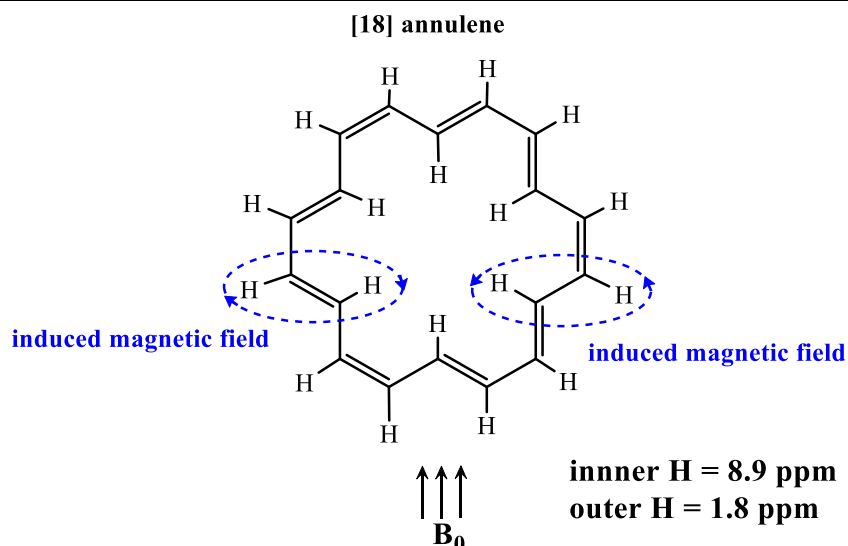


The effect of substituents on the chemical shifts of protons in aromatic rings is shown in Table. Strongly electron withdrawing groups, such as nitro or aldehyde, result in significant deshielding of the aromatic protons, the ortho and para protons being more affected than the meta protons on account of the greater transmission of electropositive character to the ortho and para carbons by mesomerism.

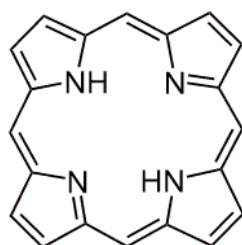
Conversely, the presence of amino groups on benzene rings causes a shielding of the ortho and para carbons because of the donation of electrons from the nitrogen lone pair to these carbon atoms, resulting in an increase in the electron density at the attached protons.

Table: Chemical shifts of <i>ortho</i> , <i>meta</i> , and <i>para</i> protons in monosubstituted benzenes			
δ (in ppm) from benzene (7.27)			
Substituent	<i>ortho</i>	<i>meta</i>	<i>para</i>
NO ₂	0.93	0.26	0.38
CHO	0.60	0.24	0.34
CO ₂ H	0.85	0.35	0.38
CO ₂ R	0.76	0.30	0.35
COR	0.57	0.25	0.30
CONH ₂	0.55	0.27	0.20
NHCOR	0.22	0.03	-0.17
Cl	0.02	0.00	-0.01
Br	0.20	0.00	-0.08
CH ₃	-0.15	-0.10	-0.10
OH	-0.43	-0.03	-0.34
OCH ₃	-0.39	-0.01	-0.35
NH ₂	-0.63	-0.15	-0.54
N(CH ₃) ₂	-0.50	-0.06	-0.50

Interestingly, aromatic compounds with inner hydrogens such as, for example, porphyrins, [18]-annulene and the ones with hydrogens over the ring are shielded by the induced magnetic field and appear scientifically upfield:



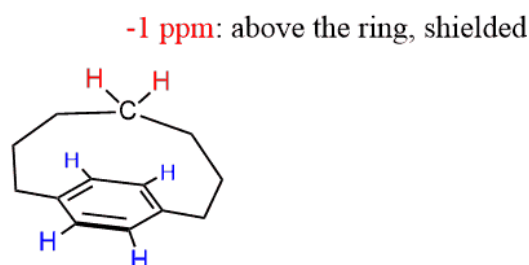
In [18] annulene, the applied and induced magnetic fields are opposite near inner protons, hence the inner protons appear at lower ppm (downfield) while the applied and induced magnetic fields are aligned near outer protons and hence appear at high ppm (upfield).



A Porphyrin

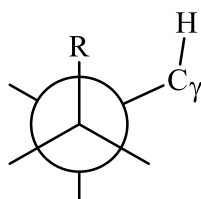
Inner hydrogens: **-2 ppm**

Outer hydrogens: **8 ppm**



5. Van der Waals Deshielding

In a rigid molecule, it is possible for a proton to occupy a sterically hindered position, and in consequence, the electron cloud of the hindering group will tend to repel, by van der Waals electrostatic repulsion, the electron cloud surrounding the proton. The proton will be deshielded and appear at higher δ values than would be predicted in the absence of the effect.



Van der Waal's deshielding

Steric crowding involving the hydrogen atom/s can result in proton-bearing carbon atoms being shielded. Steric perturbation of C-H bonds results in the electrons moving away from the hydrogen atoms and towards the carbon atoms, thus resulting in the carbon atoms (γ) being shielded (and the protons deshielded).

6. Solvent Effects

NMR spectrum of a compound measured in one solvent may be slightly different from that measured in another solvent of different polarity. Hence, it is important to mention the solvent used in an NMR spectrum.

Substances dissolved in aromatic solvents generally give signals at higher fields than when dissolved in aliphatic solvents. These effects are attributed to the diamagnetic anisotropy of aromatic rings, and are more significant when intermolecular solute-solvent associations lead to complex formation. The aromatic solvents benzene and pyridine cause changes in chemical shifts as large as 0.5 to 0.8 ppm compared to less magnetically active solvents like chloroform or acetone.

As a result of dipole-dipole or Van der Waals interactions between the solute and solvent, some orientations of the solute molecules may be more favoured than others, giving rise to shifts in the resonance frequencies of the protons of the solute molecules.

7. Temperature and Concentration

The chemical shifts of exchangeable protons such as those of OH, NH, or SH groups are markedly dependent on concentration, solvent, and temperature. Protons that can form hydrogen bonding (e.g., hydroxyl or amino protons) exhibit extremely variable absorption positions over a wide range.

Some of the deshielded protons are those attached to carboxylic acids. These protons have their resonances at 10 to 12 ppm. Both mesomeric and electronegativity effects of oxygen withdraw electrons from the acidic proton making it shifted enormously downfield.

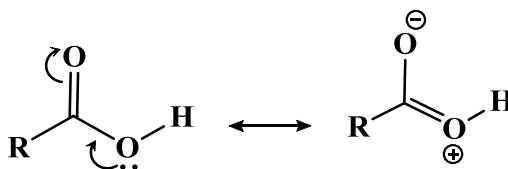


Table: Chemical shifts and characteristics of OH, NH, and SH protons		
Protons	δ values	Special features
R-OH (pure)	0.5-4.0	Can appear as a multiplet because of coupling with adjacent protons, but usually occurs as a broad singlet; position is temperature, solvent, and concentration dependent. H-bonded enols may

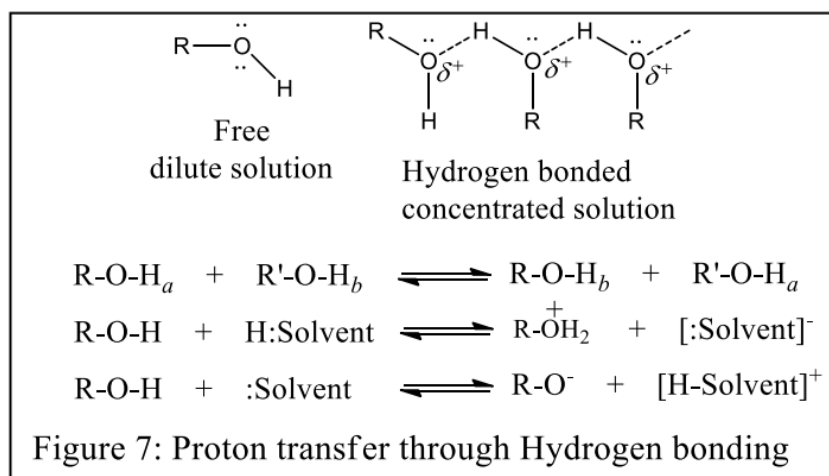
		occur from δ 11.0 to 16.0
R-OH (with a trace of acid)	4-8 (variable)	Occurs as a sharp singlet; position of line is concentration dependent
Ar-OH	4.5 (without H-bonding)	Position of signals dependent on solvent, concentration, and temperature
	9.0 (with H bonding)	
R-CO ₂ H	10-13	Position of signals largely dependent on concentration
RNH ₂ , RNHR'	2-5	Usually sharp single line but may be broadened; position is solvent dependent

8. Hydrogen Bonding

Hydrogen bonding results in downfield shifts of proton resonances from their positions in the unbonded state. The higher δ values of the hydroxyl resonance of phenols or carboxylic acids are attributed to hydrogen bonding. The amount of hydrogen bonding is often a function of concentration and temperature. Increase of temperature or dilution with carbon tetrachloride can often break up hydrogen bonds and cause an upfield shift.

The more hydrogen bonding that takes place, the more deshielded a proton becomes. The more concentrated the solution, the more molecules can come into contact with each other and form hydrogen bonds. At high dilution (no H bonding), hydroxyl protons absorb near 0.5–1.0 ppm; in concentrated solution, their absorption is closer to 4–5 ppm. Protons on other heteroatoms show similar tendencies.

Hydrogens that can exchange either with the solvent medium or with one another also tend to be variable in their absorption positions. The following equations (Figure) illustrate possible situations,



The hydrogen bonded proton being attached to a highly electronegative atom will have smaller electron density around it. Being less shielded, the field felt by such a proton will be more and hence resonance will occur downfield. The downfield shift depends upon the strength of hydrogen bonding. Intermolecular and intra- molecular hydrogen bonding can be easily distinguished as the latter does not show any shift in absorption due to the change in concentration. In case of phenols, absorption occurs between—2 to 6 δ . But if the concentration is decreased, i.e., if the volume of the solvent, say, carbon tetrachloride is increased, then the absorption for OH proton occurs upfield. In case the OH group on benzene is intramolecularly bonded with some other group in the ortho position, the absorption for OH proton may occur even at the negative tau value. For example, the OH proton in salicylic acid absorbs at -0.6τ . Enols show strong intra-molecular hydrogen bonding which is further stabilised by resonance. Due to this, a great deshielding effect is caused and absorption for such a proton occurs at the negative tau value (10.5–12 δ). It is due to the fact that acids exist as dimers as a result of hydrogen bonding. A signal for carboxylic acid proton in ethoxy acetic acid appears at 10.95 δ .

INTER AND INTRA-MOLECULAR H-BONDING

The intermolecular and intramolecular hydrogen bonding can be distinguished by the use of ^1H NMR spectroscopy. The intermolecular hydrogen bonding depends on the concentration of the sample and decreases on dilution with a non-polar solvent and also with increasing temperature. The pure ethanol show signals near 5 ppm but on dilution with non-hydrogen-bonded solvents (CCl_4 , CDCl_3 , C_6D_6), the OH signal shift upfield due to the breaking of intermolecular hydrogen bonds. The more acidic hydroxyl group of phenol also shows similar concentration dependence to that of alcohols.

The intramolecular hydrogen bonding does not depend on the concentration of the sample and hence show no change in their absorption position on dilution. The salicylates, enols of β - diketones and carboxylic acids are some classic examples of intramolecular hydrogen bonding. The hydroxyl proton of enol form of 2,4-pentanedione (4-hydroxypent-3-ene-2-one) shows a peak significantly down field (10-16 ppm) compared to the OH signal in other function groups due to strong intramolecular hydrogen bonding. Similarly, the hydroxyl proton of carboxylic acid displays a very downfield peak at about 10-13 ppm which may be attributed to the favored hydrogen-bonded dimeric association.

Chapter 4

NMR PEAK AREA

PEAK AREA AND PROTON COUNTING

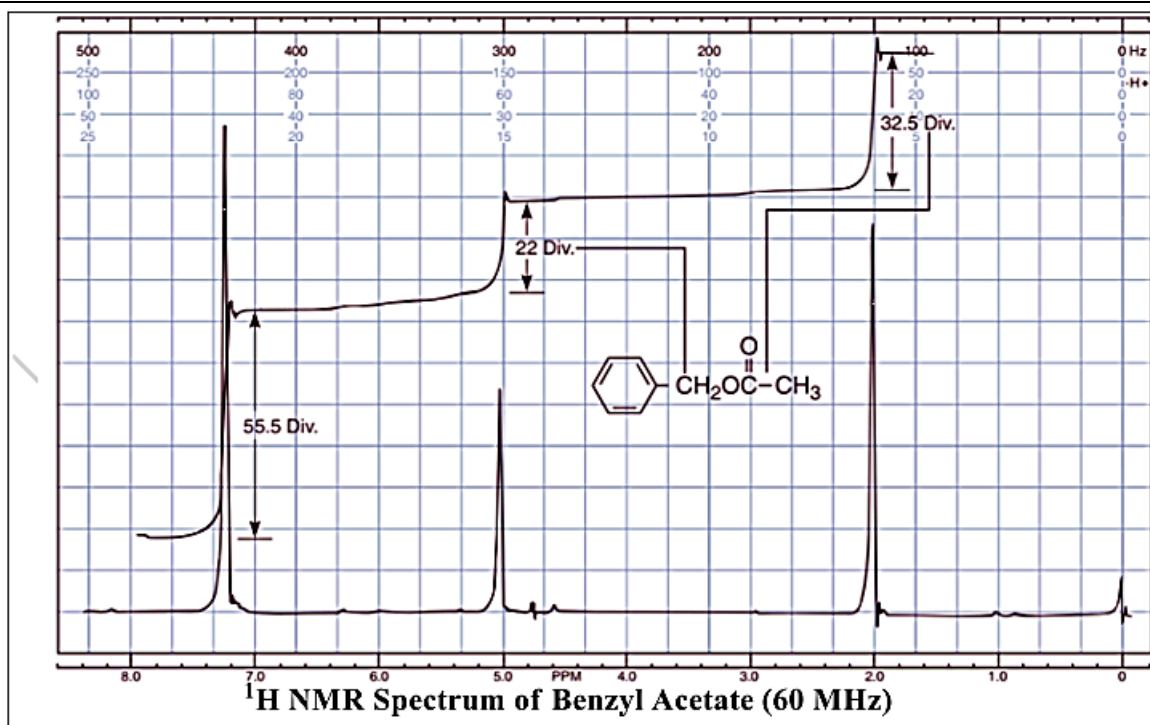
In an nmr spectrum, various peaks represent equivalent sets of protons. The size or the area of each peak tells the number of protons in each set present in the compound under investigation. The area under an nmr signal is directly proportional to the number of protons giving rise to signal. For flipping over of a proton, a quantum of energy is absorbed in the same effective magnetic field. Greater the number of protons that flip over at a particular frequency, greater will be the energy absorbed and greater is the area under the absorption peak.

Squares under each peak are simply counted and from this, the ratio between various kinds of protons is found out. These ratios are then converted into whole numbers. These whole numbers (or some multiple of them) tell the number of protons represented by the various nmr signals.

Integration

It is a process by which relative areas under spectral peaks are measured. These areas are proportional to the number of equivalent protons giving the signal. The NMR spectrum not only distinguishes how many different types of protons a molecule has, but also reveals how many of each type are contained within the molecule. In the NMR spectrum, the area under each peak is proportional to the number of hydrogens generating that peak.

Let us consider the ^1H NMR spectrum of benzyl acetate for instance. The NMR spectrometer has the capability to electronically integrate the area under each peak. It does this by tracing over each peak a vertically rising line, called the integral, which rises in height by an amount proportional to the area under the peak. The integral values are proportional to the actual number of protons represented by the peak.



It is noted from the spectrum that the height of the integral line does not give the absolute number of hydrogens. It gives the relative number of each type of hydrogen. For a given integral to be of any use, there must be a second integral to which it may be referred. The first integral rises for 55.5 divisions on the chart paper; the second, 22.0 divisions; and the third, 32.5 divisions. These numbers are relative. One can find ratios of the types of protons by dividing each of the larger numbers by the smallest number:

$$\frac{55.5 \text{ division}}{22.0 \text{ division}} = 2.52 \quad \frac{22.0 \text{ division}}{22.0 \text{ division}} = 1.00 \quad \frac{32.5 \text{ division}}{22.0 \text{ division}} = 1.48$$

The number ratio of the protons of all the types is **2.52:1.00:1.48**. If it is assumed that the peak at 5.1 ppm is really due to two hydrogens, then one can arrive at the true ratio by multiplying each figure by 2 and rounding off to 5:2:3. The area ratio of the three peaks is 5:2:3, the same as the ratio of the numbers of the three types of hydrogens in benzyl acetate.

Clearly, the peak at 7.3 ppm, which integrates for five protons, arises from the resonance of the aromatic ring protons, whereas that at 2.0 ppm, which integrates for three protons, is due to the methyl protons. The two-proton resonance at 5.1 ppm arises from the benzylic protons. Thus, it can be said that the integrals give the simplest ratio, but not necessarily the true ratio, of numbers of protons of each type.

*Chapter 5***PROTONS EQUIVALENCY**

The behaviour and appearance of NMR spectrum for a particular compound depends upon the nature of proton and their environment. The peak value (chemical shift) due to a proton in a molecule is affected by the magnetic field B^0 (induced magnetic field) produced by the circulation of neighbouring electrons. But the appearance of the absorption of peak for proton depends upon the magnetic field produced by neighbouring nuclei. Sometimes the signal appears as single line while mostly it appears as multiplet due interaction of nearby protons.

It has been observed that only non-equivalent neighbouring protons affect the absorption peak (signal) of protons while equivalent protons have no effect on the peak of a proton. The equivalency of proton may be described in two different ways: chemically equivalent protons and magnetically equivalent protons.

A Chemically Equivalent Protons

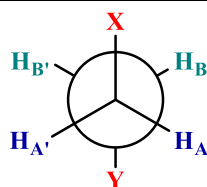
Two or more protons are defined as being chemically equivalent if their electronic environments are indistinguishable. They experience identical shielding effects, and therefore, they possess the same value of chemical shift.

B Magnetically Equivalent Protons

Two protons are defined as being magnetically equivalent if each couple equally to a third neighbouring proton; otherwise they are magnetically nonequivalent.

Chemical equivalence means chemical shift equivalence, and magnetic equivalence means coupling equivalence. Similarly, magnetically equivalent nuclei will not only be chemically equivalent but also have the same coupling constant with every other nucleus in the molecule. Therefore, the test for magnetic equivalence is valid only when the two nuclei are chemical shift equivalent. However, if the two protons are chemically nonequivalent, they must also be magnetically nonequivalent,

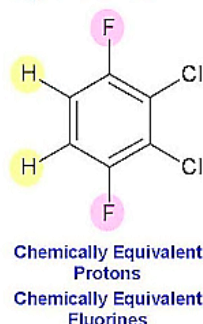
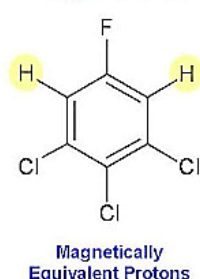
The following structure represents the most stable conformation of the molecule of the type $X-CH_2-CH_2-Y$. As a result of symmetry, H_A and $H_{A'}$ will have the same chemical shift values, as will H_B and $H_{B'}$.



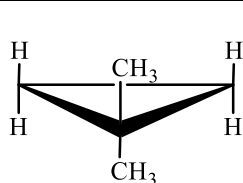
Now H_A will undergo spin-spin coupling to H_B , but (assuming no rotation around the C-C bond) this will be different from the coupling of $H_{A'}$ to H_B . Therefore, the protons H_A and $H_{A'}$ are chemically equivalent but are magnetically nonequivalent.

In the molecule on the left, the two protons are both chemically and magnetically equivalent as they both have the same coupling with the fluorine. In the molecule on the right, the two protons are chemically equivalent but not magnetically equivalent as they do not have the same coupling to each of the fluorines. Similarly, the two fluorines are chemically equivalent but not magnetically equivalent as they do not have the same coupling to each of the protons.

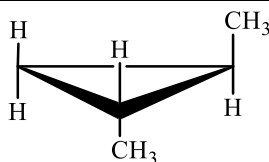
Magnetic and Chemical Equivalence



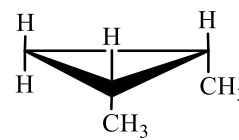
S. No	Compound	Structure	No. of Non-equivalent protons	No. of signals
1	Ethane	H_3C-CH_3	0	1
2	Ethane chloride	H_3C-CH_2-Cl	2	2
2	Propane	$H_3C-CH_2-CH_3$	2	2
2	2-chloropropane	$H_3C-CH(Cl)-CH_3$	2	2
3	1-chloropropane	$H_3C-CH_2-CH_2-Cl$	3	3



1,1-dimethylcyclopropane
2 signals

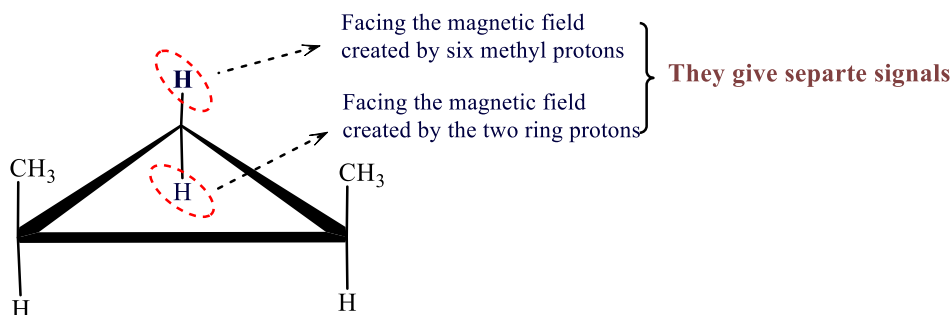


trans-1,2-dimethylcyclopropane
3 signals



cis-1,2-dimethylcyclopropane
4 signals

All magnetically equivalent protons are chemically equivalent but not all chemically equivalent protons are magnetically equivalent



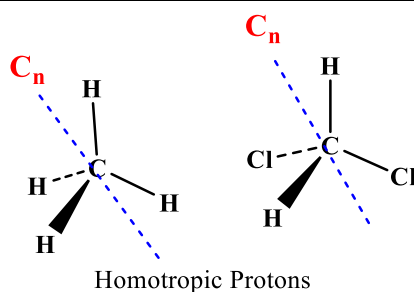
TOPICITY

Protons in identical chemical environments have identical chemical shifts. However, just because two protons have the same connectivity to the molecule does not mean they are chemically equivalent. This is related to the concept of Topicity.

Topicity is the stereochemical relationship of substituents relative to the structure to which they are attached. the stereochemical relationship between different groups in a molecule. Depending on the relationship, such groups can be heterotopic, homotopic, enantiotopic, or diastereotopic.

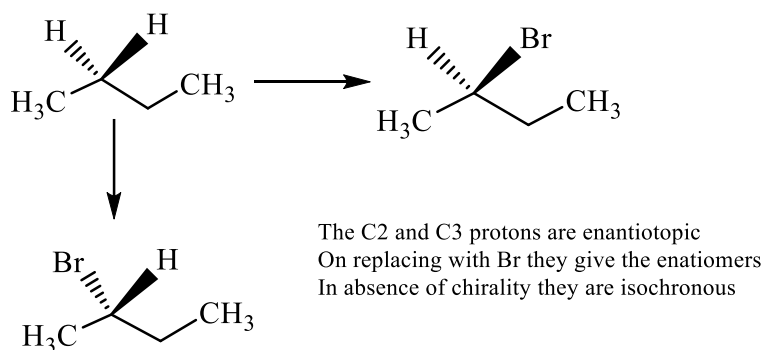
1. Homotopic Protons

Protons that are interchangeable through an axis of symmetry (C_n) are homotopic. Homotopic groups are equivalent groups. Two groups A and B are homotopic if the molecule remains the same (including stereochemically) when the groups are interchanged with the remaining parts of the molecule fixed. Homotopic atoms are always identical, in any environment. Homotopic NMR-active nuclei have the same chemical shift in an NMR spectrum. Examples: protons of CH_4 and CH_2Cl_2 .



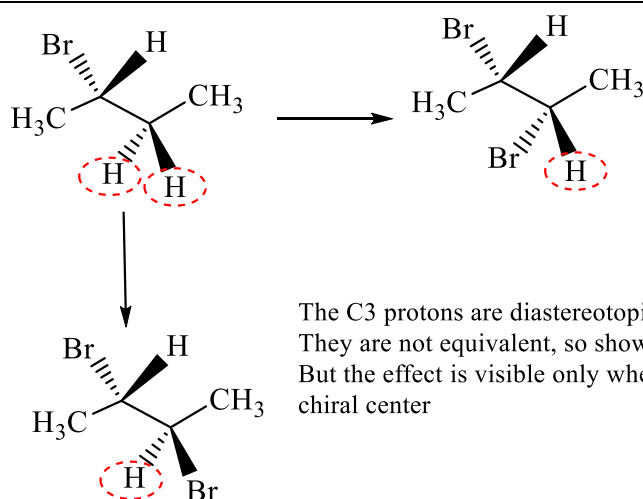
2. Enantiotopic Protons

Protons that are interchangeable through any other symmetry operation are called **enantiotopic**, and these are chemical shift equivalent only in an achiral environment. Enantiotopic refers to the relationship between two groups in a molecule which, if one or the other were replaced, give a chiral compound. The two possible compounds resulting from that replacement would be enantiomers. They are equivalent under normal conditions. They give single signal normally. In a chiral environment (solvent/reagent) they are not equivalent.



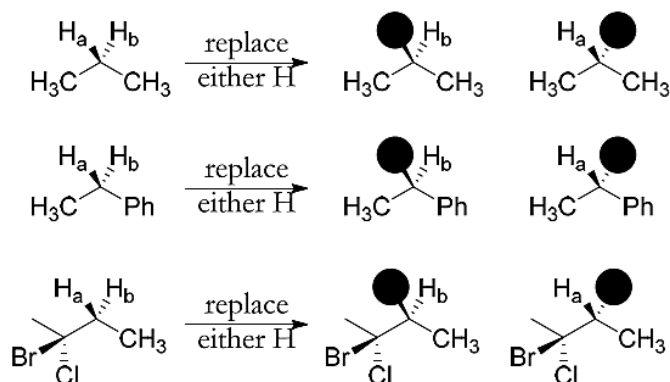
3. Diastereotopic Protons

Non-interchangeable geminal protons are diastereotopic in nature, and they are not chemical shift equivalent in any environment. The groups in a molecule which, if replaced, would generate compounds that are diastereomers. Diastereotopic groups are often, but not always, identical groups attached to the same atom in a molecule containing at least one chiral center. If two pairs of enantiotopic groups are there anyone pair can be considered as diastereotopic.



The C3 protons are diastereotopic
They are not equivalent, so show separate signals
But the effect is visible only when the groups are adjacent to a chiral center

To find the topicity relationship of two groups to each other, replace first one group, then the other group with a placeholder. If the two molecules produced are identical, then the groups are homotopic; if the molecules are enantiomers, then the groups are enantiotopic; and if the molecules are diastereomers, then the groups are diastereotopic. Groups that are diastereotopic are chemically inequivalent, so they will have a different chemical shift from each other in NMR, and will show coupling as if they were neighboring protons instead of on the same carbon atom



These molecules are identical, so H_a and H_b are homotopic (chemically equivalent)

These molecules are enantiomers, so H_a and H_b are enantiotopic (chemically equivalent in an achiral environment only)

These molecules are diastereomers, so H_a and H_b are diastereotopic (chemically inequivalent)

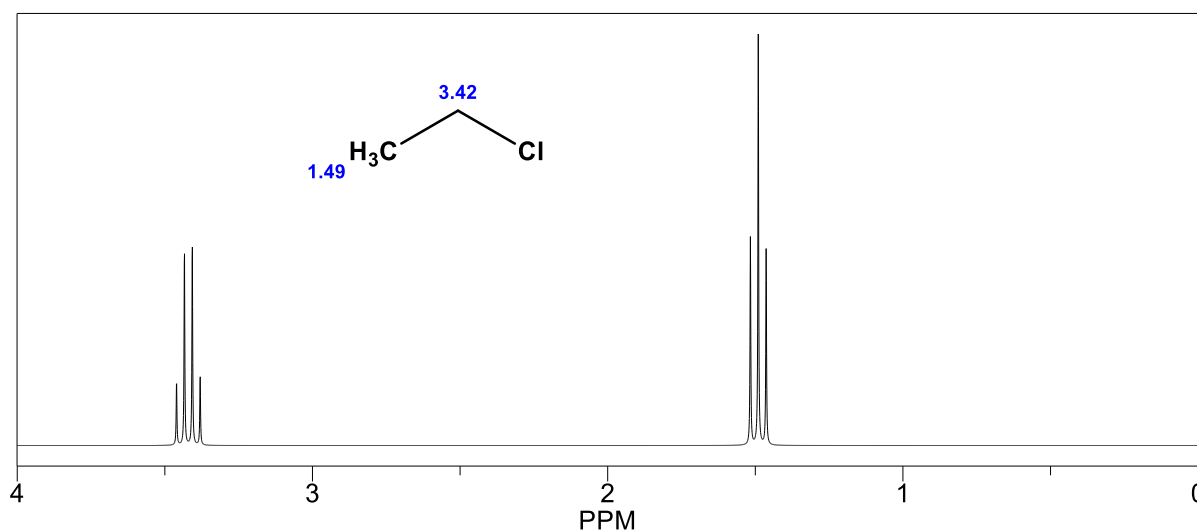
Chapter 6

SPIN-SPIN COUPLING

INTRODUCTION

Each signal in a nmr spectrum represents one kind or one set of protons in a molecule. It is found that in certain molecules, a single peak (singlet) is not observed, but instead, a multiplet (group of peaks) is observed. *This splitting of the signal is called the multiplicity of the signals.* The multiplicities of the signals provide information about the environment of the neighbouring proton/s.

Consider the ^1H -NMR spectrum of a compound, for instance $\text{CH}_3\text{CH}_2\text{Cl}$. As expected, CH_3 , and CH_2 protons appeared at different positions in the spectrum as they are nonequivalent, and have different chemical shifts.



NMR Spectrum of Ethyl Chloride

These signals, however, do not necessarily occur as single peaks (singlets). In ethyl chloride, methyl protons (CH_3) have two neighbouring equivalent protons (CH_2), and it is observed that it appears as a three-line signal (**a triplet**). Similarly, methylene protons (CH_2) has three neighbouring equivalent protons (CH_3), and the signal is split into four lines (**a quartet**).

The number of lines (multiplicity) observed in the NMR signal for a group of protons is, therefore, not related to the number of protons in that group; the multiplicity of lines is related to the number of protons in neighbouring groups.

SPIN-SPIN COUPLING

Splitting of the spectral lines arises because of a coupling interaction between the neighbouring protons attached to the same carbon or to adjacent carbon atoms, and is related to the number of possible spin orientations that these neighbours can adopt. **The phenomenon is called spin-spin splitting or spin-spin coupling.**

The multiplicity of the signal arises due to the actual magnetic field of a proton is modified by the instantaneous spin arrangement, either parallel or antiparallel, of the neighbouring proton and their attainment magnetic fields which are transmitted through the bonded electrons.

Consider a molecule of ethyl bromide ($\text{CH}_3\text{CH}_2\text{Br}$). The spin of two protons ($-\text{CH}_2-$) can couple with the adjacent methyl group ($-\text{CH}_3$) in three different ways relative to the external field. The three different ways of alignment are:

(i)	$\uparrow\uparrow$	Reinforcing	 External field
(ii)	$\uparrow\downarrow \downarrow\uparrow$	Not effecting	
(iii)	$\downarrow\downarrow$	Opposing	

Thus, a triplet of peaks results with the intensity ratio of **1:2:1** which corresponds to the distribution ratio of alignment. Similarly, the spin of three protons (CH_3-) can couple with the adjacent methylene group ($-\text{CH}_2-$) in four different ways relative to the external field.

(i)	$\uparrow\uparrow\uparrow$	Strongly Reinforcing	 External field
(ii)	$\uparrow\uparrow\downarrow \uparrow\downarrow\uparrow \downarrow\uparrow\uparrow$	Weakly Reinforcing	
(iii)	$\downarrow\downarrow\uparrow \downarrow\uparrow\downarrow \uparrow\downarrow\downarrow$	Weakly Opposing	
(iv)	$\downarrow\downarrow\downarrow$	Strongly Opposing	

Thus, quartet of peaks results with an intensity ratio of 1: 3: 3: 1 which corresponds to the distribution ratio of all the alignments.

MULTIPLICITY OF THE SIGNAL: THE N+1 RULE

The simple rule to find the multiplicity of a signal from a group of protons is the number of the neighbouring (n) equally coupled protons plus 1 (one).

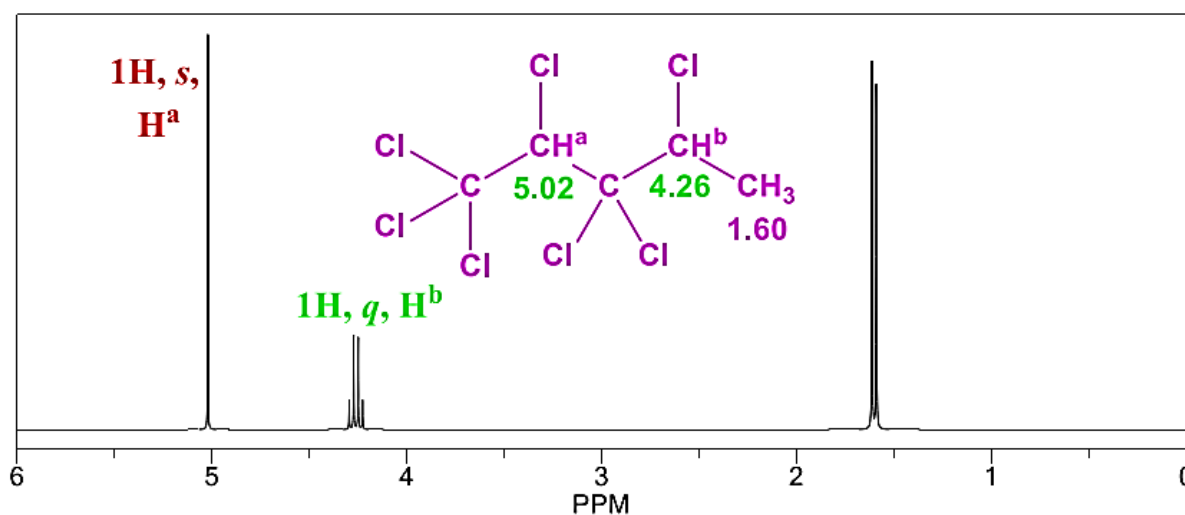
If a proton (or protons) is coupled with n other equivalent protons, then it will give rise to (n + 1) peaks. Thus, in an ethyl group, the methylene protons are coupled to three adjacent equivalent protons,

and hence their resonance is split into $(3 + 1) = 4$ lines. The methyl protons are coupled to the two equivalent methylene protons, and their resonance is, therefore, split into $(2 + 1) = 3$ lines.

A wider generalization would be $S = 2nI + 1$ where S is the number of lines formed by coupling, i.e., multiplicity, n is the number of equivalent neighbouring nuclei which are causing the coupling, and I is the spin quantum number of the nuclei causing the coupling. This generalization is correct only when the coupled nuclei have a spin of $1/2$.

Thus, in considering the multiplicity of the CH_2 protons in the $-\text{CH}_2-\text{CH}_3$ group, n will refer to the number of protons with which the CH_2 protons are coupled, which in this case is three, whereas the spin quantum number of the nuclei with which the CH_2 protons are coupled, which in this case is $1/2$.

It is to be noted that area under a one-proton signal is always the same whether it is a singlet, doublet, triplet, quartet or a multiplet. This statement is true in all cases with any fixed number of proton/s forming the signal. The spectrum in the following figure shows two one-proton signals, more downfield proton appears as a singlet, but the other as a quartet, though the area/integration under each signal is the same.



^1H -NMR Spectrum of 1,1,1,2,3,3,3-heptachloropentane

RELATIVE INTENSITIES OF LINES IN A MULTIPLET

The relative intensities of the individual lines of a multiplet correspond to the numerical coefficient of the lines in the binomial expression: $(1+x)^n$ or $(a+b)^n$.

When a proton or group of protons has one neighbouring proton, then $n = 1$, i.e., $(a + b)^1 = a + b = 1a + 1b$, the ratio of the coefficients is 1:1. When $n = 2$, i.e., $(a + b)^2 = a^2 + 2ab + b^2$, the ratio of the

coefficients is 1:2:1. When $n = 3$, i.e., $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$, the ratio of the coefficients is 1:3:3:1.

S#	$(a+b)^n$	Solution	Lines Intensities
1	$(a+b)^1$	$a+b$	1:1
2	$(a+b)^2$	$a^2+2ab+b^2$	1:2:1
3	$(a+b)^3$	$a^3+3a^2b+3ab^2+b^3$	1:3:3:1
4	$(a+b)^4$	$a^4+4a^3b+6a^2b^2+4ab^3+b^4$	1:4:6:4:1

PASCAL'S TRIANGLE

The theoretical line intensity ratios of multiplets derived from the $(1 + x)^n$ rule can be easily verified by following the entries in the mathematical mnemonic device called **Pascal's triangle**. Each entry in the triangle is the sum of the two entries above it.

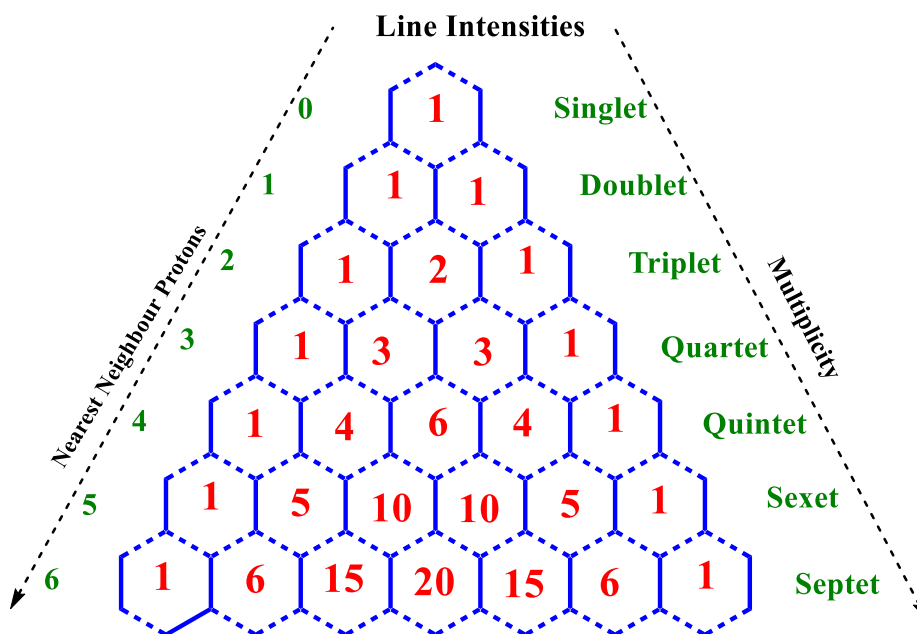


Figure: Pascal Triangle

The intensities of the outer peaks of a multiplet such as a septet are so small compared to the inner peaks that they are often obscured in the baseline of the spectrum.

COUPLING CONSTANT

The distance between the centres of the two adjacent peaks in a multiplet is usually constant and is called coupling constant(J).

The coupling constant is a measure of how strongly a proton is affected by the spin states of its neighboring protons. The spacing between the adjacent peaks of a multiplet is measured on the same scale as the chemical shift, and the coupling constant (J) is always expressed in Hertz (Hz) and is determined by the following formula:

$$J_{\text{Hz}} = \Delta\text{ppm} \times \text{instrument frequency}$$

Where Δppm is the difference in ppm of two peaks for a given proton.

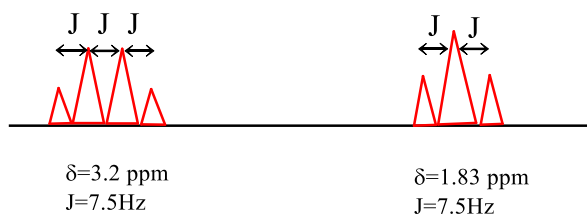
For example, if the spacing between the peaks is 0.2 ppm and the frequency of the spectrophotometer is 30 Hz, then the value of J constant is $(0.2 \text{ ppm} \times 30 \text{ MHz}) = 6 \text{ Hz}$. The same proton with have $\Delta V = 0.1 \text{ ppm}$ in a 60 MHz spectrometer and the value of J constant $(0.1 \text{ ppm} \times 60 \text{ MHz} = 6 \text{ Hz})$.

In general coupling constant is designated as $^nJ_{ab}$. The superscript indicates the number of bonds through which interaction occurs. For example, 1J , 2J , 3J means that the one, two or three bonds are involved in coupling interaction. The subscript (ab) tells about the nuclei (H_a and H_b) which are taking part in the coupling.

The value of coupling constant remains the same irrespective of the strengths of the applied magnetic field. The J values for proton–proton coupling can range from **0** to **20** Hz, but most commonly coupling constants fall in the range of 0–10 Hz. A value of $J = 0$ means that there is no significant interaction with neighbouring protons, and thus the absorption is not affected by the spin states of neighbouring protons. This normally occurs when there are more than three bonds separating different types of protons.

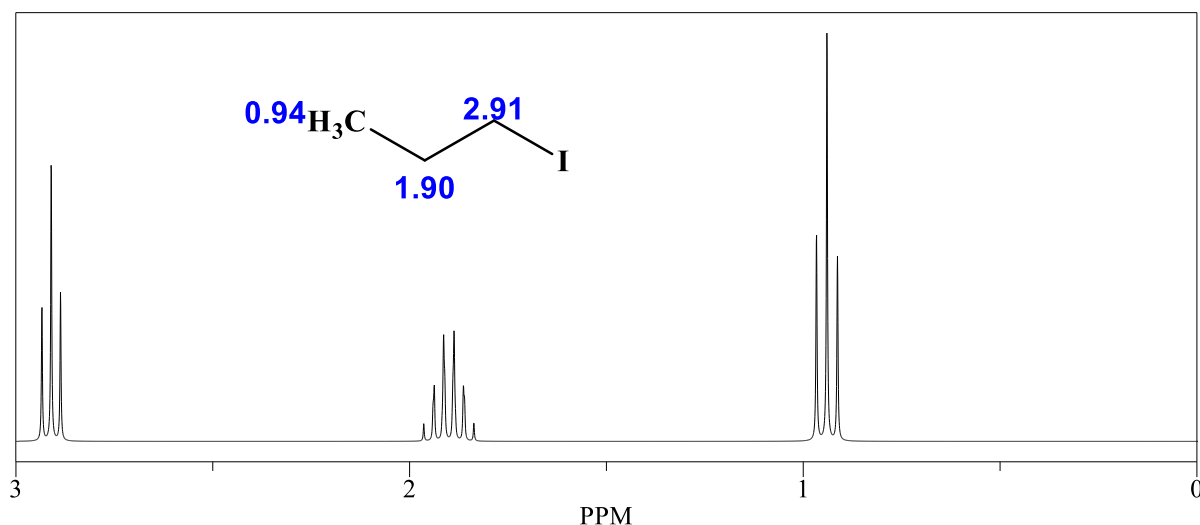
The interaction between protons is reciprocal— if two protons are coupled, they are coupled equally and $J_{ab} = J_{bc}$. That is, if H_a is split by H_b by some amount, say $J=6 \text{ Hz}$, then H_b must be split by H_a by $J = 6 \text{ Hz}$.

In the NMR spectrum of ethyl iodide ($\text{CH}_3\text{CH}_2\text{I}$), the methyl protons give rise to a triplet centred at 1.83 ppm, and the methylene protons give a quartet centred at 3.20 ppm, the coupling constant between methyl and methylene protons is 7.5 Hz.



The quartet due to $\text{—CH}_2\text{—}$ is split up to the same extent by —CH_3 protons as the splitting of triplet due to —CH_3 by $\text{—CH}_2\text{—}$ protons.

In the NMR spectrum of 1-iodopropane, $\text{CH}_3^c\text{—CH}_2^b\text{—CH}_2^a\text{—I}$, the signal for the 'b' types of protons is under the influence of protons 'a' as well as protons 'c'. Here, $J_{ab} = 6.8$ Hz and $J_{bc} = 7.3$ Hz. Under the influence of the five adjacent protons, the signal for $\text{—CH}_2^b\text{—}$ protons will be split into a sextet. If J_{ab} were equal to J_{ac} , a symmetrical sextet would have been observed. In case of 1-iodopropane, J_{ab} is close enough to J_{ac} , and hence the signal for $\text{—CH}_2^b\text{—}$ protons will resemble a sextet.



On the other hand, in the NMR spectrum of pure ethyl alcohol, $\text{CH}_3^a\text{—CH}_2^b\text{—OH}^c$, $J_{ab} = 5.0$ Hz and $J_{bc} = 7.2$ Hz. Since the two coupling constants are much different, we do not observe a quintet, expected for 'b' protons in pure ethyl alcohol. Actually, a group of eight lines, $(3+1)(1+1) = 8$, are observed in the multiplet due to 'b' protons. Quintet would have been observed if J_{ab} were equal to J_{bc} .

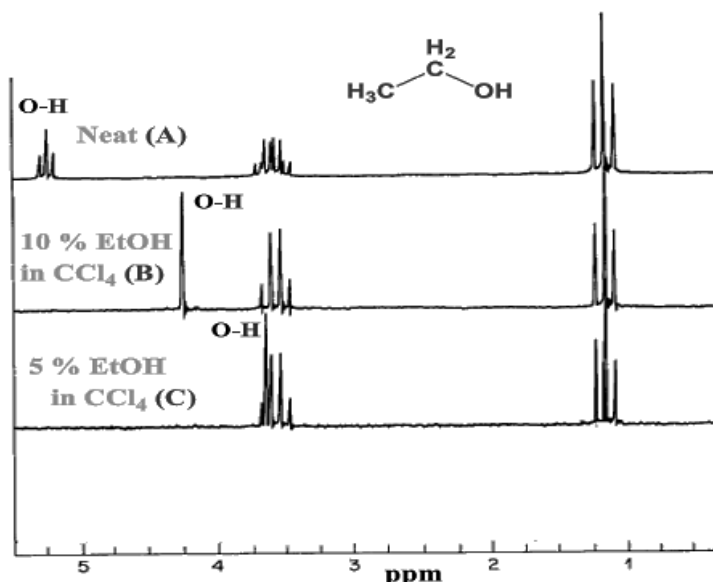


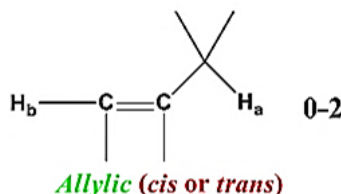
Figure: The NMR spectrum of Ethanol at various concentrations

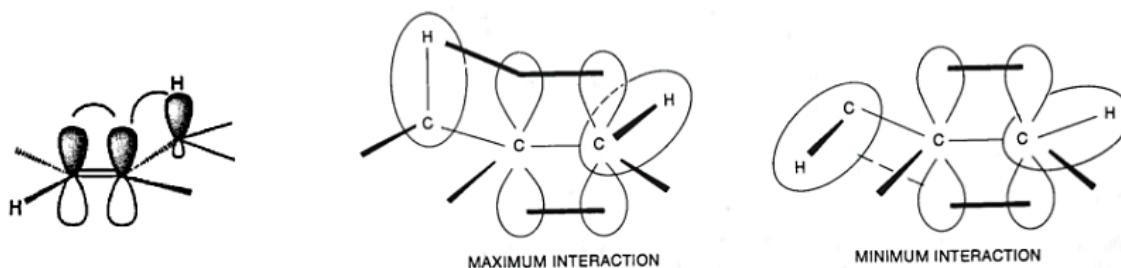
TYPES OF COUPLINGS AND THEIR J VALUES

1. **Geminal Coupling**, involving protons on $-\text{CH}_2-$ groups, is strong, 2J being typically 12-18 Hz, but it will only be observed where the geminal protons have different chemical shift positions.
2. **Vicinal coupling** (three bonds separating the protons) varies from $^3J = 0$ to $^3J = 18$ Hz in rigid systems, but in freely rotating carbon chains (alkyl groups) it is usually around 8 Hz.
3. **Trans coupling** in alkenes (J , 13-18 Hz) is stronger than *cis* coupling (J , 10-12 Hz). Typical values, 16 Hz and 8 Hz, respectively.

4. Allylic coupling,

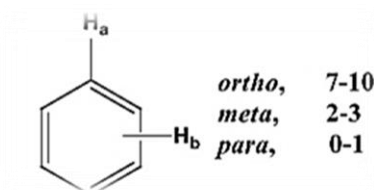
Four-bond ($^4J_{\text{HH}}$) couplings between hydrogens on carbon atoms α to the double bond and hydrogens at the opposite end of the double bond called allylic couplings. Apparently, orbital overlap between the π orbital of the double bond and the hybrid sp^3 -C-H orbital enable the nuclear interactions that result in the coupling





As in allyl chloride, is the four-bond coupling to be met in nonaromatic molecules and is very small (4J , 0-2 Hz). The analogous coupling in aromatic systems (for example, between the methyl protons and the ortho protons in the ring) is also very small.

5. **Aromatic coupling** depends on whether the coupling protons are ortho, meta or para to each other, and in simple cases the coupling constant is definitive in deciding the orientation; thus, $^3J_{\text{ortho}}$, 7-10 Hz; $^4J_{\text{meta}}$, 2-3 Hz; $^5J_{\text{para}}$, 0-1 Hz.



FACTORS AFFECTING THE VALUES OF COUPLING CONSTANT (J)

Coupling constant is a measure of the effectiveness of spin-spin coupling. The value of coupling constant depends partly on the number of covalent bonds through which protons may interact and partly upon the structural and stereochemical relationships between the coupled protons.

Spin-spin coupling constants are not easy to predict theoretically, and depend on a number of factors:

- (i) the hybridization of the atoms involved in the coupling;
- (ii) the bond angles;
- (iii) the dihedral angles;
- (iv) the C - C bond length;
- (v) substituent effects (electronegativity, neighbouring π -bond and lone pair effects)

In the following lines these factors are discussed.

1. Geminal Coupling($^2J_{HH}$)

Protons attached to the same carbon are called as geminal protons and these are separated by two bonds. If geminal protons are chemically non-equivalent then only they show coupling. The geminal coupling ($^2J_{HH}$) constant values are usually negative or near to zero.

a). Hybridisation

Geminal coupling ($^2J_{HH}$) in general depends on the hybridization of carbon atom and the bond angle. When s-character is increased, the absolute value of geminal coupling constant is decreased (move to positive value). Thus, the order of geminal coupling constant is:

$$^2J_{sp^3} > ^2J_{sp^2} > ^2J_{sp}$$

The value of coupling constant increases with increase in s-character of the hybrid orbital of C-atom bearing geminal hydrogens.

Compounds	Hybridisation	J_{gem}
CH ₄	SP ³ (25%-s character)	-12.4 Hz
$\begin{array}{c} H > C = C < H \\ H > C = C < H \end{array}$	SP ² (33.33% -s character)	+2.5 Hz

b). Electronegativity The increase in electronegativity of the atom or group, which withdraws sigma electrons, increases the value of coupling constant. For example, the value of coupling constant for methyl chloride is – 10.8 cps whereas it is – 9.4 cps for methyl fluoride.

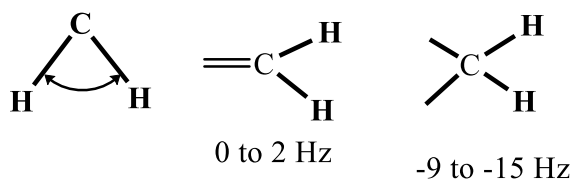
Compounds	J_{gem}
$\begin{array}{c} +\delta \quad \quad -\delta \\ CH_3 - Cl \end{array}$	-10.8 Hz
$\begin{array}{c} +\delta \quad \quad -\delta \\ CH_3 - F \end{array}$	-9.4 Hz

The value of J decreases if an electronegative substituent withdraws electrons from the π bonds. For example, J_{gem} is +2.3 cps for ethene whereas it is – 3.3 cps for vinyl fluoride.

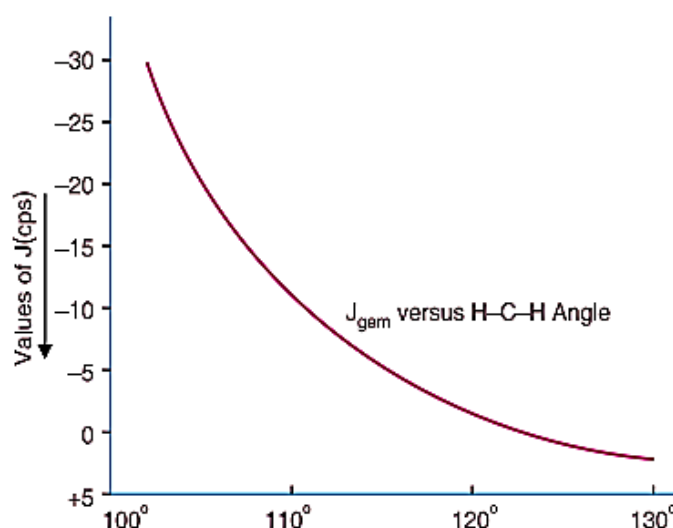
Compounds	J_{gem}
$\begin{array}{c} H > C = C < H \\ H > C = C < H \end{array}$ (ethene)	+2.5 Hz
$\begin{array}{c} H > C = C < H \\ F > C = C < H \end{array}$ (Fluoro ethene)	-3.3 Hz

c). Bond Angle

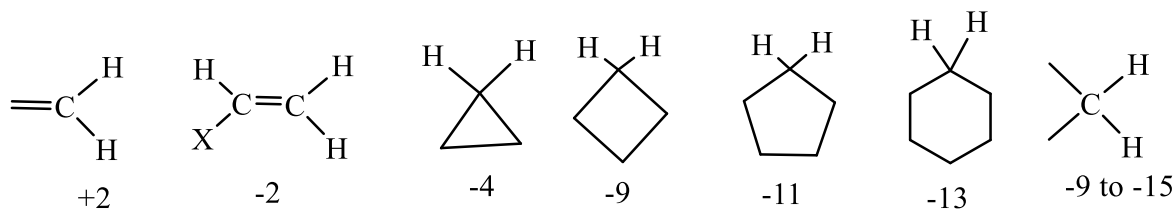
The bond angle (HCH) also gives rise to change in $^2J_{HH}$ value. In general, the value of 2J coupling constant increases as the bond angle between the coupling protons decreases. As the angle decreases, the two atoms come closer and hence the interaction between the electron spin become greater.

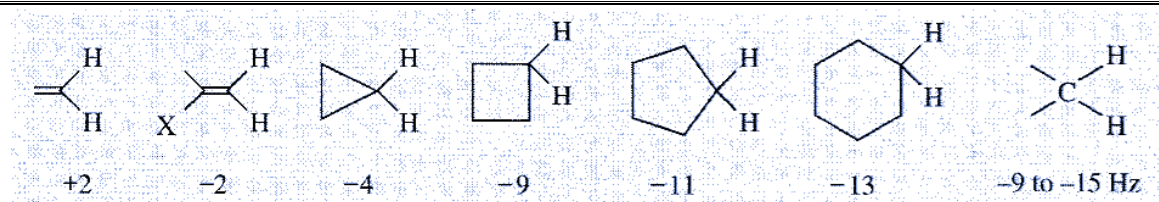


A plot showing the relationship between the values of J versus the bond angle is shown as under:



Therefore, geminal coupling constant can also help in determining the ring size. As the ring size increases, the absolute value of geminal coupling constant also increases. As the CCC bond angle in the ring increases (from 3 membered to 6 membered), the complementary HCH bond angle decreases (from 3 membered to 6 membered) and consequently the geminal coupling constant increases.





2. Vicinal Coupling ($^3J_{\text{HH}}$)

Protons attached to the adjacent carbon atoms are called as vicinal protons and thus are separated by three bonds. Vicinal coupling constant always has the positive value and depends on the dihedral angle (angle of rotation, HCCH), the carbon-carbon bond length, and the effects of electronegative atoms.

a). Electronegativity

The electronegativity of attached substituents alters the value of vicinal (vic) coupling, as it does that of geminal coupling. The more electronegative the substituent the smaller the value of J_{vic} , so that in unhindered alkanes the value is = 8 Hz, and in halogeno ethanes it is lowered to 6-7 Hz.

b). Bond Angle

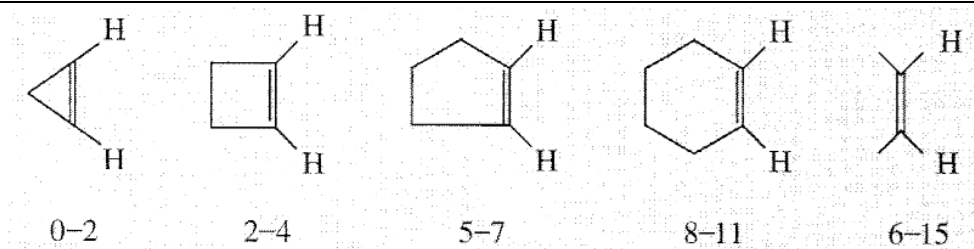
For vicinal protons, the value of coupling constant varies with the dihedral angle. Martin Karplus derived an equation which relates the vicinal coupling constant with the dihedral angle by a mathematical equation as given below. The Karplus equation describes how the coupling constant between two protons is affected by the dihedral angle between them. The Karplus equation describes how the coupling constant between two protons is affected by the dihedral angle between them. The equation follows the general format of $J = A\cos^2\phi + B\cos\phi + C$, with the exact values of A, B and C dependent on several different factors.

The dependence of J_{vic} with dihedral angle (ϕ) can be obtained qualitatively from Karplus equations:

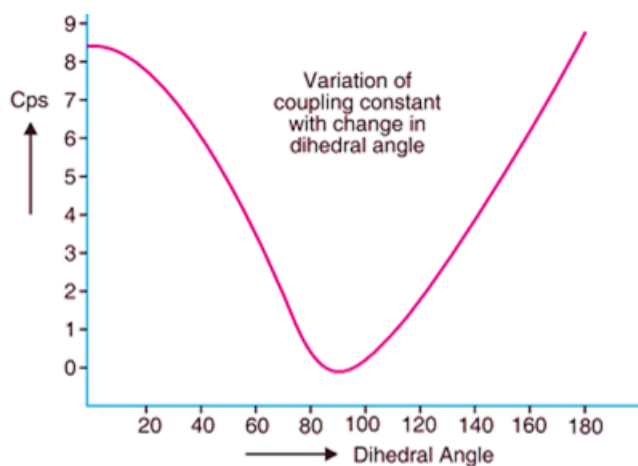
$$\phi \text{ between } 0^\circ \text{ and } 90^\circ ; \quad J_{\text{vic}} = 8.5\cos^2\phi - 0.28$$

$$\phi \text{ between } 90^\circ \text{ and } 180^\circ ; \quad J_{\text{vic}} = 9.5\cos^2\phi - 0.28$$

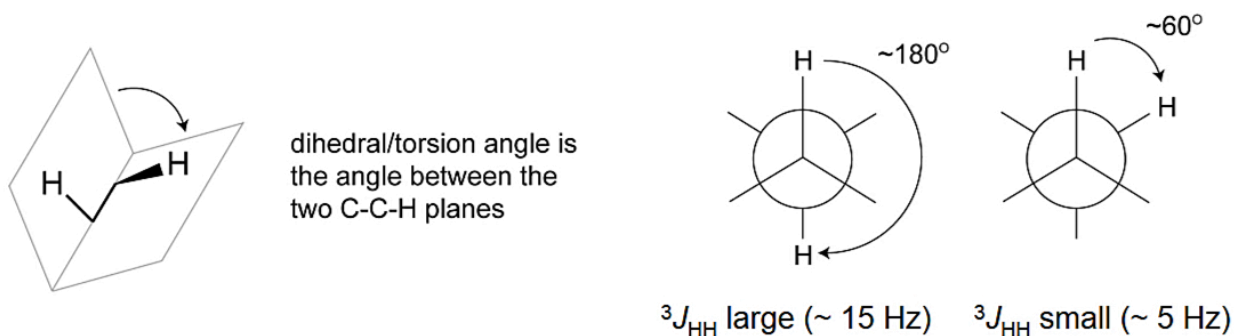
The valence angle also changes the value of $^3J_{\text{HH}}$ coupling constant. Valence angle is related with ring size. In general, as the valence angle increases, the coupling constant also increases. The coupling constant increases with the decrease of bond length. The vicinal coupling constant decreases with increasing the electro-negativity of an atom attached to the system.



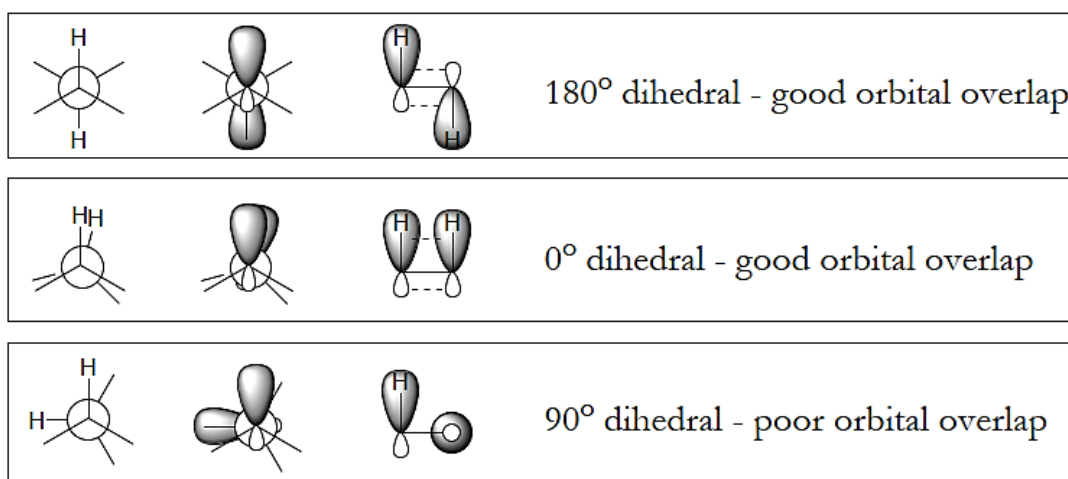
A plot for the values of J versus the dihedral angle for vicinal protons is shown as under.



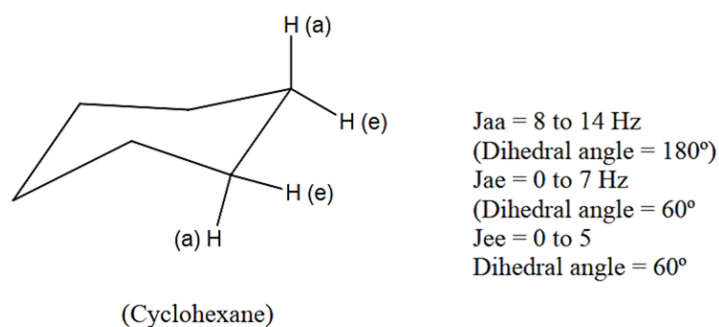
When the dihedral angle is 0° or 180° , the coupling constant is found to be maximum. It is slightly negative when dihedral angle is 90° .



The side-side overlapping of the two C-H bond orbitals is maximum at 0° , where the C-H orbitals are parallel and is minimum at 90° , where they are perpendicular. At 180° , overlapping with the back lobes of the orbital occurs.



In case of cyclopentanes, where the dihedral angle for cis protons is 0°, J_{vic} is found to be 8 Hz whereas for trans protons where the dihedral angle is 90°, J_{vic} is observed to be nearly equal to zero.



Thus, from the value of the coupling constant it is possible to determine the configuration and conformation of a system.

Function	J_{ab}/Hz
	6-8
	10-12
	13-18
	4-10

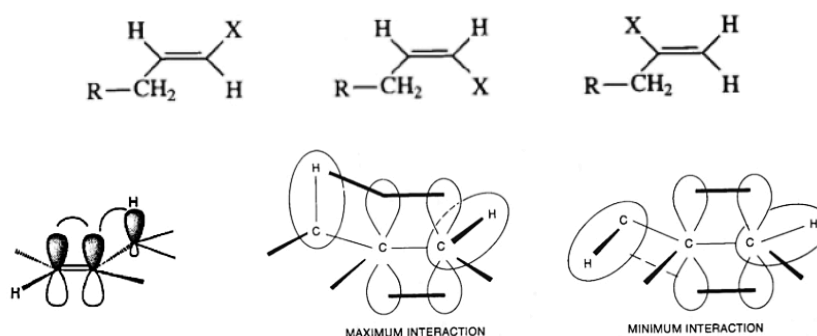
A. Long Range Coupling

The magnitude of J decreases sharply with the distance. It is very less through more than three covalent bonds but in some cases the coupling is observable even in case of more than three bonds.

Such coupling is most common in systems such as olefins, acetylenes, aromatics, hetero-aromatics, and strained ring systems (small or bridged rings). This kind of coupling which occurs beyond three bonds, are called as long-range coupling.

a). Allylic coupling

In addition to the germinal and vicinal coupling, alkenes also show small coupling between protons substituted on carbons α to the double bond and those on the opposite end of the double bond. This four-bond coupling is called allylic coupling. Allylic coupling constant are very less about 0-3 Hz. The π -electrons of the double bond apparently help in coupling the concerned nuclei and the magnitude of this depends upon the extent of overlap of the carbon-hydrogen σ - bond with the π bond. This type of coupling also occurs in acetylenes.

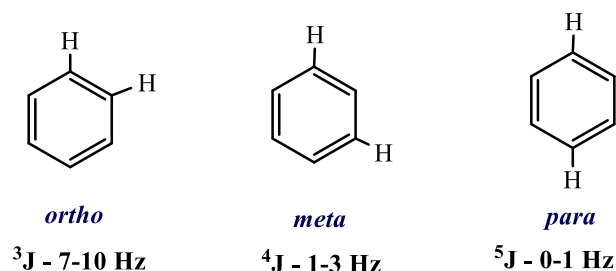


b). Homoallylic Systems

In some alkenes, coupling can occur between the C-H σ -bonds on either side of the double bond. Homoallylic coupling is weaker than the allylic coupling because it occurs over a greater distance (five bonds). This type of coupling (5J) is generally very small or even non-existent in most molecules but sometimes it appears in NMR spectrum.

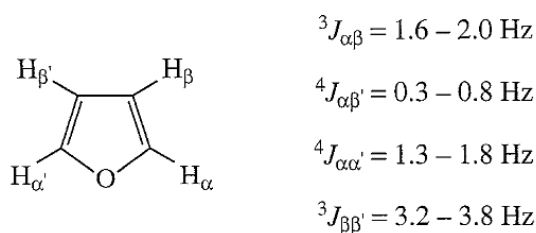
c). Aromatic coupling

In aromatic systems also, coupling occurs beyond the adjacent carbon atoms (more than three bonds). The ortho, meta and para protons can all interact but the ortho coupling is maximum, followed by meta and then para. In fact, the para coupling is rarely observed.



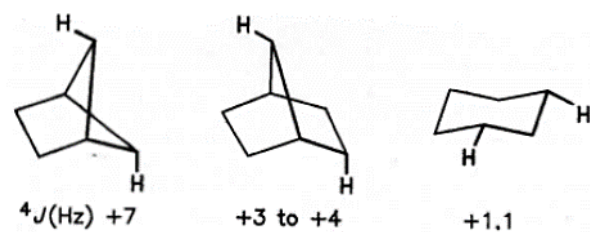
d). Hetero-aromatic coupling

The hetero-aromatic systems also show long range coupling analogues to the aromatic systems. For example, in furan coupling occurs between all of the ring protons. The values of the coupling constant are given as follows:



e). Rigid systems

The long-range coupling occurs in compounds without π -systems also but it is less common. This unusual long-range coupling in saturated systems occurs through an arrangement of atoms in the form of 'W' in which hydrogens occupy the end positions. The magnitude of coupling constant is usually small except in highly strained ring systems. Apparently, these rigid strained arrangements somehow have a favorable geometry for the overlaps involved. Most of the W type couplings give rise to the peak broadening rather than apparent splitting.



MULTIPLETS IN COMPLEX COUPLING

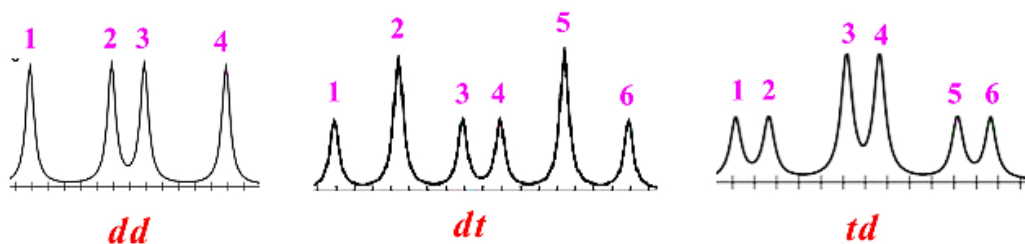
In naming multiplets in a complex coupling the biggest coupling constant determines the “first name” of the multiplet and the smallest coupling constant determine the “last name”. For example, a doublet of triplets (dt) is six-line pattern with one large coupling and two equal small couplings.

Few patterns of multiplets are shown are given in the following table.

Abbreviation	Multiplet	No. of lines
dd	doublet of doublets	4 lines
dt	doublet of triplets	6 lines
td	triplet of doublets	6 lines

dq	doublet of quartets	8 lines
qd	quartet of doublet	8 lines

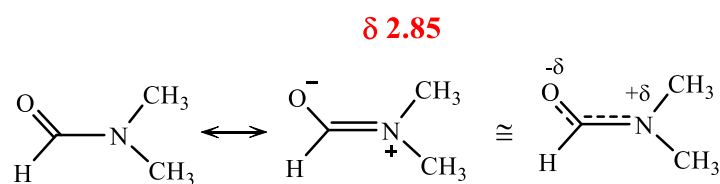
The following figure shows the pattern in various multiplets:



RESTRICTED ROTATION

The isomers which arise due to rotation about single bond are called conformational isomers or conformers. The presence of conformations in a solution cannot be detected by the nmr spectroscopy. But at temperatures much below room temperature, the rate of interconversion of rotational isomers is usually diminished and nmr absorption resulting from each may be observed.

At room temperature, a neat sample of dimethylformamide (DMF) shows two CH_3 peaks because the rate of rotation around the hindered “partial double bond” is slow. At $\sim 123^\circ$, the rate of exchange of the two CH_3 groups is rapid enough so that the two peaks merge.



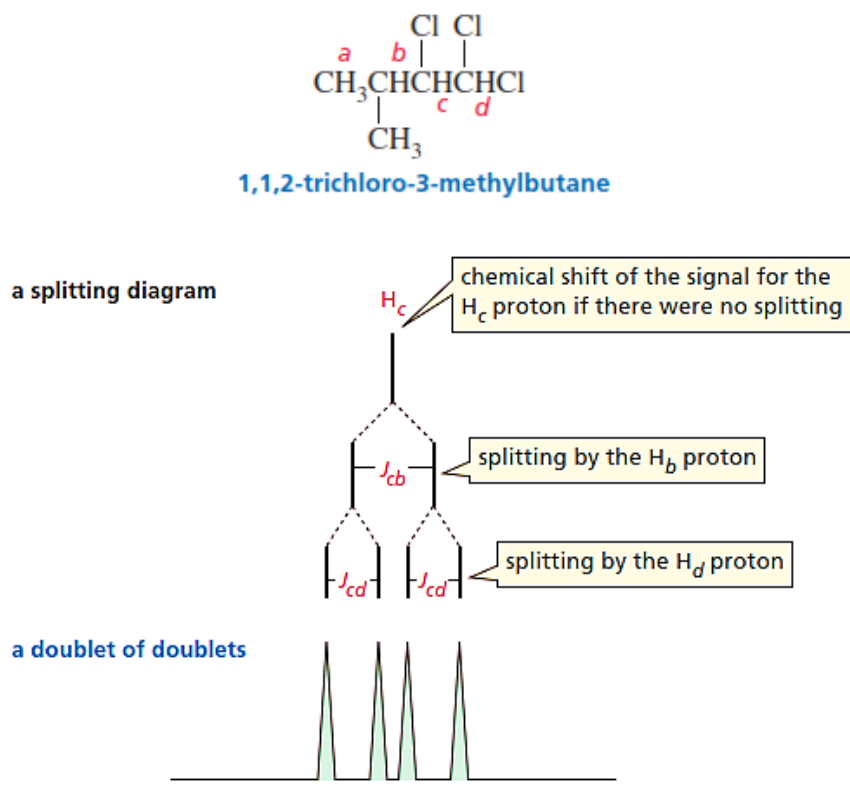
Restricted Rotation in Dimethyl Formamide

DMF is represented by two resonance forms as shown in Figure, and the result of conjugation between the carbonyl group and the nitrogen nonbonding pair is to increase the double-bond character of the C-N bond sufficiently to restrict the rotation at room temperature; one methyl group is cis to oxygen, the other is trans.

SPLITTING DIAGRAMS

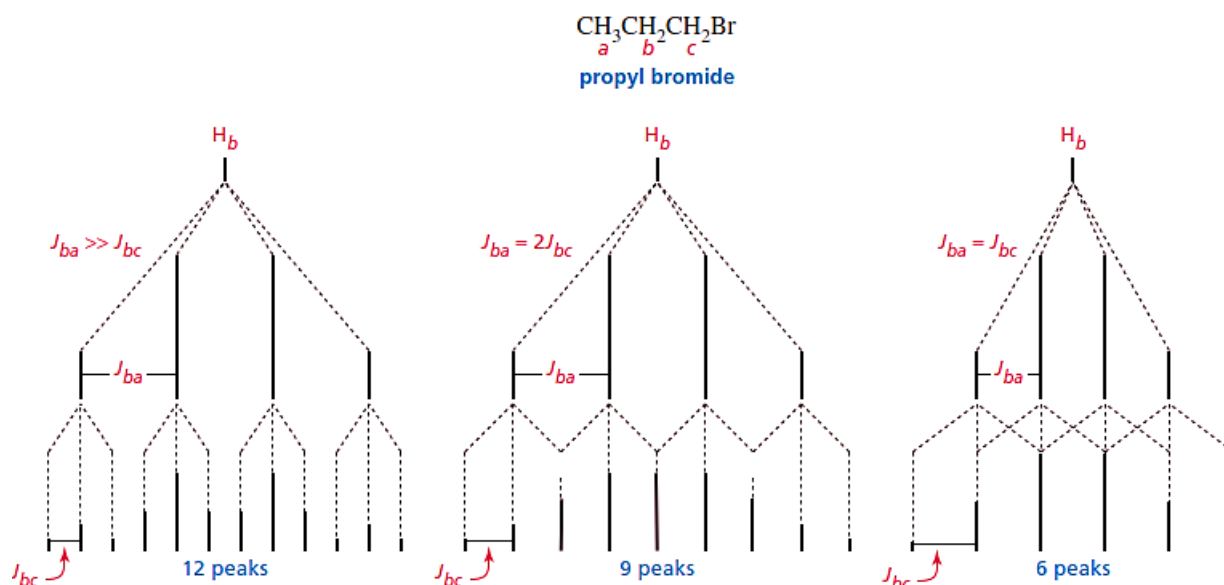
The splitting pattern obtained when a signal is split by more than one set of protons can best be understood by using a splitting diagram. In a splitting diagram (or splitting tree), the NMR peaks are shown as vertical lines and the effect of each of the splitting is shown one at a time.

For example, a splitting diagram is shown in figure for splitting of the signal for the proton of 1,1,2-trichloro-3-methylbutane into a doublet of doublets by the H_b and H_d protons.



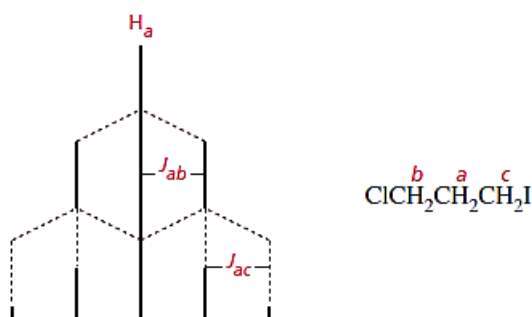
The signal for the protons of propyl bromide is split into a quartet by the protons, and each of the resulting four peaks is split into a triplet by the protons.

How many of the 12 peaks are actually seen depends on the relative magnitudes of the two coupling constants, J_{ba} and J_{bc} . For example, the figure shows that there are 12 peaks when J_{ba} is much greater than J_{bc} , 9 peaks when $J_{ba} = 2 J_{bc}$ and only 6 peaks when $J_{ba} = J_{bc}$. The number of peaks actually observed depends on how many overlaps with one another.



When peaks overlap, their intensities add together. We would expect the signal for the protons of 1-chloro-3-iodopropane to be triplet of triplets (split into nine peaks) because the signal would be split into a triplet by the protons and each of the resulting peaks into a triplet by the protons. The signal, however, is a quintet.

Finding that the signal for the protons of 1-chloro-3-iodopropane is a quintet indicates that J_{ab} and J_{ac} have about the same value. The splitting diagram shows that a quintet results if $J_{ab} = J_{bc}$.



We can conclude that when two different sets of protons split a signal, the multiplicity of the signal should be determined by using the $n+1$ rule separately for each set of hydrogens when the coupling constants for the two sets are different. When the coupling constants are similar, however, the multiplicity of a signal can be determined by treating both sets of adjacent hydrogens as if they were equivalent. In other words, the $n+1$ rule can be applied to both sets of protons simultaneously.

FIRST ORDER AND HIGHER ORDER COUPLING

First-Order Coupling

The spin-spin coupling in which the energy differences between the spin states are equal results in first-order splitting.

In general, first-order spectra will only arise if the separation between multiplets (the chemical shift difference between signals, expressed in Hz) is much larger than the coupling constant, J ; if $\Delta\nu \geq 8J$, then fairly unperturbed spectra will arise.

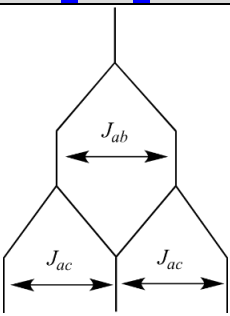
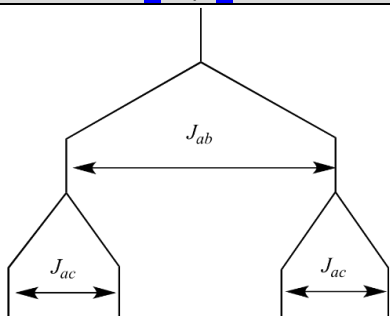
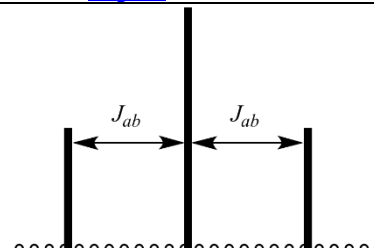
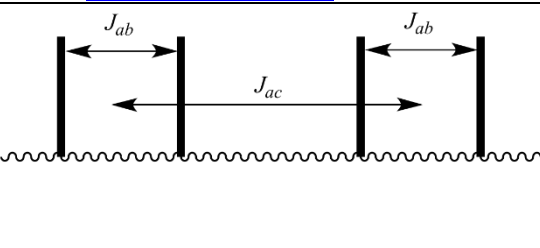
$$\frac{\Delta\nu}{J} \geq 8$$

Non-First Order Coupling

The spin-spin coupling in which the energy differences between the nuclear spin states are not equal.

In case of non-first order coupling, the $n+1$ rule is not obeyed; complex splitting patterns (non-first-order splitting) often result. High-order splitting pattern takes place when chemical shift difference in Hertz is much less or the same of the order of magnitude as the J coupling.

$$\frac{\Delta\nu}{J} \leq 10$$

First order coupling $J_{ab} = J_{ac}$	Non-first-order coupling $J_{ab} \neq J_{ac}$
	
<u>First-order splitting</u> results The <u>$n+1$ rule</u> is obeyed A <u>triplet</u> is observed	<u>Non-first-order splitting</u> results The <u>$n+1$ rule</u> is not obeyed A <u>doublet of doublets</u> is observed
	

The splitting pattern depends on the magnetic field. The second-order splitting at the lower field can be resolved into first-order splitting pattern at the high field.

In a simple first-order multiplet, the number of peaks is determined by the number of coupled neighbouring protons with the same (or very slightly different) coupling constants. The multiplicity then is ' $n + 1$ ' where n being the number of neighbouring equally coupled protons, and Pascal's triangle hold in this case.

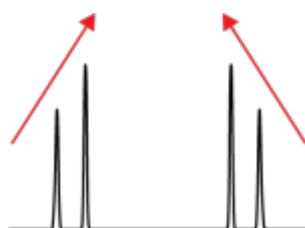
The relative intensities of the peaks of a first-order multiplet also depend on n . The doublet peaks ($n = 1$) appear in the ratio 1:1, triplets ($n = 2$) are 1:2:1, and quartets ($n = 3$) are 1:3:3:1, and so forth.

Requirements for a simple, First-Order Multiplet

1. The ratio $\Delta\nu/J$ must be larger than about 8; $\Delta\nu$ is the distance between the midpoints of the coupled multiplets. J is the coupling constant.
2. The number of peaks in the multiplet is ' $n + 1$ ', where n is the number of neighbouring protons with the same coupling constant.
3. The distance in Hz between the individual peaks of a simple, first-order multiplet represents the coupling constant.
4. The simple first-order multiplet is centrosymmetric, with the most intense peak(s) central.

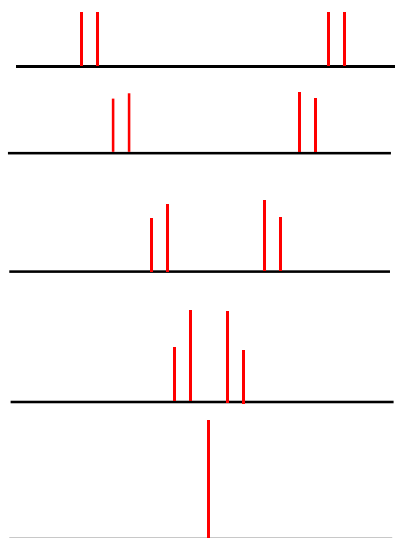
ROOFING EFFECT

When there is a strong coupling effect (interaction of the magnetic fields) between hydrogen atoms, the peaks can become distorted as their chemical shifts get closer together (second order coupling). The parts of the peak closest to each other become larger while the outer parts become smaller. The signals appear to lean towards each other, and this is known as '**roofing**'. This effect can be helpful in data interpretation to help identify neighboring proton peaks in a spectrum.



If two signals are coupled to each other and have very similar (but not identical) chemical shifts, another effect will appear: second-order coupling. This means that the peaks appear to "lean" toward

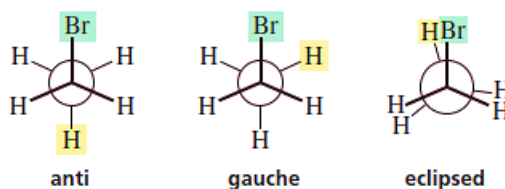
each other – the peaks on the outside of the coupled pair are shorter, and the peaks on the inside are taller.



As the chemical shifts of H_a and H_b become more and more similar, the coupling between them becomes more second-order and the peaks lean more.

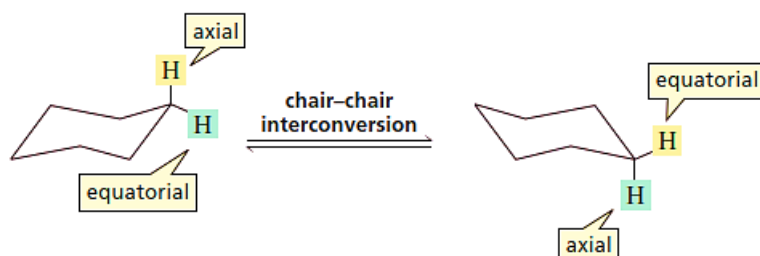
TIME DEPENDENCE OF NMR SPECTROSCOPY

The three methyl hydrogens of ethyl bromide give rise to one signal in the NMR spectrum because they are chemically equivalent due to rotation about the bond. At any one instant, however, the three hydrogens can be in quite different environments: One can be anti to the bromine, one can be gauche to the bromine, one can be eclipsed with the bromine.

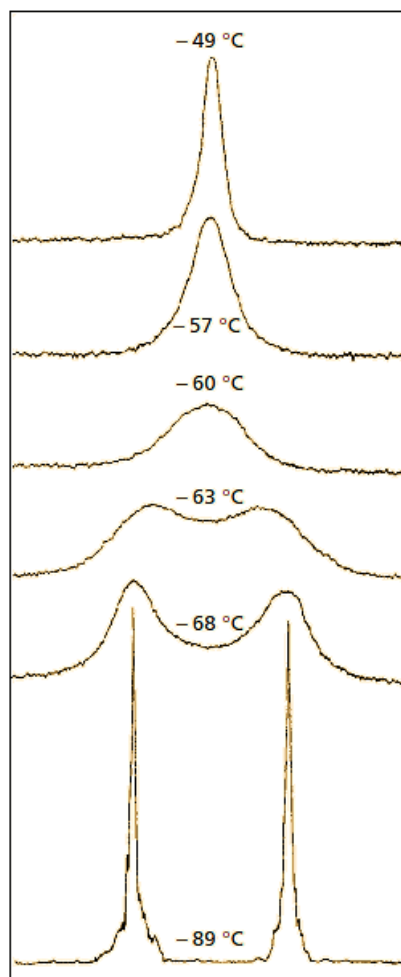


Because each of the three methyl hydrogens has the same average environment, we see one signal for the methyl group in the NMR spectrum.

Similarly, the NMR spectrum of cyclohexane shows only one signal, even though cyclohexane has both axial and equatorial protons. There is only one signal because the interconversion of the chair conformers of cyclohexane occurs too rapidly at room temperature to be detected by the NMR spectrometer. Because axial protons in one chair conformer are equatorial protons in the other chair conformer, all the protons in cyclohexane have the same average environment on the NMR time scale, so the NMR spectrum shows one signal.



The rate of chair–chair interconversion is temperature-dependent—the lower the temperature, the slower is the rate of interconversion. has 11 deuteriums, which means that it has only one hydrogen. NMR spectra of taken at various temperatures are shown in Figure.



Cyclohexane with only one hydrogen was used for this experiment in order to prevent splitting of the signal, which would have complicated the spectrum. Deuterium signals are not detectable in ^1H NMR, and splitting by a deuterium on the same or on an adjacent carbon is not normally detectable at the operating frequency of an ^1H NMR spectrometer.

At room temperature, the ^1H NMR spectrum of shows one sharp signal, which is an average for the axial proton of one chair and the equatorial proton of the other chair. As the temperature decreases,

the signal becomes broader and eventually separates into two signals, which are equidistant from the original signal. At -89°C two sharp singlets are observed because at that temperature, the rate of chair–chair interconversion has decreased sufficiently to allow the two kinds of protons (axial and equatorial) to be individually detected on the NMR time scale.

SURVEY OF TYPICAL ^1H NMR ABSORPTIONS BY TYPE OF COMPOUND

A Alkanes

Alkanes can have three different types of hydrogens (methyl, methylene, and methine), each of which appears in its own region of the NMR spectrum.

CHEMICAL SHIFTS		
R-CH_3	0.7–1.3 ppm	Methyl groups are often recognizable as a tall singlet, doublet, or triplet even when overlapping other CH absorptions.
$\text{R-CH}_2\text{-R}$	1.2–1.4 ppm	In long chains, all of the methylene (CH_2) absorptions may be overlapped in an unresolvable group.
R_3CH	1.4–1.7 ppm	Note that methine hydrogens (CH) have a larger chemical shift than those in methylene or methyl groups.
COUPLING BEHAVIOUR		
$-\text{CH}-\text{CH}-$	$^3J \approx 7\text{--}8\text{ Hz}$	In hydrocarbon chains, adjacent hydrogens will generally couple, with the spin–spin splitting following the $n+1$ Rule.

B Alkenes

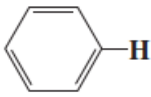
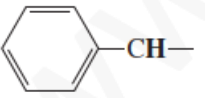
Alkenes have two types of hydrogens: vinyl (those attached directly to the double bond) and allylic hydrogens (those attached to a carbon, the carbon atom attached to the double bond). Each type has a characteristic chemical shift region.

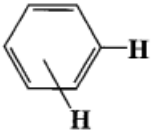
CHEMICAL SHIFTS

C=C-H	4.5–6.5 ppm	Hydrogens attached to a double bond (vinyl hydrogens) are deshielded by the anisotropy of the adjacent double bond.
C=C-C-C-H	1.6–2.6 ppm	Hydrogens attached to a carbon adjacent to a double bond (allylic hydrogens) are also deshielded by the anisotropy of the double bond, but because the double bond is more distant, the effect is smaller.
COUPLING BEHAVIOUR		
H-C=C-H $\begin{array}{c} \text{---C=C---H} \\ \\ \text{H} \end{array}$	$^3J_{\text{trans}} \approx 11\text{--}18 \text{ Hz}$ $^3J_{\text{cis}} \approx 6\text{--}15 \text{ Hz}$ $^2J \approx 0\text{--}3 \text{ Hz}$	The splitting patterns of vinyl protons may be complicated by the fact that they may not be equivalent even when located on the same carbon of the double bond
$\begin{array}{c} \text{---C=C---C---H} \\ \\ \text{H} \end{array}$	$^4J \approx 0\text{--}4 \text{ Hz}$	When allylic hydrogens are present in an alkene, they may show long-range allylic coupling to hydrogens on the far double bond carbon as well as the usual splitting due to the hydrogen on the adjacent (nearest) carbon.

C. Aromatic Compounds

Aromatic compounds have two characteristic types of hydrogens: aromatic ring hydrogens (benzene ring hydrogens) and benzylic hydrogens (those attached to an adjacent carbon atom).

CHEMICAL SHIFTS		
	6.5–8.0 ppm	Hydrogens attached to an aromatic (benzenoid) ring have a large chemical shift, usually near 7.0 ppm. They are deshielded by the large anisotropic field generated by the electrons in the ring's π system.
	2.3– 2.7 ppm	Benzylic hydrogens are also deshielded by the anisotropic field of the ring, but they are more distant from the ring, and the effect is smaller.
COUPLING BEHAVIOUR		

	${}^3J_{ortho} \approx 7-10 \text{ Hz}$ ${}^4J_{meta} = 2-3 \text{ Hz}$ ${}^5J_{para} = 0-1 \text{ Hz}$	It is often possible to determine the positions of the substituents on the ring from these splitting patterns and the magnitudes of the coupling constants.
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D. Alkynes

Terminal alkynes (those with a triple bond at the end of a chain) will show an acetylenic hydrogen, as well as the α hydrogens found on carbon atoms next to the triple bond. The acetylenic hydrogen will be absent if the triple bond is in the middle of a chain.

CHEMICAL SHIFTS		
$\text{C}\equiv\text{C}-\text{H}$	1.7–2.7 ppm	The terminal or acetylenic hydrogen has a chemical shift near 1.9 ppm due to anisotropic shielding by the adjacent π bonds.
$\text{C}\equiv\text{C}-\text{CH}-$	1.6–2.6 ppm	Protons on a carbon next to the triple bond are also affected by the π system.
COUPLING BEHAVIOUR		
$\text{H}-\text{C}\equiv\text{C}-\text{C}-\text{H}$	${}^4J = 2-3 \text{ Hz}$	“Allylic coupling” is often observed in alkynes, but is relatively small.

E. Alkyl Halides

In alkyl halides, the α hydrogen (the one attached to the same carbon as the halogen) will be deshielded.

CHEMICAL SHIFTS		
$-\text{CH}-\text{I}$	2.0– 4.0 ppm	The chemical shift of a hydrogen atom attached to the same carbon as a halide atom will increase (move further downfield). This deshielding effect is due to the electronegativity of the attached halogen atom. The extent of the shift is increased as the electronegativity of the attached atom increases, with the largest shift found in compounds containing fluorine.
$-\text{CH}-\text{Br}$	2.7– 4.1 ppm	
$-\text{CH}-\text{Cl}$	3.1– 4.1 ppm	
$-\text{CH}-\text{F}$	4.2– 4.8 ppm	
COUPLING BEHAVIOUR		
$-\text{CH}-\text{F}$ $-\text{CH}-\text{CF}-$	${}^2J \approx 50 \text{ Hz}$ ${}^3J \approx 20 \text{ Hz}$	Compounds containing fluorine will show spin–spin splitting due to coupling between the fluorine and the hydrogens on either the

		same or the adjacent carbon atom. ^{19}F has a spin of $\frac{1}{2}$. The other halogens (I, Cl, Br) do not show any coupling.
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F. Alcohols

In alcohols, both the hydroxyl proton and the α hydrogens (those on the same carbon as the hydroxyl group) have characteristic chemical shifts.

CHEMICAL SHIFTS		
C–OH	0.5–5.0 ppm	The chemical shift of the -OH hydrogen is highly variable, its position depending on concentration, solvent, and temperature. The peak may be broadened at its base by the same set of factors.
CH–O–H	3.2–3.8 ppm	Protons on the α carbon are deshielded by the electronegative oxygen atom and are shifted downfield in the spectrum.
COUPLING BEHAVIOUR		
CH-OH	<i>No coupling (usually), or $^3J = 5\text{ Hz}$</i>	Because of the rapid chemical exchange of the -OH proton in many solutions, coupling is not usually observed between the -OH proton and those hydrogens attached to the α carbon.

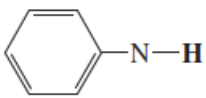
G. Ethers

In ethers, the α hydrogens (those attached to the α carbon, which is the carbon atom attached to the oxygen) are highly deshielded.

CHEMICAL SHIFTS		
R–O–CH–	3.2–3.8 ppm	The hydrogens on the carbons attached to the oxygen are deshielded due to the electronegativity of the oxygen.
COUPLING BEHAVIOUR		
R-O-CH ₃		The methoxy peak is unsplit and stands out as a tall, sharp singlet.

H. Amines

Two characteristic types of hydrogens are found in amines: those attached to nitrogen (the hydrogens of the amino group) and those attached to the α carbon (the same carbon to which the amino group is attached).

CHEMICAL SHIFTS		
$\text{R}-\text{N}-\text{H}$	0.5–4.0 ppm	Hydrogens attached to a nitrogen have a variable chemical shift depending on the temperature, acidity, amount of hydrogen bonding, and solvent.
$-\text{CH}-\text{N}-$	2.2–2.9 ppm	The α hydrogen is slightly deshielded due to the electronegativity of the attached nitrogen.
	3.0–5.0 ppm	This hydrogen is deshielded due to the anisotropy of the ring and the resonance that removes electron density from nitrogen and changes its hybridization.
COUPLING BEHAVIOUR		
$-\text{N}-\text{H}$	$^1J \approx 50 \text{ Hz}$	Direct coupling between a nitrogen and an attached hydrogen is not usually observed, but is quite large when it occurs. More commonly this coupling is obscured by quadrupole broadening by nitrogen or by proton exchange.
$-\text{N}-\text{CH}$ $\text{C}-\text{N}-\text{H}$ $ $ H	$^2J \approx 0 \text{ Hz}$ $^3J \approx 0 \text{ Hz}$	This coupling is usually not observed. Due to chemical exchange, this coupling is usually not observed.

I. Nitriles

In nitriles, only the α hydrogens (those attached to the same carbon as the cyano group) have a characteristic chemical shift.

CHEMICAL SHIFTS		
$-\text{CH}-\text{C}\equiv\text{N}$	2.1–3.0 ppm	The α hydrogens are slightly deshielded by the cyano group.

J. Aldehydes

Two types of hydrogens are found in aldehydes: the aldehyde hydrogen and the α hydrogens (those hydrogens attached to the same carbon as the aldehyde group).

CHEMICAL SHIFTS		
$\text{R}-\text{CHO}$	9.0–10.0 ppm	The aldehyde hydrogen is shifted far downfield due to the anisotropy of the carbonyl group ($\text{C}=\text{O}$).

$\text{R}-\text{CH}-\text{CH}=\text{O}$	2.1–2.4 ppm	Hydrogens on the carbon adjacent to the C=O group are also deshielded due to the carbonyl group, but they are more distant, and the effect is smaller.
COUPLING BEHAVIOUR		
$-\text{CH}-\text{CHO}$	$^3J \approx 1-3 \text{ Hz}$	Coupling frequently occurs between the aldehyde hydrogen and hydrogens on the adjacent carbon, but J is small.

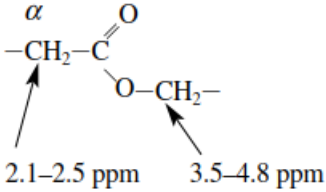
K. Ketones

Ketones have only one distinct type of hydrogen atom—those attached to the α carbon.

CHEMICAL SHIFTS		
$\begin{array}{c} \text{R}-\text{CH}-\text{C}=\text{O} \\ \\ \text{R} \end{array}$	2.1–2.4 ppm	The α hydrogens in ketones are deshielded by the anisotropy of the adjacent C=O group.

L. Esters

Two distinct types of hydrogen are found in esters: those on the carbon atom attached to the oxygen atom in the alcohol part of the ester and those on the α carbon in the acid part of the ester (that is, those attached to the carbon next to the C=O group).

CHEMICAL SHIFTS		
		The α hydrogens in esters are deshielded by the anisotropy of the adjacent (C=O) group. Hydrogens on the carbon attached to the single-bonded oxygen are deshielded due to the electronegativity of oxygen.

M. Carboxylic Acids

Carboxylic acids have the acid proton (the one attached to the -COOH group) and the α hydrogens (those attached to the same carbon as the carboxyl group).

CHEMICAL SHIFTS		
$\text{R}-\text{COOH}$	11.0–12.0 ppm	This hydrogen is deshielded by the attached oxygen, and it is highly acidic. This (usually broad) signal is a very characteristic peak for carboxylic acids.

$-\text{CH}-\text{COOH}$	2.1–2.5 ppm	Hydrogens adjacent to the carbonyl group are slightly deshielded.
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N. Amides

CHEMICAL SHIFTS		
$\text{R}(\text{CO})-\text{N}-\text{H}$	5.0–9.0 ppm	Hydrogens attached to an amide nitrogen are variable in chemical shift, the value being dependent on the temperature, concentration, and solvent.
$-\text{CH}-\text{CONH}-$	2.1–2.5 ppm	The α hydrogens in amides absorb in the same range as other acyl (next to $\text{C}=\text{O}$) hydrogens. They are slightly deshielded by the carbonyl group.
$\text{R}(\text{CO})-\text{N}-\text{CH}$	2.2–2.9 ppm	Hydrogens on the carbon next to the nitrogen of an amide are slightly deshielded by the electronegativity of the attached nitrogen.
COUPLING BEHAVIOUR		
$-\text{N}-\text{H}$	$^1J \approx 50 \text{ Hz}$	In cases in which this coupling is seen (rare) it is quite large, typically 50 Hz or more. In most cases either the quadrupole moment of the nitrogen atom or chemical exchange decouples this interaction.
$-\text{N}-\text{CH}-$ $-\text{N}-\text{CH}-$ $\quad $ $\quad \text{H}$	$^2J \approx 0 \text{ Hz}$ $^3J \approx 0 \text{ Hz}$	Usually not seen Usually not seen

O. Nitroalkane

In nitroalkanes, α hydrogens, those hydrogen atoms that are attached to the same carbon which the nitro group is attached, have a characteristically large chemical shift.

CHEMICAL SHIFTS		
$-\text{CH}-\text{NO}_2$	4.1–4.4 ppm	Deshielded by the nitro group.

SIMPLIFICATION OF SECOND ORDER SPECTRA

The complete analysis of an NMR spectrum becomes difficult when signals overlap and hence useful information is buried due to complexity of the spectrum. There are some important methods to simplify an NMR spectrum to get the maximum information. The most common methods are discussed below:

1. High Field Strength

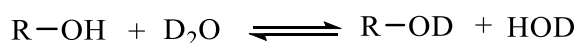
Chemical shift measured in Hertz is directly proportional to the applied magnetic field, whereas the coupling constant (Hz) remains the same. Thus, on increasing the field strength, the value of $\Delta\nu/J$ will increase which further gives the more separation between individual multiplets, overlapped at lower field strength. Thus, the NMR instruments with higher field strength give better resolution and therefore easily interpretable spectra.

2. Solvent Effects

Chemical shift depends on the solvent used to record NMR spectra. The interaction between the solvent molecules and the solute molecule may shift the resonance position of a proton. If the solvent is non-polar such as hydrocarbons, there will only be weak (van der Waals or London forces) interactions between solvent and solute molecules and hence the chemical shift of the protons in a molecule will remain same or very minute changes will occur. If the solvent is polar in nature such as acetone, chloroform, DMSO etc., strong dipole interactions between solvent and solute molecules would be there especially when the solute molecules also contains polar bonds. Because of these strong interactions, the observed chemical shift will shift to a greater extent with respect to that taken in non-polar solvent. If the solvent has a strong diamagnetic anisotropy (e.g., benzene, pyridine) the interaction between the compound and the anisotropic field of the solvent may give rise to a remarkable shifting in the resonance position.

3. Deuterium Exchange Reactions

Deuterium exchange is the most elementary and very useful technique for assigning the protons of OH, NH₂, SH etc groups which are exchangeable with deuterium.



Due to the above reactions, the peak due to the alcoholic hydrogen will disappear and a signal due to HOD proton will appear in the spectrum. For detecting the protons exchangeable with D₂O, either the spectrum is recorded in D₂O or a few drops of D₂O are added to the sample before recording the

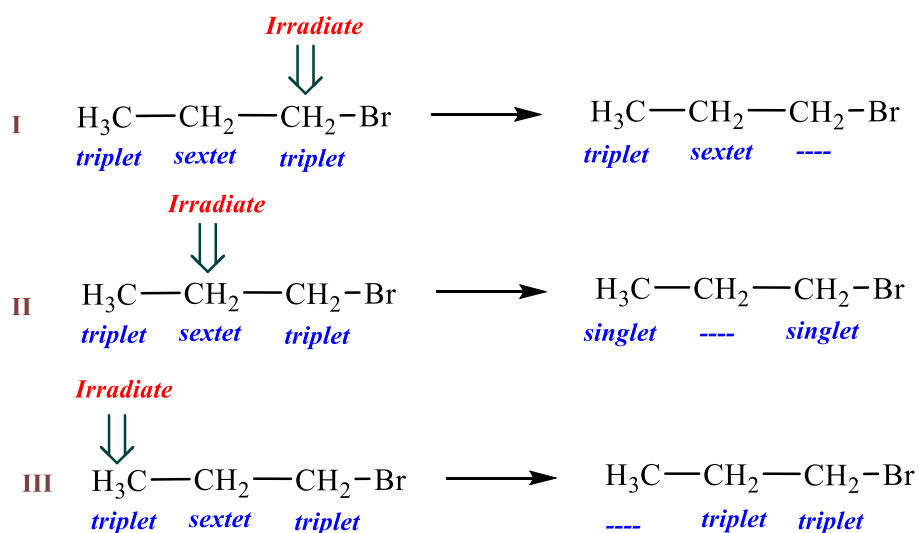
NMR spectrum. The deuterium is not detected in the ^1H NMR as it absorbs at different field strength and hence there will not be any coupling/splitting due to the proton which has been exchanged with deuterium.

4. Spin-Spin Decoupling (Double Irradiation or Double Resonance)

The spin-spin coupling between neighboring nuclei results in to the signals splitting causing different multiplets, however in certain cases, the splitting patterns are very complex. The spin-spin decoupling technique is used to simplify of the NMR spectra.

In spin decoupling, a second strong radiofrequency is applied while the spectrum is scanned in usual fashion. This radiofrequency can be set to irradiate a particular group of equivalent hydrogens in the molecule. Irradiation with high energy radio waves causes rapid transition among their nuclear spin states. Thus, the neighboring protons cannot see different spin states of the irradiated protons and consequently any spin-spin interactions are effectively removed and the NMR spectrum is simplified.

Consider an example of 1-bromopropane to understand the concept of spin-spin decoupling.



Double Irradiation Results for 1-Bromopropane

The above figure shows the effects of double irradiation of the α -CH₂, β -CH₂ and γ -CH₂ protons. If the protons at α -carbon are irradiated, the signal due to these protons will be collapsed and hence no coupling will be observed due to these protons. Therefore, the protons at β -carbon will be observed as quartet due to coupling with methyl protons only, and the methyl protons will appear as a triplet due to coupling with adjacent methylene protons. The irradiation of the β -CH₂ and γ -CH₂ protons and the changes in the signals due to other protons can be explained in a similar way.

5. Nuclear Overhauser Effect (NOE)

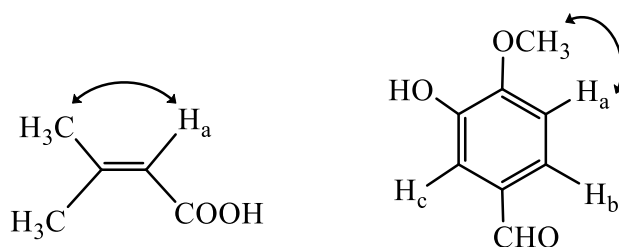
The Nuclear Overhauser Effect was first introduced by Overhauser in 1953. This effect is a change in the intensity of a signal due to a particular proton on double irradiation of nearby protons within the molecule. In this case less intense radiation is used than the spin-spin decoupling or double irradiation. An increase of few percent in the intensity of absorption is observed between protons which are in close proximity. It is generally observed over very short distances, 2-4 Å.

NOE and spin-spin coupling are two different concepts. NOE is the interaction of the nuclei through space and the number of intervening bonds between the concerned protons has no significance, whereas spin-spin coupling takes place via the bonds separating the nuclei.

NOE is associated with dipolar relaxation mechanism. A spin excited nucleus may undergo spin relaxation via the transfer of its spin energy to the nuclei in close proximity. The efficiency of this process is directly related to the distance between the two nuclei. If we irradiate one of the nuclei, the intensity of the nucleus closet in space to the irradiated nucleus will be enhanced in comparison to those far removed. Thus, NOE may be useful in studying the stereochemical relationship between molecules.

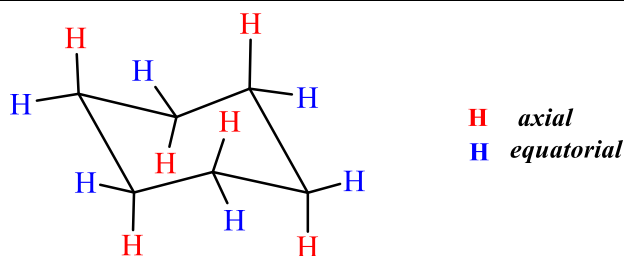
To understand the phenomenon of NOE let's take an example of 3-methyl-2-butanoic acid. If the vinyl proton H_c is irradiated, the intensity (integral value) of the cis-methyl group becomes greater than that due to the trans-methyl group. Similarly, on irradiation of the cis-methyl group, the intensity of the H_c increases, whereas the irradiation of the trans methyl group, which is at more distant, does not affect the intensity of the H_c signal.

Similarly, in isovanillin, irradiation of OCH_3 protons, the integral of the H_a proton (doublet) is increased due to NOE.

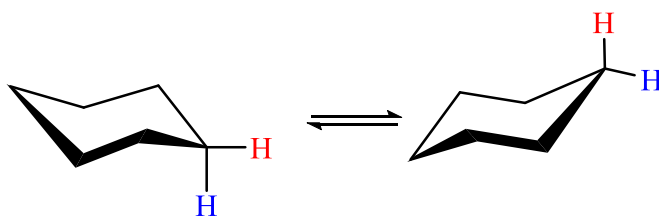


6. Variable Temperature NMR

Protons in cyclohexane are of two types; axial and equatorial. Axial protons occupy a different electronic environment than the equatorial protons. That means axial and equatorial protons are not chemically equivalent hence one can expect two signals in the 1H NMR spectrum; one for the six axial protons and the other for the six equatorial protons. The NMR spectrum of cyclohexane, however, shows only a single sharp peak at δ 1.4.



This can be explained by the fact that at room temperature the rate of ring flipping is high. The NMR spectrometer is too slow to acquire a spectrum of the individual conformations of cyclohexane and only detect the average electronic environment of the protons and hence only one signal is observed. The single peak observed has a chemical shift in between the actual chemical shifts of the axial and the equatorial protons. The rate of ring flipping can be slowed down by decreasing the temperature. If the sample is cool down to -100°C , ring flipping occurs at a very slow rate, and separate signals are observed for the axial and equatorial protons.



Ring Flipping in Cyclohexane